

Stability and nonlinear dynamics of flowing membranes with protein inclusions

From verified eigenvalues to snap-through, flutter, and collective patterns

Integrated research report | Six computational rounds | 4 October 2026

Abstract

We develop a stability and nonlinear-dynamics framework for IRENE and use it to determine how a membrane responds to static pulling and tangential flow around anchored proteins. The generalized eigenproblem is obtained by differentiating the mixed finite-element residual and is tested against exact spectra, cylinder-wake instability, independent plate equations, time integration, and mesh refinement. The analysis exposes a missing vertical protein force condition and a nonconvergent boundary-force diagnostic in the inspected implementation. With a physically force-free protein, the reference flow threshold moves from approximately 16 to 27.45 in the dimensionless drive. Obstacle drag lowers upstream tension, producing compressive stress and a subcritical buckling transition. Direct and adjoint projections predict the backward branch and the two-thirds imperfection law without fitting their coefficients. Nonlinear dynamics develops an upstream pit, lifts the protein, and reaches an overhang. Tangential friction instead permits a saturated source of upstream waves. Static ring pulling reveals why a stable height-controlled shape can snap under force control and why confined tether nucleation costs substantially more than the long-tube force. Mobile curvature proteins couple demixing, advection, and buckling; periodic anchored arrays produce density-dependent long-wave instabilities and terrace states in homogenized equations. This report integrates governing equations, numerical verification, mechanisms, nonlinear outcomes, and all visual assets first added in 2026 into a single account. It identifies which conclusions are resolved and which require further physical, numerical, or collaborative checks.

Contents

1	Questions, physical setting, and research progression	4
1.1	Geometry, material parameters, and control variables	4
1.2	How the six rounds fit together	4
2	Governing geometry and a stability operator from IRENE	4
2.1	Mixed fields and geometric constraints	5
2.2	Dynamic residual, mass matrix, and sign convention	5
2.3	Boundary reduction, shift-invert, and spectral coverage	5
2.4	Elementary exact tests	5
3	Verification: exact membrane spectra and cylinder flow	6
3.1	Flat annulus: exact spectrum and mesh refinement	7
3.2	Cylinder base state and spatial resolution	8
3.3	Complex spectra, competing branches, and onset	10
3.4	Independent time stepping and vortex shedding	12
3.5	Combined benchmark assessment	18
4	Vertical force, boundary work, and physical protein constraints	19
4.1	Virtual work and the missing boundary moment	19
4.2	A conserved vertical stress and consistent reactions	20
4.3	A free boundary is not a force-free rigid inclusion	20
5	Static ring pulling and control-ensemble stability	21
5.1	Relaxed energy and the force-controlled zero mode	21
5.2	Profiles and geometric confinement	23
6	Tether nucleation and continuation beyond a height graph	25
6.1	Confinement-dependent pulling forces	25
6.2	Arclength formulation and endpoint conditions	26
7	From preliminary flow calculations to corrected onset predictions	29
7.1	What the original-rim calculations established	29
7.2	An illustrative slip problem and its numerical limits	32
7.3	Corrected conditions and the quantitative verification ledger	34
8	Flow-driven buckling: compression and the plate reduction	35
8.1	Exact plate operator about the flat base state	35
8.2	Independent mixed discretization and energetic stiffness	36
8.3	Compression and critical-mode geometry	36
8.4	Beyond the first eigenmode	37
9	Outer boundaries, domain size, and where forcing matters	38
9.1	Size, width, and outer-height constraints	38
9.2	Tension and screening: what does and does not become local	39
9.3	Direct and adjoint forcing sensitivity	39
10	Several anchored proteins and an approximate common drag scale	39
10.1	Collective buckling of finite groups	39
10.2	Total finite-patch drag versus dilute per-protein drag	40
11	A computed theory of subcritical snap-through	40
11.1	Continuation and the amplitude equation	41
11.2	Residual derivatives and slaved second-order fields	41
11.3	Fold construction and Koiter sensitivity	41

11.4	The force-free branch and the revised deformation scenario	43
12	Nonlinear collapse, bottlenecks, and a protein that rises	44
12.1	BDF2 stepping and a force-free height solve	44
12.2	Clamped and force-free trajectories	44
13	Parametric surfaces, overhangs, and the limits of remeshing	47
13.1	General frame, mesh gauge, and comparison with Monge dynamics	47
13.2	Mesh regularity and transfer experiments	49
13.3	Established invagination and unresolved tube evolution	49
14	Tangential friction and a saturated source of upstream waves	50
14.1	Follower forcing and the divergence-flutter map	50
14.2	Nonlinear saturation and the below-threshold restart	51
15	Consistent equations for mobile curvature proteins	53
15.1	Energy, density convention, and nonlinear virtual power	53
15.2	Analytical dispersion and relative advection	54
16	Demixing, transported domains, and the treatment of outflow	55
16.1	Energy relaxation, domains, and a source-driven plume	55
16.2	Transport scales and boundary artifacts	58
16.3	Final mobility regimes with absorbing outflow	59
17	Mechanochemical mode merger and nonlinear runaway	60
17.1	Boundary-specific onsets and destabilizing cubic coefficients	60
17.2	A candidate degenerate double-zero organizing centre	60
17.3	What the nonlinear trajectories resolve	61
18	A recruited protein wake acts as an imperfection	62
18.1	Projected forcing, fold shifts, and snap sharpness	62
19	Periodic anchored arrays and long-wave collective onset	63
19.1	Bloch perturbations and density-controlled thresholds	63
20	Nonlinear lattice stresses and one-dimensional terraces	65
20.1	Tilted cells, rigid-rim geometry, and frozen-flow closure	65
20.2	A convected slope equation and density-dependent criticality	66
21	Two-dimensional terraces and finite patches	67
21.1	Anisotropic coefficients and transverse stability	67
21.2	Convective versus absolute instability	68
22	Physical scales, synthesis, and reproducibility	69
22.1	Dimensional force, velocity, and time scales	69
22.2	Established conclusions and remaining scientific questions	69
22.3	Implementation and evidence map	69
22.4	Reproduction of numerical results and document sources	70

1 Questions, physical setting, and research progression

The project began with a practical question: how can a stationary surface computed by IRENE be classified as stable or unstable? Answering it changed the physical interpretation of both protein pulling and flow-driven deformation. The work then progressed from verification to force balance, corrected onset calculations, nonlinear snap-through, strongly deformed surfaces, and mechanochemical and collective extensions. The presentation below follows those dependencies rather than the order in which files were created.

Membrane shape and membrane flow are coupled through curvature, incompressibility, and surface stress. A protein inclusion supplies both a geometric constraint and an obstacle to tangential motion. Steady deformation alone cannot establish whether a computed membrane persists under perturbations. This distinction is especially important when a force–displacement curve softens or when flow generates compressive stress.

IRENE provides a finite-element formulation for viscous fluid surfaces [1]. The large-deformation protein study of Ferraro and Castellana [2] motivates two questions addressed here: which ring-pulling equilibria are stable, and whether sufficiently strong flow destabilizes a membrane around an inclusion. We extend the steady solver with linear spectra, continuation, adjoint projections, and nonlinear dynamics. The central result is a flow-induced snap, with friction-driven flutter as a stable dynamic alternative.

1.1 Geometry, material parameters, and control variables

The membrane has bending rigidity κ , surface viscosity η , and imposed reference tension σ_0 . In the Monge representation its surface is $\mathbf{X} = (x, y, z)$, with $g_{ij} = \delta_{ij} + z_{,i}z_{,j}$, area element $dA = \sqrt{g}dx dy$, mean curvature $\mathcal{H} = \frac{1}{2}g^{ij}b_{ij}$, and Gaussian curvature $\mathcal{K}$. The no-flow energy is

$$E[z] = \int_{\Omega} (2\kappa\mathcal{H}^2 + \sigma_0) dA. \quad (1)$$

The anchored inclusion is a rigid disc of radius r_0 : anchoring fixes its lateral position, while its vertical height may be prescribed or force-free. The contact slope is denoted $s = \tan \alpha$; it is not time. For flow through a square of side L , the drive and tension parameters are $SL = \eta v_0 L / \kappa$ and $\Gamma = \sigma_0 L^2 / \kappa$. The reference problem has $L = 100r_0$, $\Gamma = 25$, no slip on the inclusion, and zero contact slope. Normal friction $\zeta > 0$ gives a relaxation time $\tau_n = \zeta r_0^4 / \kappa$. In-plane friction b , when present, is a separate coupling with screening length $\ell_b = \sqrt{\eta/b}$.

The central mechanism and corrected onset are developed in Sections 7–9; the nonlinear snap, collapse, fold, and flutter in Sections 11–14. Sections 15–18 extend the dynamics to mobile curvature proteins, and Sections 19–21 treat collective anchored arrays. The preceding methods and static results provide the force conventions and validation on which those interpretations depend.

1.2 How the six rounds fit together

The initial work established a general eigenvalue tool and verified equilibrium and flow calculations. The corrected physical boundary conditions then made the onset problem comparable with an independent plate discretization. Round 3 expanded the analysis to multiple proteins, weakly nonlinear snap theory, flutter maps, periodic arrays, and no-flow tube continuation. Rounds 4 and 5 developed force-free post-snap dynamics, consistent mobile-protein equations, nonlinear lattice closures, saturated flutter, and a parametric description of folding. Round 6 tested two-dimensional terraces, the merger of mobile-protein modes, source-induced folds, and remeshing limitations.

Two distinctions recur throughout. First, a boundary condition is part of the physical problem: a free rim, a rigid force-free protein, and a prescribed protein height are different constraints. Second, linear onset, local nonlinear criticality, and the eventual large-amplitude state are separate questions. A subcritical cubic coefficient does not identify a terminal shape, and a stable reduced-equation terrace does not prove stability of the unreduced three-dimensional-fluid problem. The figures are therefore placed with the calculation they support and with its relevant limitations. All 84 tracked visual assets first added in 2026 are included; animation storyboards accompany their original bundled GIFs.

2 Governing geometry and a stability operator from IRENE

The method treats stability as a property of the actual constrained residual. This avoids substituting a guessed shape equation for a mixed viscous-surface discretization and makes boundary and force inconsistencies visible in the spectrum.

2.1 Mixed fields and geometric constraints

For a height function $z(x^1, x^2)$, the tangent vectors, metric, inverse metric, normal, and curvature are

$$\mathbf{e}_i = \mathbf{e}_i^{\text{flat}} + z_{,i} \mathbf{e}_z, \quad g_{ij} = \delta_{ij} + z_{,i} z_{,j}, \quad g = 1 + |\nabla z|^2, \quad (2)$$

$$g^{ij} = \delta_{ij} - \frac{z_{,i} z_{,j}}{g}, \quad \mathbf{n} = \frac{(-z_{,1}, -z_{,2}, 1)}{\sqrt{g}}, \quad b_{ij} = \frac{z_{,ij}}{\sqrt{g}}, \quad (3)$$

$$\mathcal{H} = \frac{1}{2} g^{ij} b_{ij}, \quad \mathcal{K} = \frac{\det b}{\det g}. \quad (4)$$

IRENE avoids a fourth-order scalar element by treating $\boldsymbol{\omega} = \nabla z$ and $\mu = \mathcal{H}$ as independent mixed fields. The flow unknowns are $(\mathbf{v}, w, \sigma, z, \boldsymbol{\omega}, \mu)$, where σ enforces surface incompressibility, $\nabla_i v^i - 2\mathcal{H}w = 0$. The flow element in the inspected source is continuous P_2 for the two tangential velocity components and continuous P_1 for $w, \sigma, z, \boldsymbol{\omega}, \mu$. The protein extension adds continuous P_2 fields (φ, m) . The independent plate discretization instead uses mixed $P_2 \times P_2$ for (z, m_Δ) , with $m_\Delta = \Delta z$. These choices should not be conflated.

The weak residual includes tangential and normal force balance, kinematics, the gradient and curvature constraints, and boundary penalty/Nitsche terms. Automatic differentiation differentiates all retained terms, including geometry and boundary terms. Thus the spectrum is of the actual discretized residual, not a separately guessed continuum operator.

2.2 Dynamic residual, mass matrix, and sign convention

The convention used by the generic solver is

$$\mathbf{B}(\Psi)\dot{\Psi} + \mathcal{R}(\Psi) = 0, \quad \mathbf{A} = -D\mathcal{R}(\Psi_*), \quad \mathbf{A}x = \lambda \mathbf{B}_* x. \quad (5)$$

For the flow formulation, $\mathcal{R}$ is the corrected flow residual. Its time form is

$$\mathcal{M}(\hat{\Psi}, \nu) = \int_{\Omega} [\rho \hat{v}^i \nu_{v_i} + \rho \hat{w} \nu_w + \hat{z} \nu_z] \sqrt{g_*} d^2 x. \quad (6)$$

The protein problem adds $\hat{\varphi} \nu_\varphi$. At $\rho = 0$ the velocity fields are slaved; adding ζw to the normal force equation supplies the required overdamped shape relaxation. Algebraic fields have no arbitrary mass. Their zero rows in $\mathbf{B}$ enforce the constraints in every finite eigenmode.

For the no-flow ring, the implementation explicitly sets $\mathcal{R} = -F_{\text{IRENE}}$ and uses $\mathcal{M} = \frac{1}{2} \zeta \int \hat{z} \nu_z d^2 x$. The factor 1/2 follows from IRENE's half-normal-force weak form. Applying $\mathbf{A} = -DF_{\text{IRENE}}$ directly to this ring form would reverse the stability sign. The report consistently uses $\mathcal{R}$, not an ambiguous bare Jacobian.

2.3 Boundary reduction, shift-invert, and spectral coverage

Perturbations satisfy homogeneous essential conditions. The source replaces constrained rows in $\mathbf{A}$ by identity and those in $\mathbf{B}$ by zero; these constraints correspond to infinite generalized eigenvalues and map to zero under shift-invert. Rigid-rim degrees of freedom are reduced with a prolongation matrix $\mathbf{P}$: $\mathbf{A}_r = \mathbf{P}^T \mathbf{A} \mathbf{P}$, $\mathbf{B}_r = \mathbf{P}^T \mathbf{B} \mathbf{P}$. One height is retained per rigid protein rather than one independent height at each rim node.

The generalized non-Hermitian problem is solved by SLEPc Krylov–Schur with shift-invert and MUMPS factorization [6]. A typical tolerance is 10^{-10} . A real build can search near a complex shift $s = a + ib$ using the real block matrix

$$\begin{pmatrix} \mathbf{A} - a\mathbf{B} & b\mathbf{B} \\ -b\mathbf{B} & \mathbf{A} - a\mathbf{B} \end{pmatrix}. \quad (7)$$

Residuals and independent spectral shifts matter: a real shift near zero alone can miss a Hopf pair. The cylinder solver scans shifts $0.1 + 0.3ki$, $k = 0, \dots, 5$. Membrane searches target the much smaller membrane frequencies. Sorting the returned pairs by real part is not, by itself, proof that every potentially leading mode has been found.

2.4 Elementary exact tests

Before interpreting any membrane eigenvalue, the generic machinery was checked on problems with known real and complex spectra. These checks distinguish a failure to differentiate or solve the generalized problem from an error specific to membrane geometry. An incompressibility test also compared the mixed formulation with the correct mass-aware discrete projector: substituting a Euclidean projector changes the problem and cannot serve as a reference.

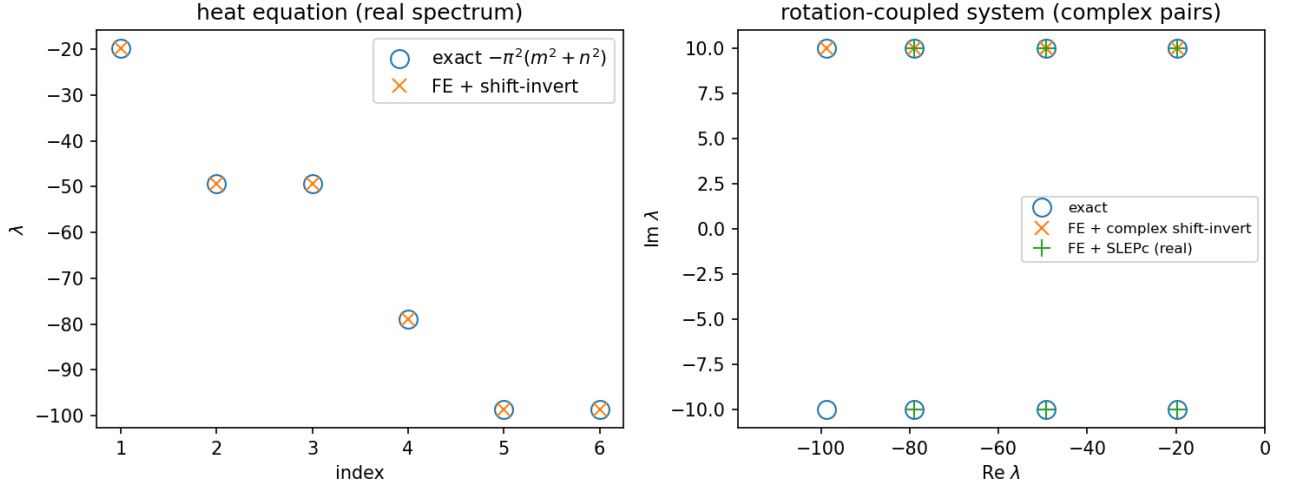


Figure 1: Generic eigensolver checks on a heat equation and a rotation-coupled system with complex eigenpairs. Both real and complex spectral calculations are tested against exact values.

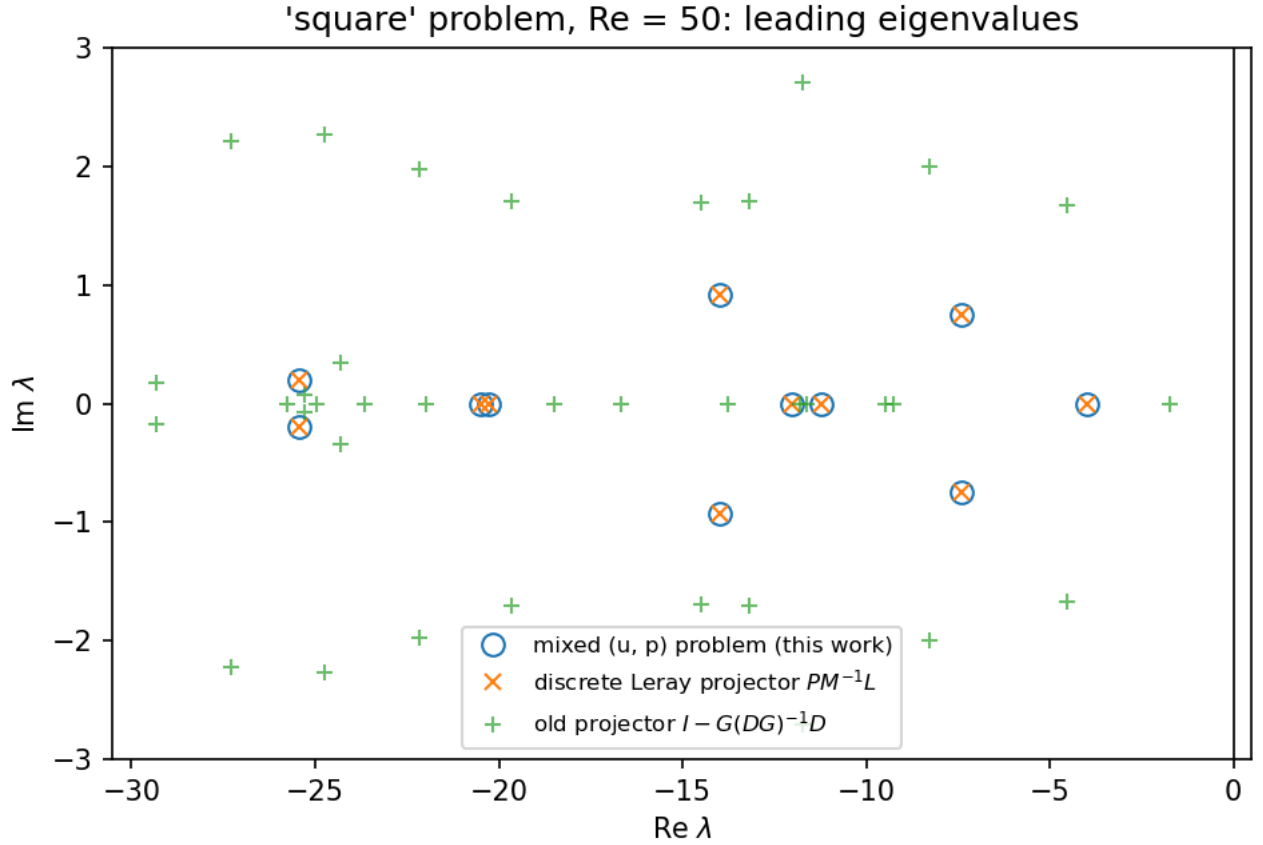


Figure 2: Mixed incompressible eigenproblem compared with the correct discrete Leray projector. A Euclidean projector is a different operator and produces different spectra.

3 Verification: exact membrane spectra and cylinder flow

Verification was layered. The flat annulus tests the bending and tension spectrum with geometric boundary conditions, while cylinder flow tests mixed constraints, complex eigenvalues, and agreement with nonlinear dynamics. Neither benchmark alone validates every large-deformation membrane calculation.

3.1 Flat annulus: exact spectrum and mesh refinement

For a flat membrane with both heights and radial slopes fixed, $\zeta z_t = -\kappa \Delta^2 z + \sigma_0 \Delta z$. A separated perturbation $f_m(r)e^{im\theta+\lambda t}$ has radial basis

$$f_m = aJ_m(kr) + bY_m(kr) + cI_m(qr) + dK_m(qr), \quad \Lambda = \kappa k^4 + \sigma_0 k^2, \quad q^2 = k^2 + \sigma_0/\kappa, \quad (8)$$

with $\lambda = -\Lambda/\zeta$. The determinant of the four basis values and their radial derivatives at r_0 and R vanishes. The FE spectra recover the $m \geq 1$ double degeneracy and twenty eigenvalues across $m = 0, \dots, 6$. The maximum relative discrepancy falls from 1.2×10^{-3} to 2.3×10^{-4} when the minimum mesh size falls from $0.05r_0$ to $0.025r_0$.

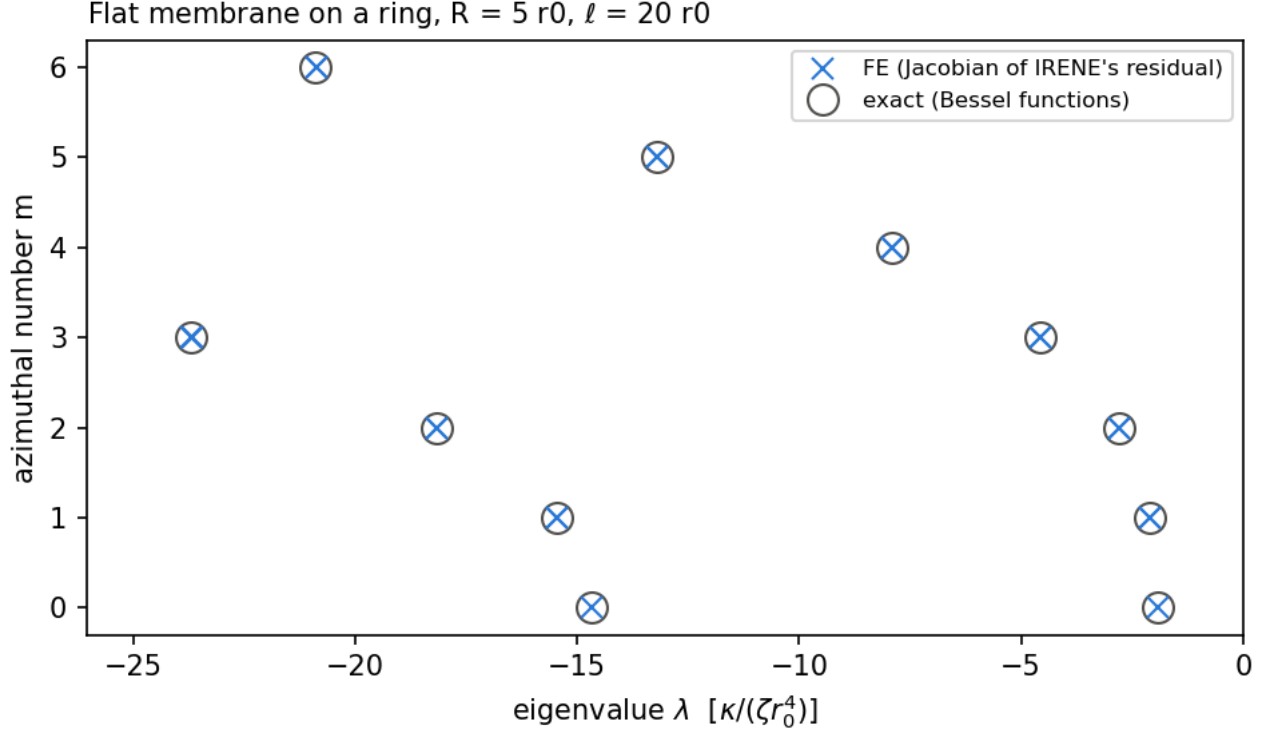


Figure 3: Coarse-mesh flat-annulus eigenvalues by azimuthal number compared with the exact Bessel spectrum. This is a separate asset from the fine-mesh check with the same original basename.

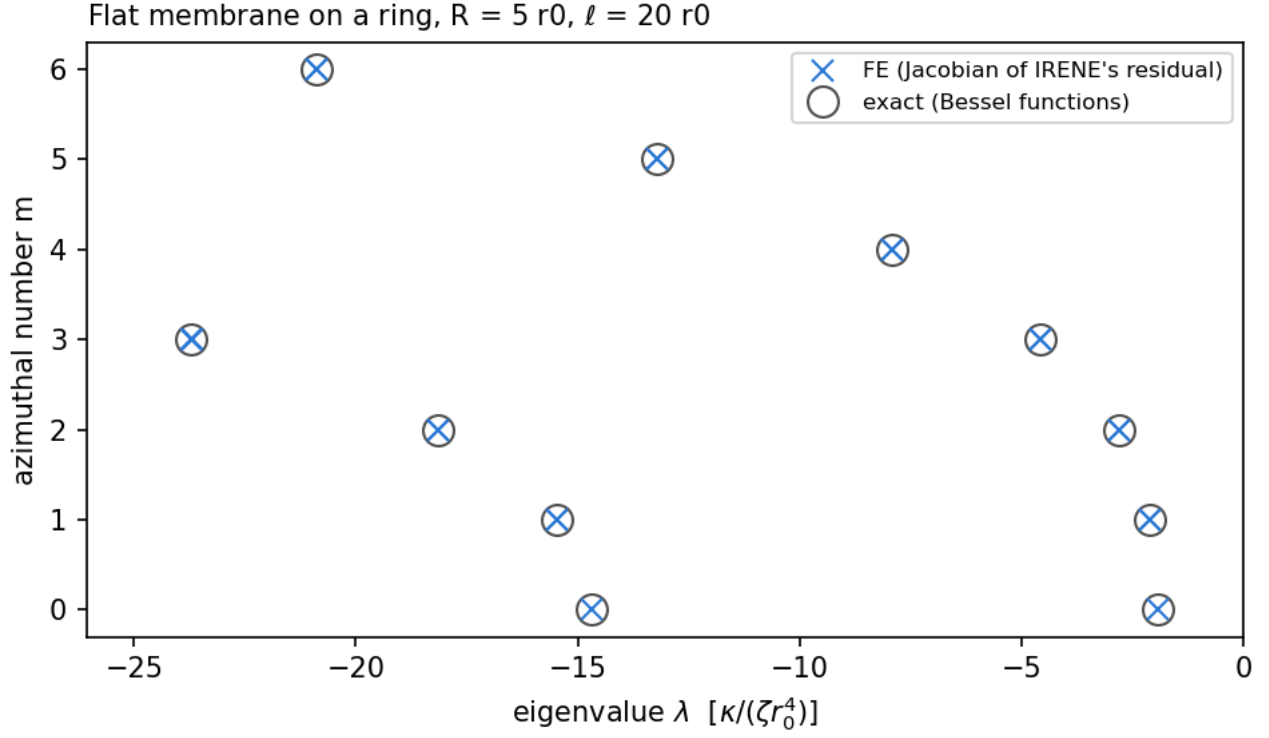


Figure 4: Fine-mesh flat-annulus spectrum. Refinement reduces the maximum discrepancy to about 2.3 times ten to the minus four across the tested twenty eigenvalues.

3.2 Cylinder base state and spatial resolution

The cylinder benchmark separates the stationary flow solver from the stability solver. A symmetric stationary wake remains a legitimate base state even above the time-dependent shedding threshold. Three meshes resolve the obstacle and wake; drag and recirculation length are compared independently of the eigenvalues.

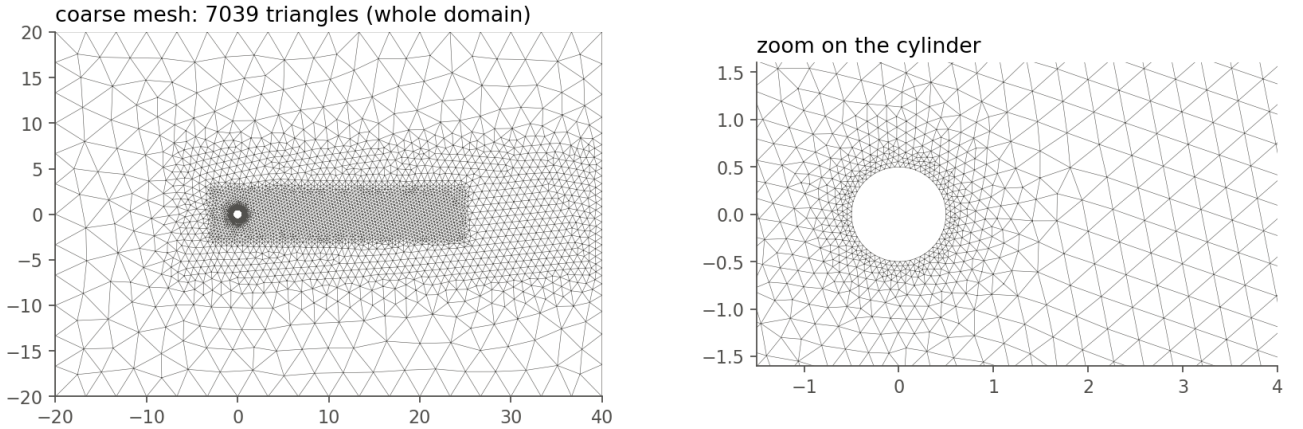


Figure 5: Coarse cylinder mesh, including a close view of the obstacle. This is the first of the three benchmark resolutions.

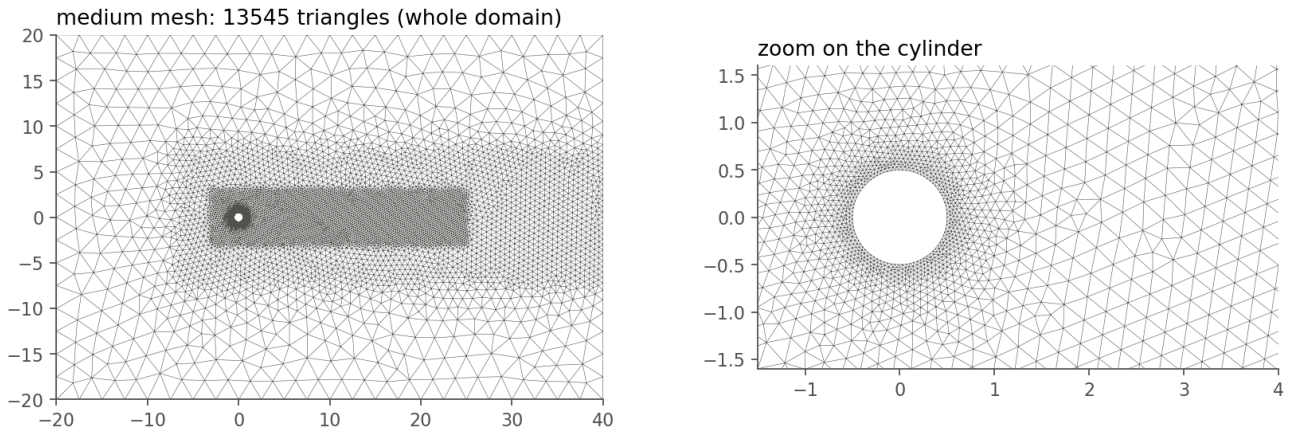


Figure 6: Intermediate cylinder mesh and obstacle refinement. It supplies the middle point of the convergence sequence.

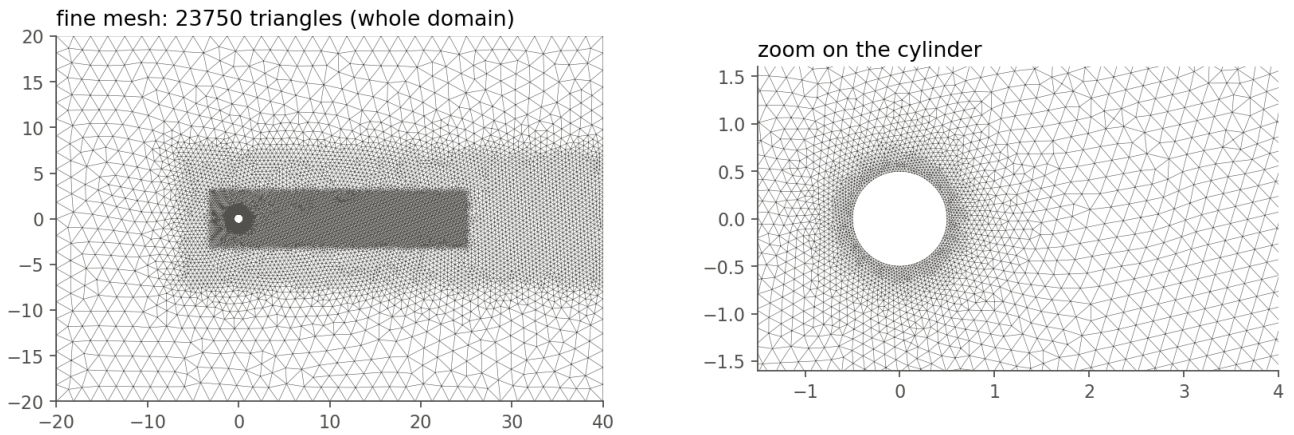


Figure 7: Fine cylinder mesh and obstacle refinement. It gives the reported benchmark onset near Reynolds number 46.32.

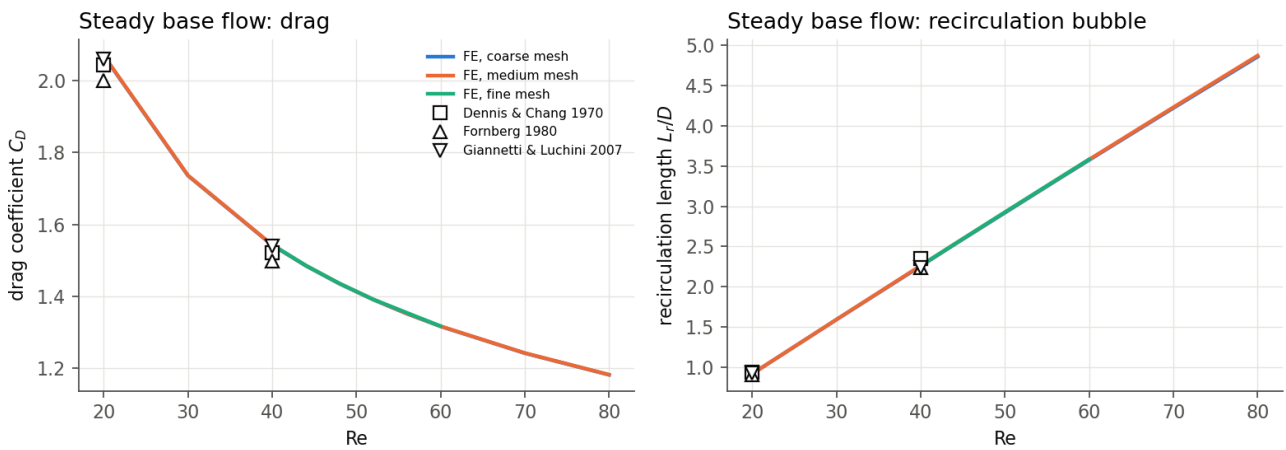


Figure 8: Cylinder drag and recirculation length compared with reference values. This tests the base solver independently of the eigensolver.

Left: the steady solution w^* (it exists at every Re , like a pencil balanced on its tip). Right: the perturbation shape that decides whether w^* survives (amplitude arbitrary). The verdict is the sign of $Re \lambda$; what is actually observed above Re_c (vortex shedding) is in `dns_vorticity.png`.

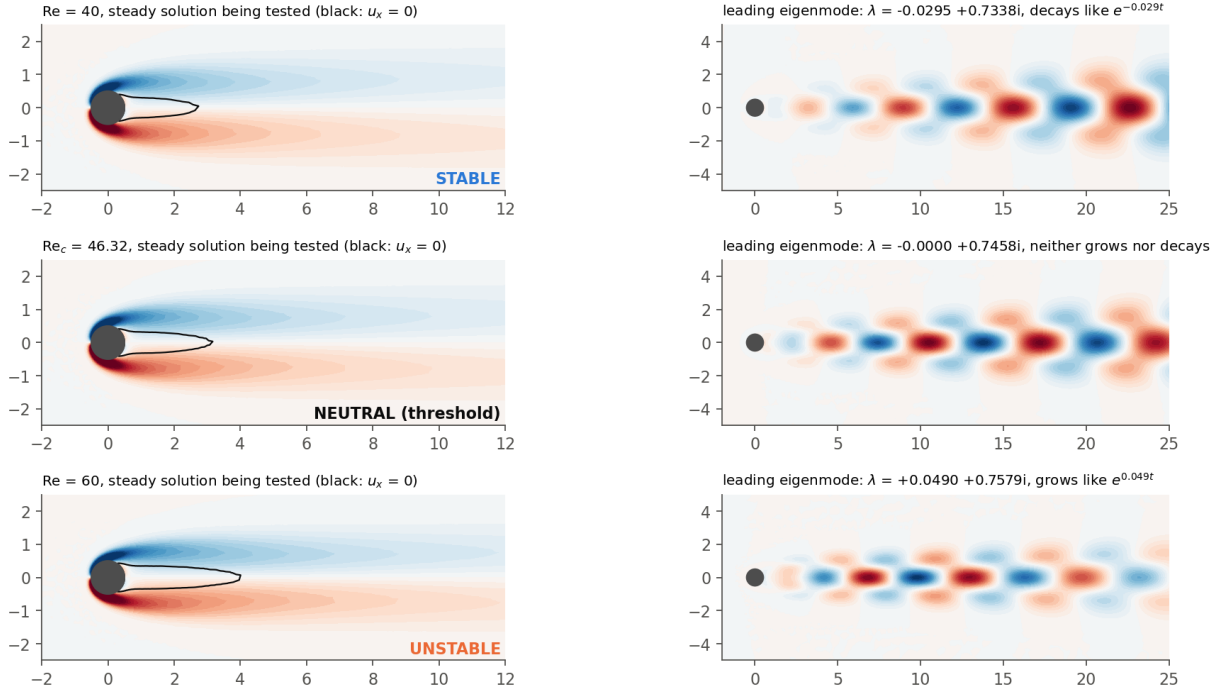


Figure 9: Steady cylinder base flows and leading perturbations below, near, and above the Hopf threshold. The unstable symmetric steady solution is a valid base state even when nonlinear dynamics sheds vortices.

3.3 Complex spectra, competing branches, and onset

The eigensolver searches several complex shifts because a search confined to the real origin can miss the oscillatory pair. The trajectories and competing branches below explain how the first Hopf crossing was selected. The critical Reynolds number is 46.46, 46.35, and 46.32 on the three meshes, close to the approximately 46.7 reference in Ref. [5]. The remaining difference is a finite-domain and discretization comparison, not a universal eigensolver tolerance.

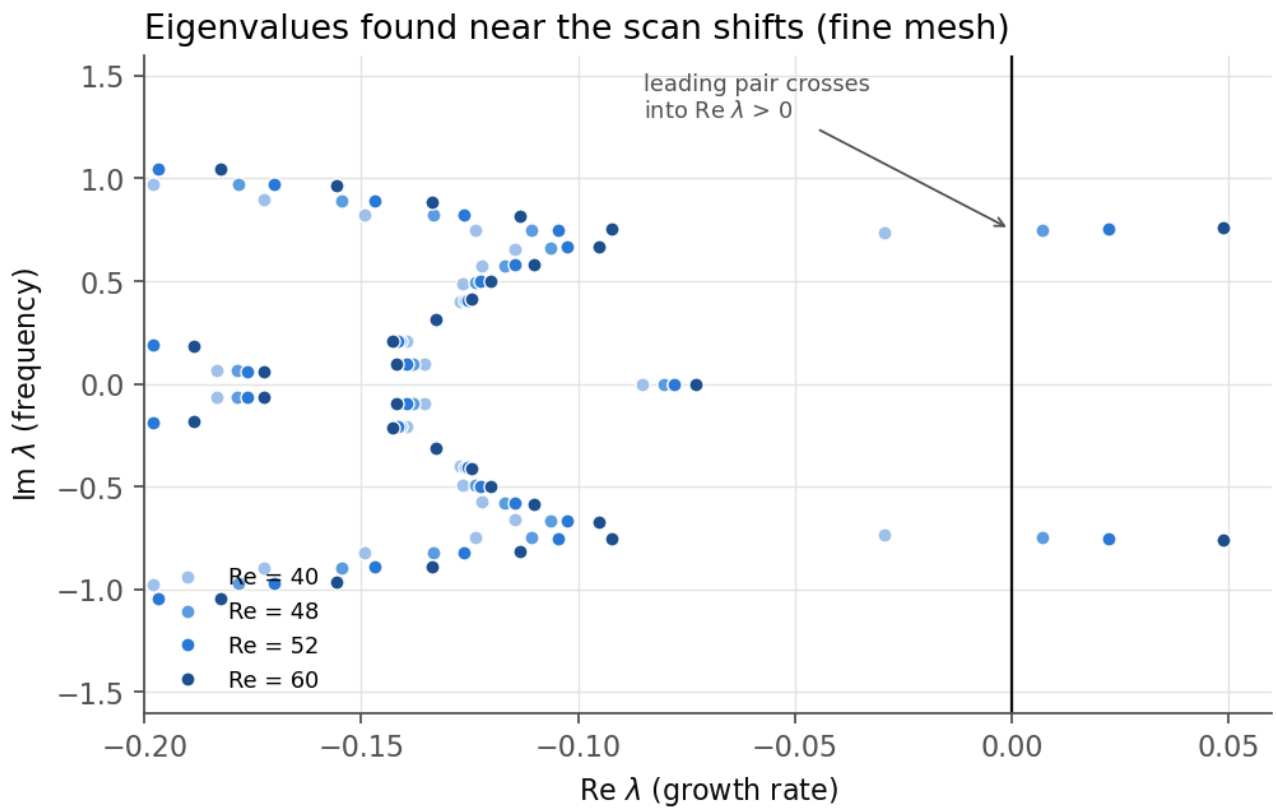


Figure 10: Cylinder spectra near the sampled complex shifts. The Hopf pair and dense damped cluster illustrate why a real shift alone is insufficient.

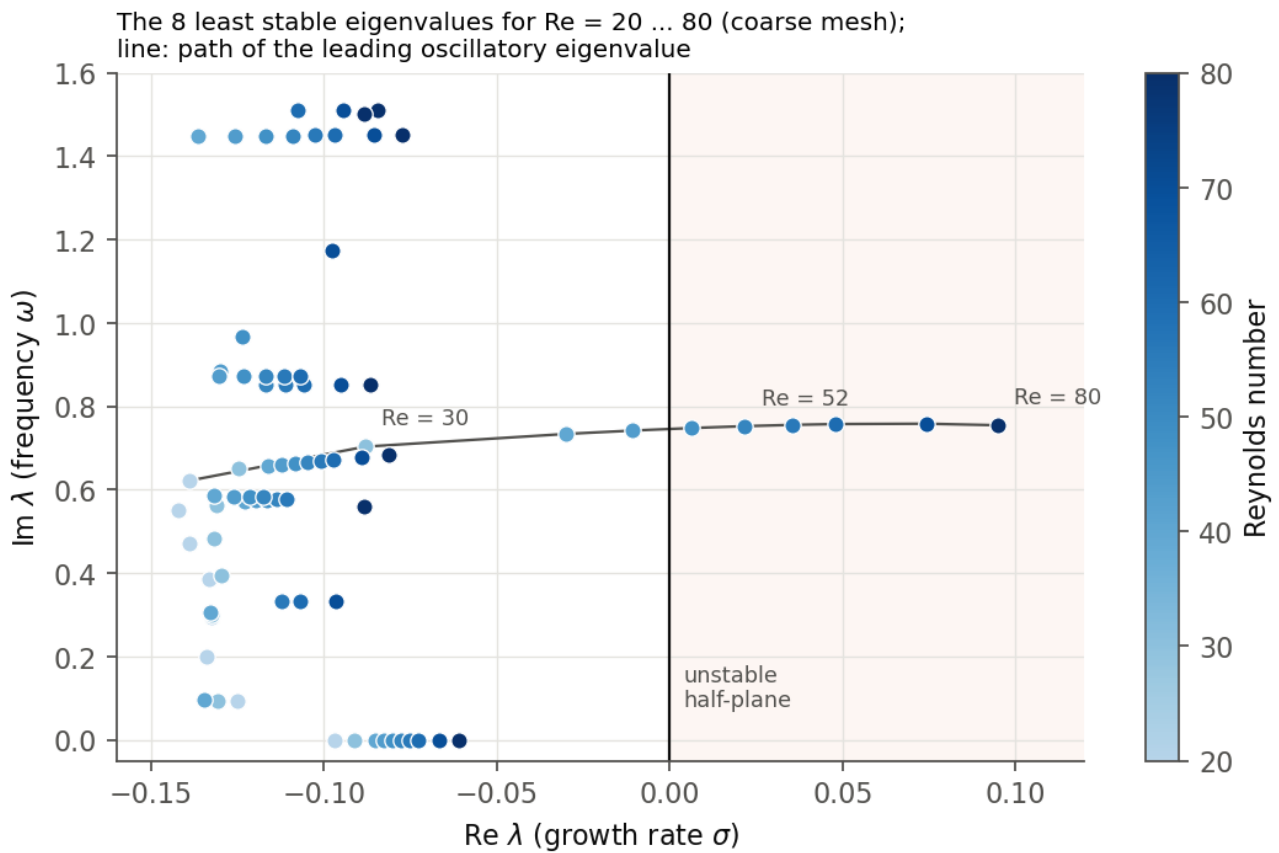


Figure 11: Cylinder eigenvalue trajectories as Reynolds number varies. The relevant oscillatory pair can be missed by searching only near the origin.

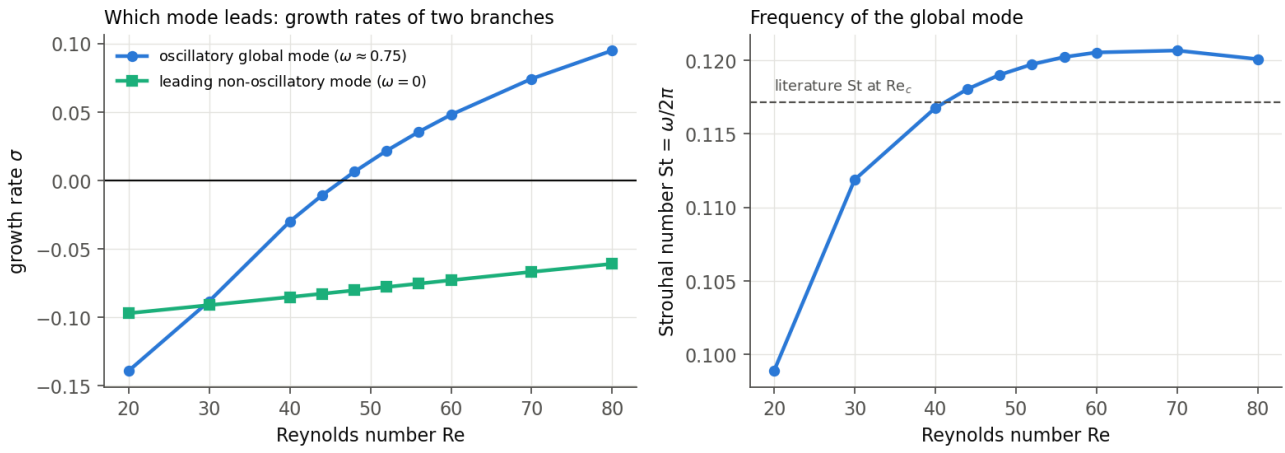


Figure 12: Competing cylinder growth-rate branches and the leading Hopf Strouhal number. Selecting the least damped returned eigenpair requires covering the relevant complex-frequency range.

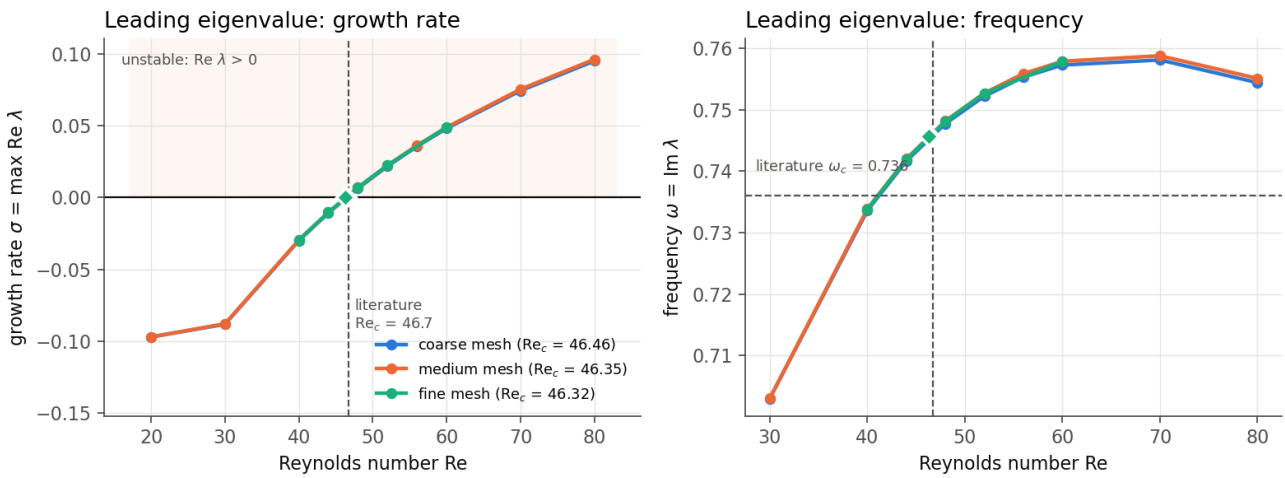


Figure 13: Leading cylinder growth rate and frequency versus Reynolds number on three meshes. The first instability occurs near Reynolds number 46.3.

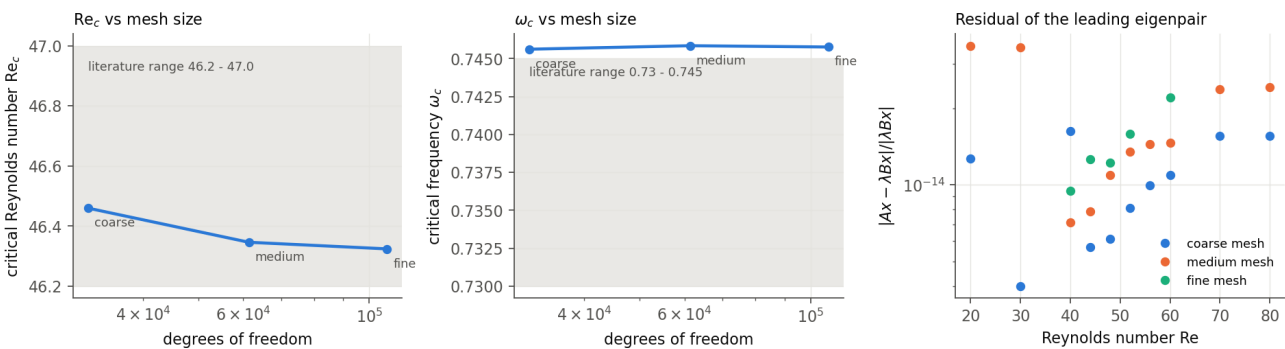


Figure 14: Cylinder onset, frequency, and eigenpair residuals under mesh refinement. The residual checks numerical solution of the discrete problem; mesh changes test its approximation to the continuum.

3.4 Independent time stepping and vortex shedding

Small perturbations are first fitted during their linear interval, when a computed eigenvalue should predict the growth rate and frequency. Subsequent saturation is a different nonlinear result. Below onset the wake returns to its symmetric state; above onset it develops vortex shedding. The growth-rate comparisons are within a few percent, and the frequency comparisons within about 0.2 percent in the recorded runs.

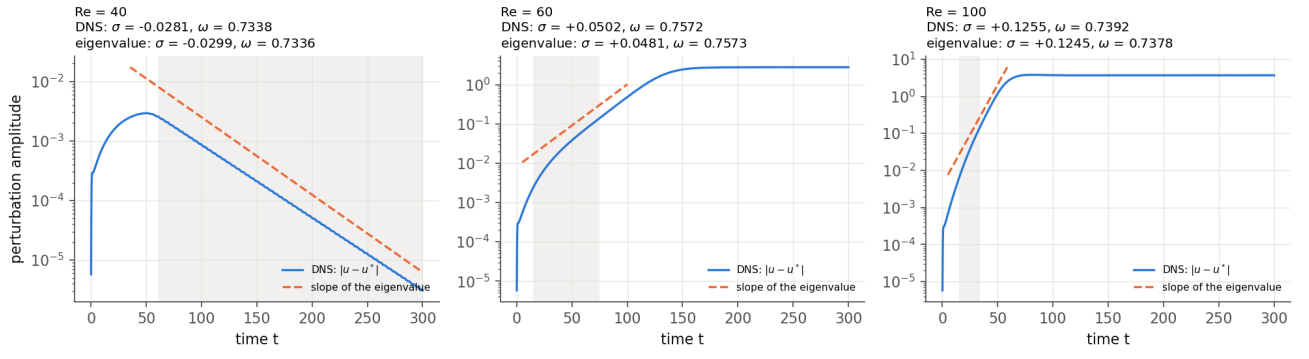


Figure 15: Perturbation amplitude in cylinder DNS compared with exponential eigenvalue predictions. Growth is fitted in the linear interval before nonlinear saturation.

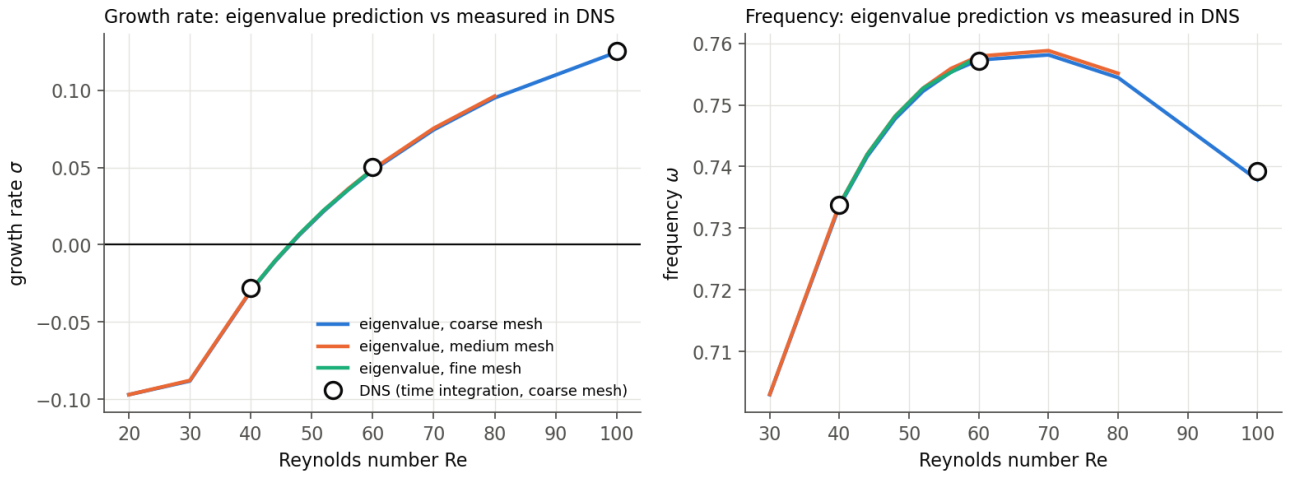


Figure 16: Cylinder growth rates and frequencies from eigenanalysis and nonlinear time integration. Agreement is strongest for frequency and within a few percent for the fitted growth rates.

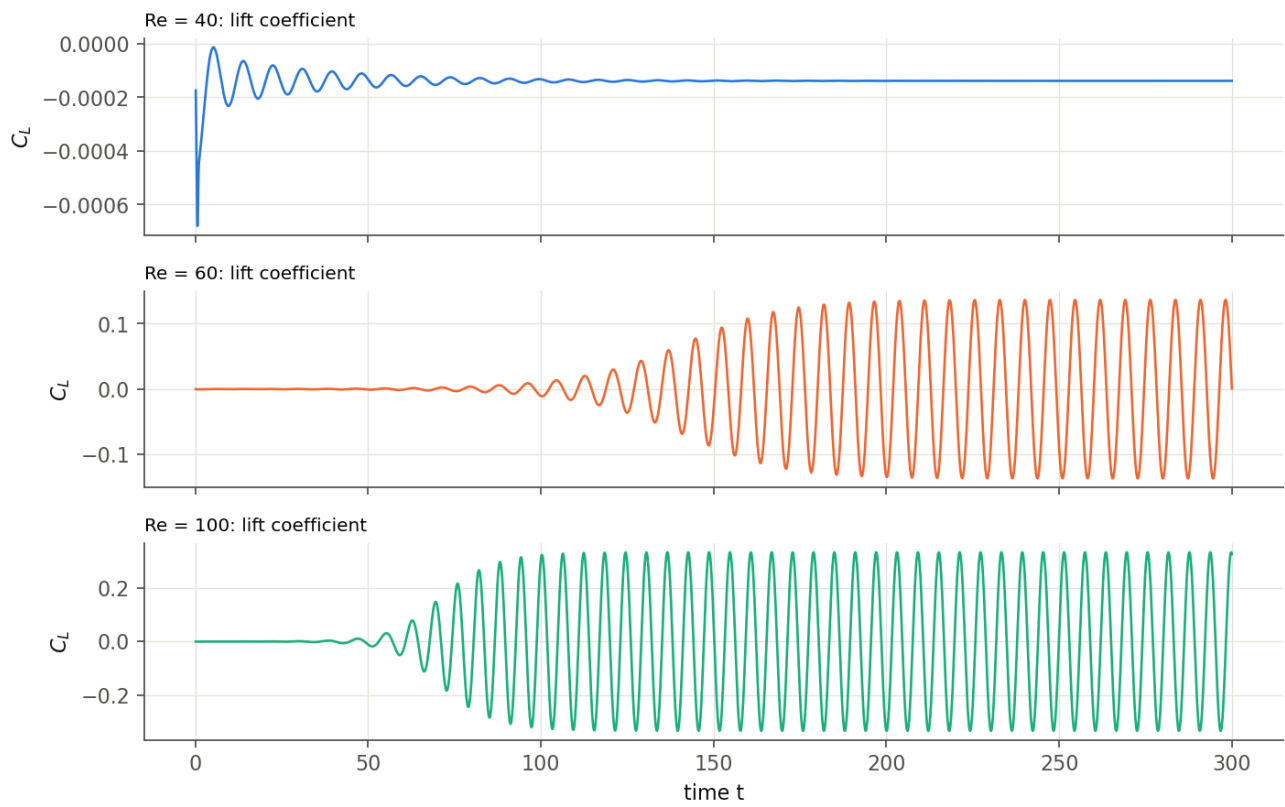


Figure 17: Lift time series at Reynolds numbers 40, 60, and 100. The low-Reynolds-number perturbation decays; the unstable cases grow and saturate in vortex shedding.

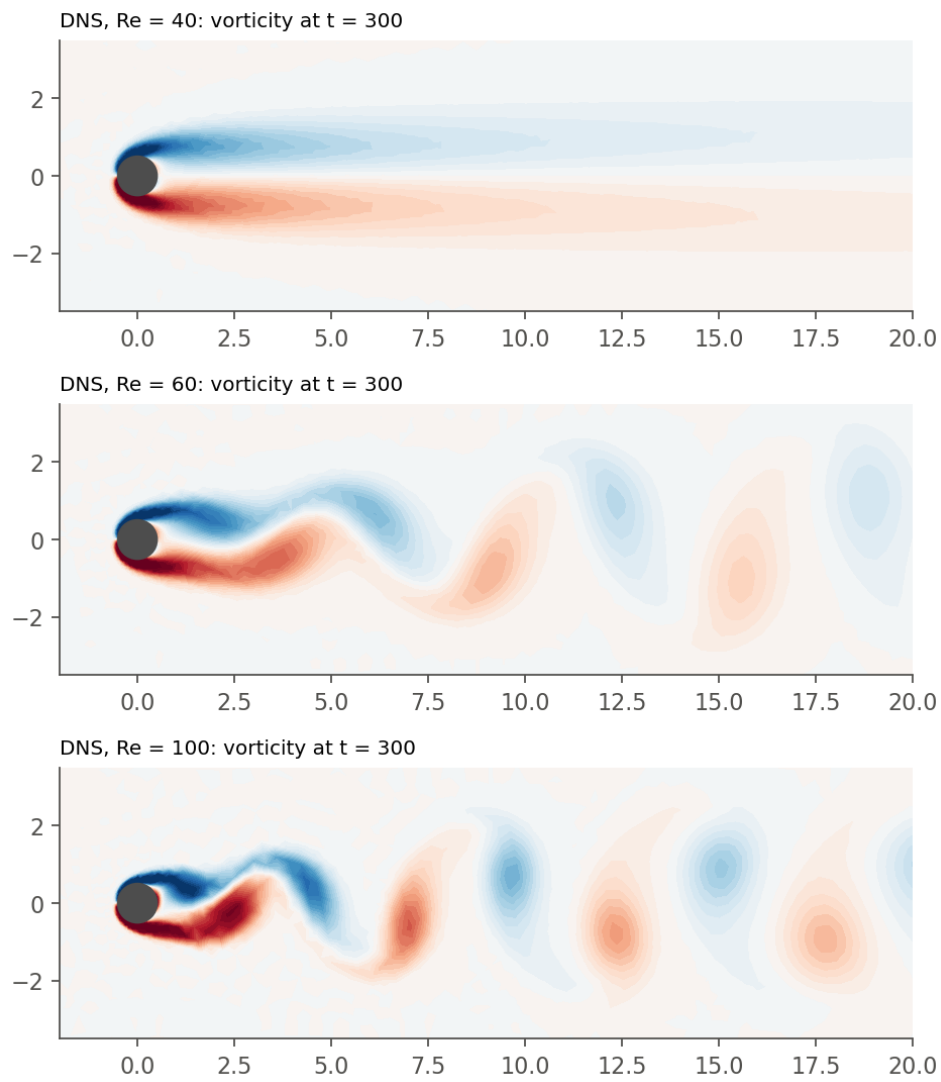


Figure 18: Nonlinear cylinder-wake snapshots for the benchmark runs. The vortex street above onset provides an independent dynamical check of the computed Hopf instability.

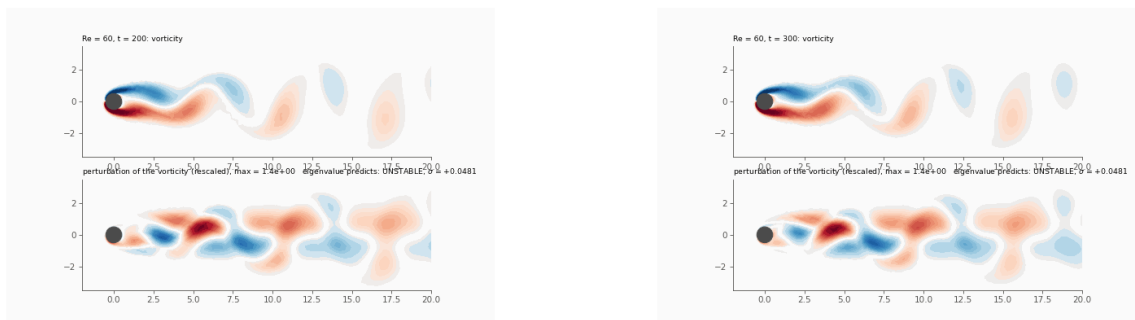


Figure 19: Cylinder benchmark below onset: an initial perturbation decays to a steady symmetric wake. Four ordered frames are selected by frame index; the spacing is not assumed uniform in physical time. The original GIF is included in the source package.



Frame 1/151

Frame 51/151



Frame 101/151

Frame 151/151

Figure 20: Cylinder benchmark above onset: growing perturbations develop into a vortex street. Four ordered frames are selected by frame index; the spacing is not assumed uniform in physical time. The original GIF is included in the source package.

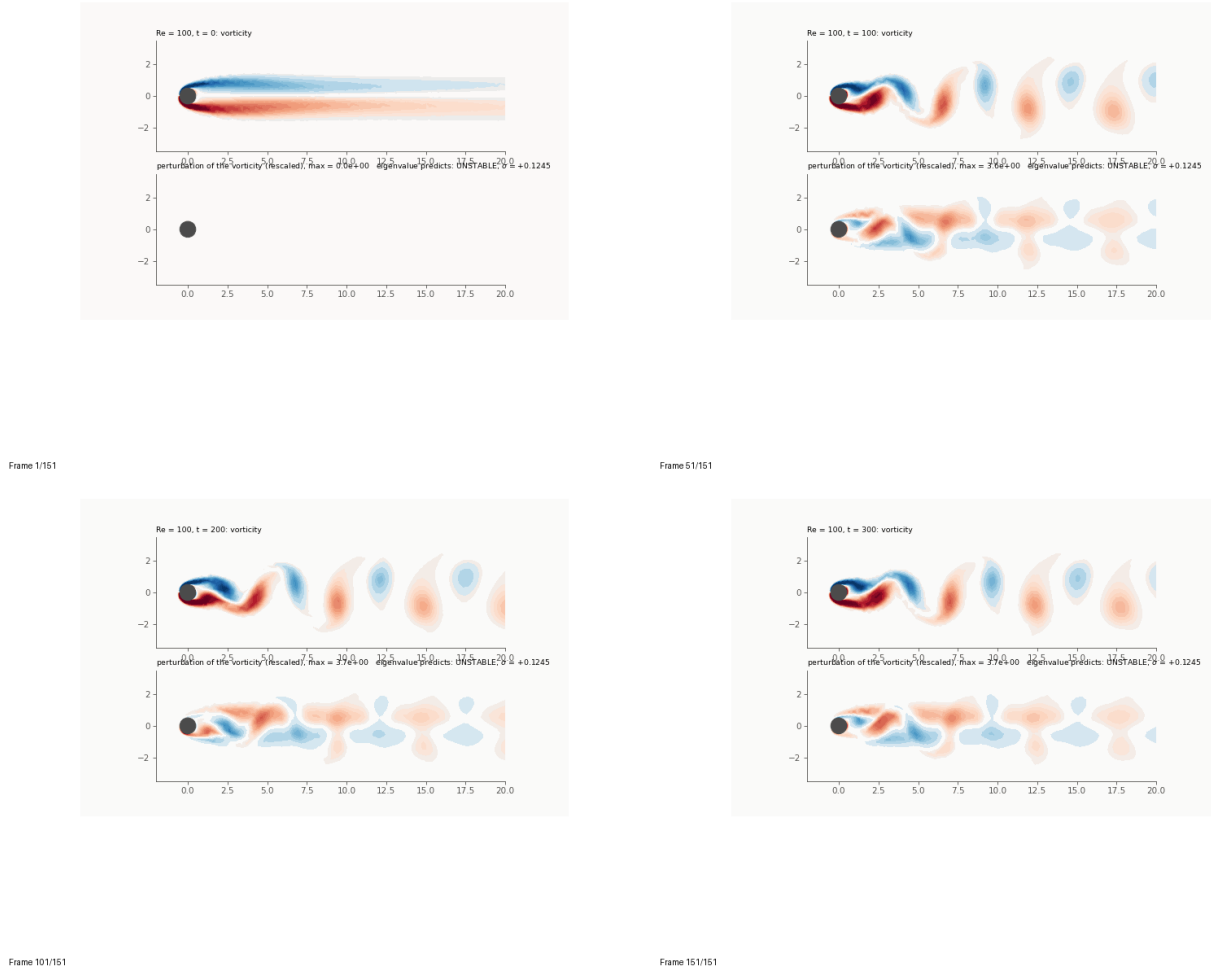


Figure 21: Cylinder benchmark at Reynolds number one hundred: nonlinear saturated vortex shedding. Four ordered frames are selected by frame index; the spacing is not assumed uniform in physical time. The original GIF is included in the source package.

3.5 Combined benchmark assessment

The stability map combines the independent classifications, and the summary connects the spectral crossing with the contrasting time-dependent outcomes. Figures 22 and 23 collect the benchmark evidence on which the later membrane stability workflow rests.

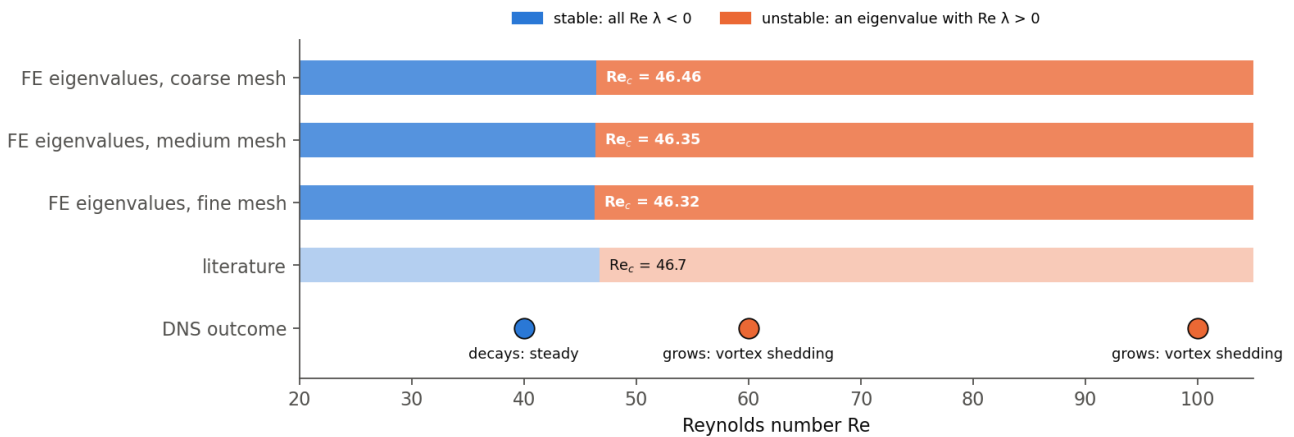


Figure 22: Cylinder stable/unstable classification from eigenvalues and independent DNS outcomes. Threshold differences across meshes are small but remain measurable.

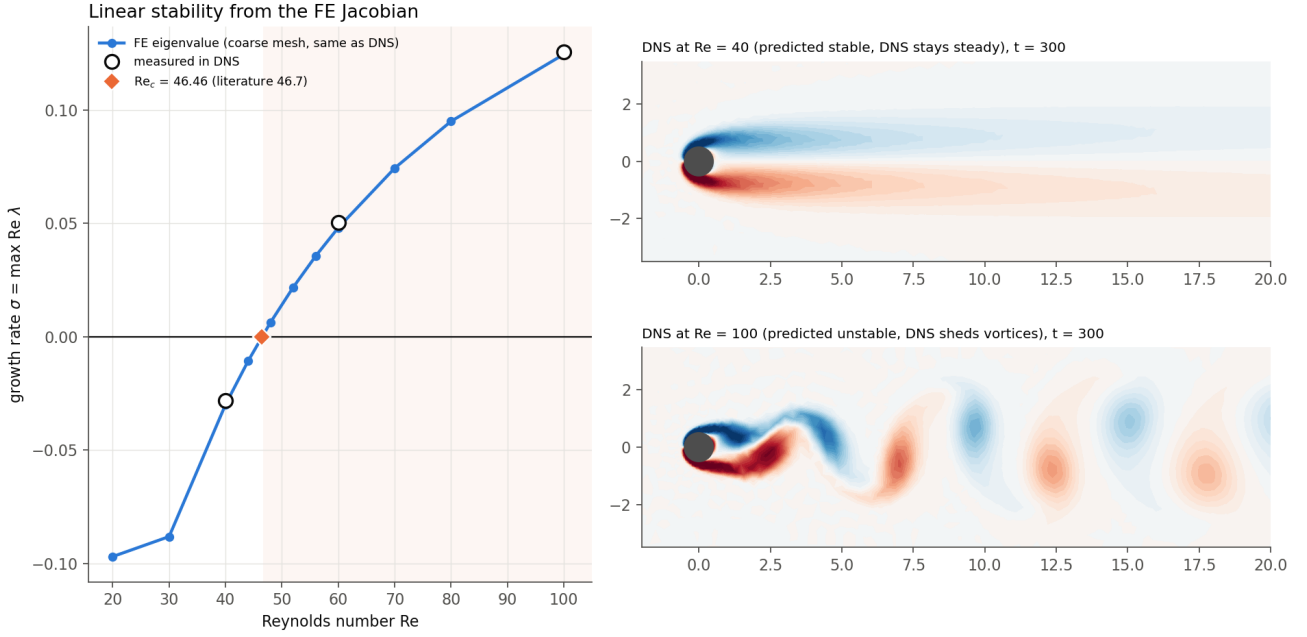


Figure 23: Overview of the cylinder benchmark: leading growth rate, threshold, and contrasting nonlinear outcomes below and above onset.

4 Vertical force, boundary work, and physical protein constraints

Stability analysis exposed formulation issues that are easy to overlook in a stationary shape calculation. The central question is which variations are admissible at a protein boundary and what force equation closes the protein height.

4.1 Virtual work and the missing boundary moment

For a rigid vertical displacement δh , the normal virtual displacement is $\psi = \delta h n_z$, which varies along a tilted surface even though δh is constant. The bending first variation contains a boundary moment contribution of the form

$$\oint 2\kappa \mathcal{H} n_{\partial\Omega}^i \nabla_i \psi \, d\ell. \quad (9)$$

The implemented tension-plus-bending line-force diagnostic omits this contribution to vertical work. Adding it improves a coarse-mesh comparison to about 6%, but does not solve the numerical issue of normal curvature derivatives evaluated at a weakly imposed rim. The sign of the contribution depends on boundary orientation; the code fixes it by comparison with $F_m = -E'_*(h)$.

At $R = 5r_0$, $s = 0.5$, $h = r_0$, the line integral changes from about -1.15 to $-0.48\kappa/r_0$ on refinement, whereas virtual work gives $-4.89\kappa/r_0$. Small-deformation exact tests give virtual-work errors 3×10^{-5} – 1.6×10^{-4} on the fine mesh, while the line-force errors can increase. A catenoid has $\mathcal{H} = 0$, so its moment term vanishes, yet the recorded line integral still fails to converge. This separates the missing physical term from the boundary-derivative discretization error. At very low tension the catenoid force is tiny relative to bending-error scales: the fine virtual-work relative error is about 1.7%, not uniformly 10^{-5} . The report does not overstate this limiting test.

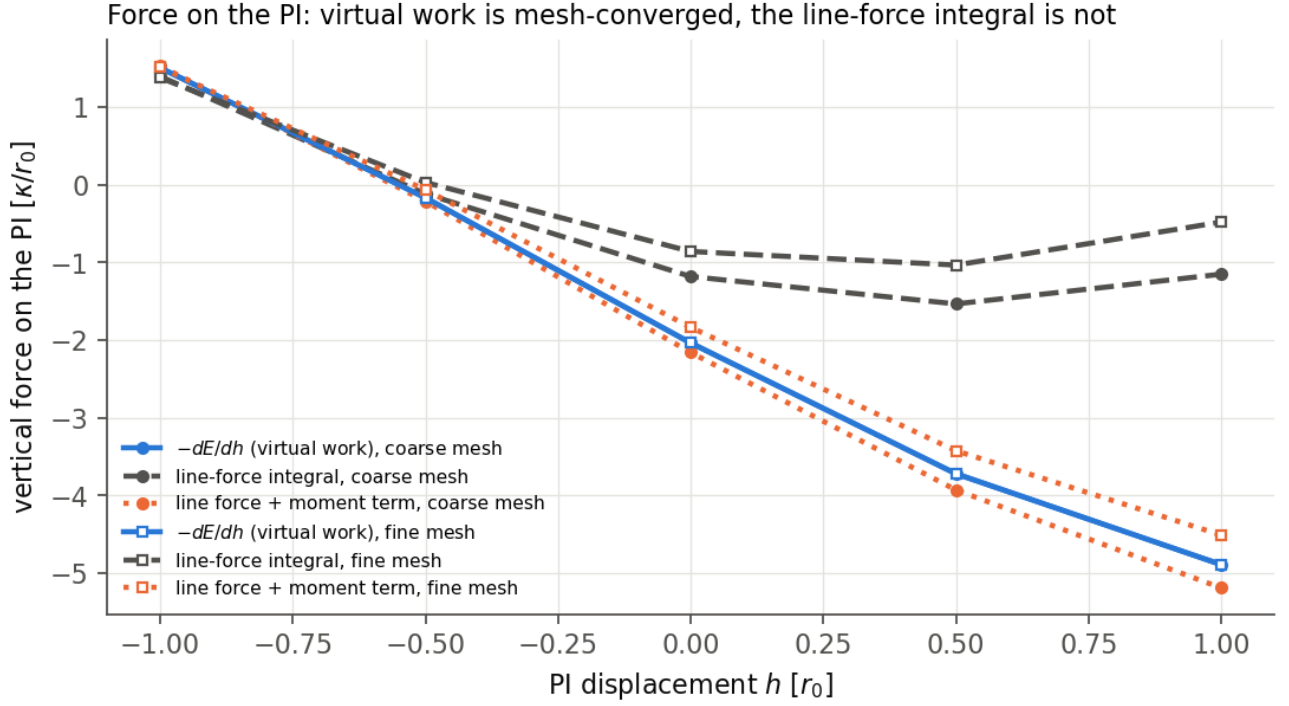


Figure 24: Ring force diagnostic under refinement. Virtual work converges; the original boundary line integral does not. Adding the moment term improves a coarse result but does not remove the boundary-derivative discretization error.

4.2 A conserved vertical stress and consistent reactions

Let $L_b = 2\kappa\mathcal{H}^2\sqrt{g}$ in base coordinates and $z_i = \partial_i z$. Vertical translation invariance gives the current

$$J_b^i = \frac{\partial L_b}{\partial z_i} - \partial_j \frac{\partial L_b}{\partial z_{ij}}, \quad \frac{\partial L_b}{\partial z_{ij}} = 2\kappa\mathcal{H} \left(\delta_{ij} - \frac{z_i z_j}{g} \right). \quad (10)$$

The total vertical flux in the flow calculation is

$$J^i = J_b^i + \frac{\sigma z_i}{\sqrt{g}} + 2\eta\sqrt{g} d^{ij} z_j. \quad (11)$$

For a smooth cutoff $\theta = 1$ around the inclusion and zero beyond an enclosing annulus,

$$F_m = - \int_{\Omega} J^i \partial_i \theta \, d^2 x. \quad (12)$$

At a steady force-balanced state this is independent of the chosen cutoff. The implementation uses cosine-squared radial tapers over several annuli; their spread measures discretization consistency. In dynamics, normal-friction/body-force terms must also enter the force accounting. The force-free height is determined by $F_m(h, v_0) = 0$, or by the corresponding consistent virtual-reaction equation on a parametric surface. Normal friction is distributed over the membrane, not an extra unspecified mass of the protein.

4.3 A free boundary is not a force-free rigid inclusion

Removing a height Dirichlet condition does not automatically produce a force-free rigid protein. A physical free disc requires constant rim height and zero total vertical force. For the original `square_b` eigenmode the recorded force is 1.9×10^{-4} in its mode normalization, and the work -5.1×10^{-6} matches the extra boundary term with ratio 0.995. The corrected linear problem ties both height and kinematic normal variables at the flat rim, removes the inappropriate bending-flux term, and obtains the global vertical balance from the reduced row. Non-flat continuation instead uses the flux of Eq. (12).

These results establish an implementation-level diagnostic. They should be discussed with M. Castellana together with the precise force diagnostic used for Fig. 5 and the setup of the published flow figure. They do not authorize an unsupported statement that every steady state or every force calculation in the cited paper is wrong.

The original tangential boundary terms also admit growing dynamic modes at rest. Restoring dissipative natural tractions removes them. Finally, the implemented boundary line-force diagnostic fails the recorded refinement tests and omits a moment contribution to virtual work. Reliable ring forces are obtained from $-dE/dh$; flow forces use a conserved stress flux or a consistent reaction. These findings concern the implemented conditions and diagnostics, not the validity of IRENE as a general framework. Their implications for the published calculations require discussion with its authors.

These diagnostics motivate the revised calculations in the subsequent sections. They also set the scope of the comparison with the earlier large-deformation study [2]: the inspected force and rim implementations require joint review with M. Castellana and the supervisor before publication. The code-level evidence is reported precisely, without treating it as a blanket assessment of IRENE or of every result in that study.

5 Static ring pulling and control-ensemble stability

The no-flow annulus provides a transparent first physical application. It separates stability at prescribed protein height from stability under an externally prescribed force, before adding the nonuniform stresses of a flowing membrane.

For an annulus $r_0 < r < R$, the outer height and slope are fixed, while the protein imposes height h and contact slope. Across the computed branches for $R/r_0 = 5, 10, 30$ and $0 \leq \tan \alpha \leq 2$, all admissible height-controlled growth rates remain negative. Additional large-ring branches extend the force comparison to $R/r_0 = 100$. The axisymmetric mode is usually least stable. A tilt mode can become least stable near the steep ends of small-ring branches, but no non-axisymmetric crossing was found. This is a statement about the sampled branches and the resolved Monge range, not a global convexity theorem.

5.1 Relaxed energy and the force-controlled zero mode

For a relaxed branch $q_*(h)$, $E_q(q_*, h) = 0$. Differentiating gives $q'_* = -E_{qq}^{-1} E_{qh}$, and therefore

$$E''_* = E_{hh} - E_{hq} E_{qq}^{-1} E_{qh}. \quad (13)$$

If the fixed-height Hessian is positive, the full dead-load Hessian is positive precisely when this scalar is positive. The external holding force is $P = E'_*$, whereas the repository plots the membrane force $F_m = -E'_*$. Accordingly $dF_m/dh < 0$ is the stable force-controlled segment. This sign convention resolves the apparent contradiction between a plotted force minimum and a positive upward pulling-force maximum.

The mixed dynamic matrix itself need not look symmetric in Euclidean nodal coordinates because constraints and row scales remain present. Energetic stability concerns the reduced Hessian on admissible shape variations. Negative relaxation rates on the sampled equilibria imply local stability under the chosen control, not global convexity of the nonlinear energy or exclusion of every remote branch.

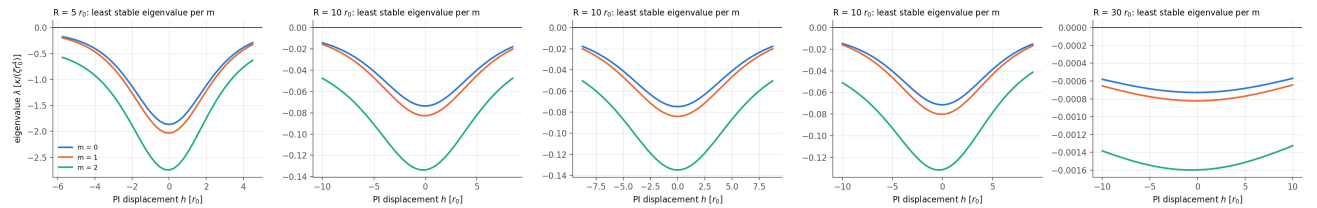


Figure 25: Ring relaxation spectra versus protein height, resolved by azimuthal number. No growing fixed-height mode is found in these sampled branches; spectra soften as the neck becomes steep.

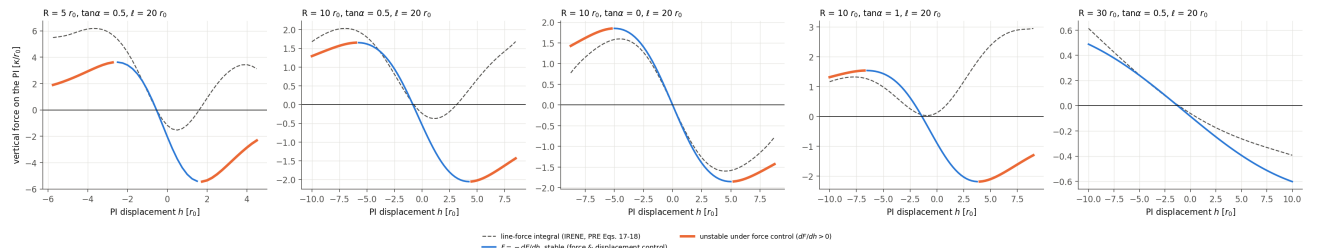


Figure 26: Ring forces and fixed-height eigenvalues for several radii. Dashed original line-force curves are retained as diagnostics; extrema and force-control conclusions use virtual work.

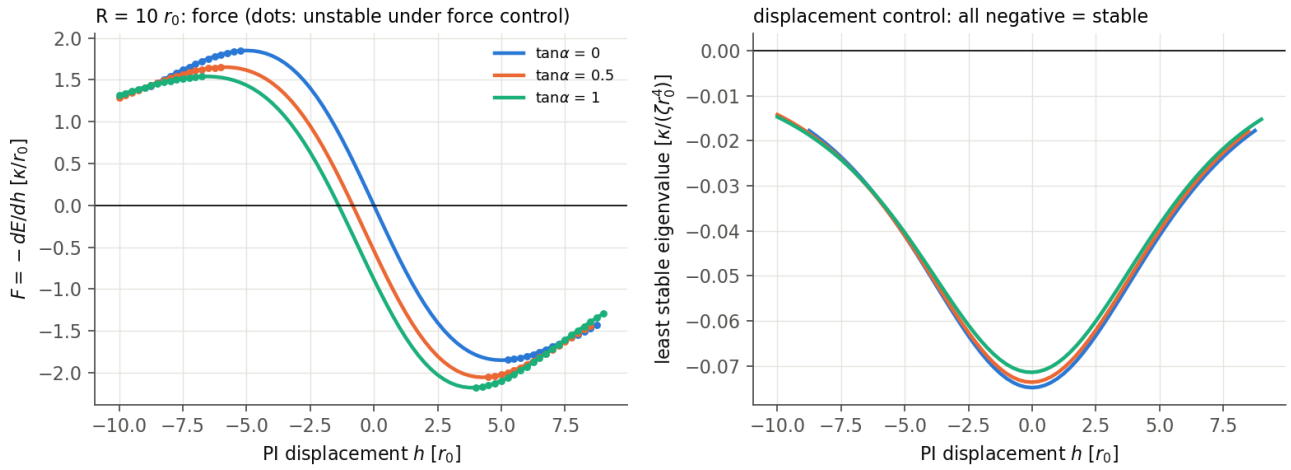


Figure 27: Contact-slope dependence of ring force and leading relaxation rate at outer radius ten. Force extrema shift, while the sampled height-controlled branches remain stable.

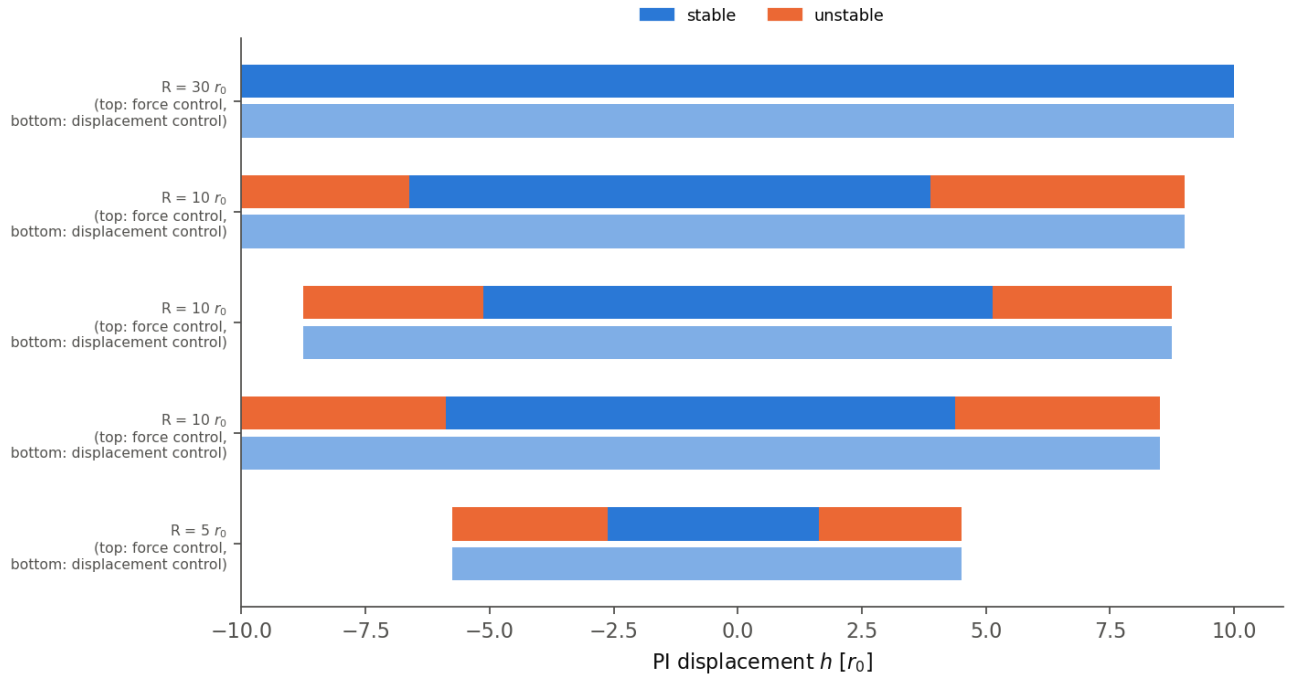


Figure 28: Ring control-ensemble summary. Force-controlled softening segments are unstable even when the fixed-height shape spectrum remains stable.

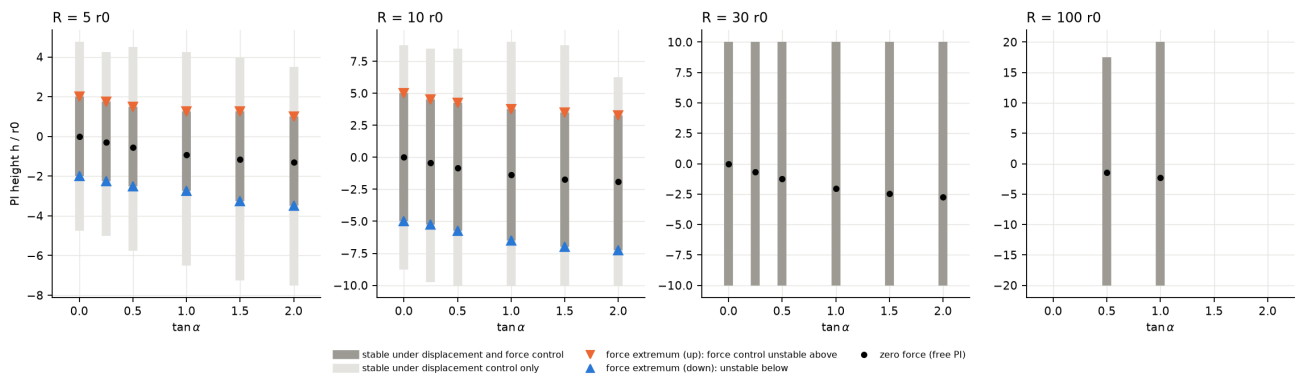


Figure 29: Height/contact-slope ring stability diagram over the computed windows. The force-control boundaries are virtual-work force extrema; height-controlled branches remain stable. Grey endpoints indicate limits of the sampled branch.

5.2 Profiles and geometric confinement

The profile families reveal what is changing while the fixed-height state remains stable. Zero contact slope retains up-down reflection symmetry; imposed slope biases the rest height and the two force barriers. Small outer rings force a steep neck over a short radial interval. The larger ring profiles have a wider deforming region but do not yet display a long-tube plateau in their Monge windows. These geometries explain why height alone is a poor measure of proximity to force-controlled failure.

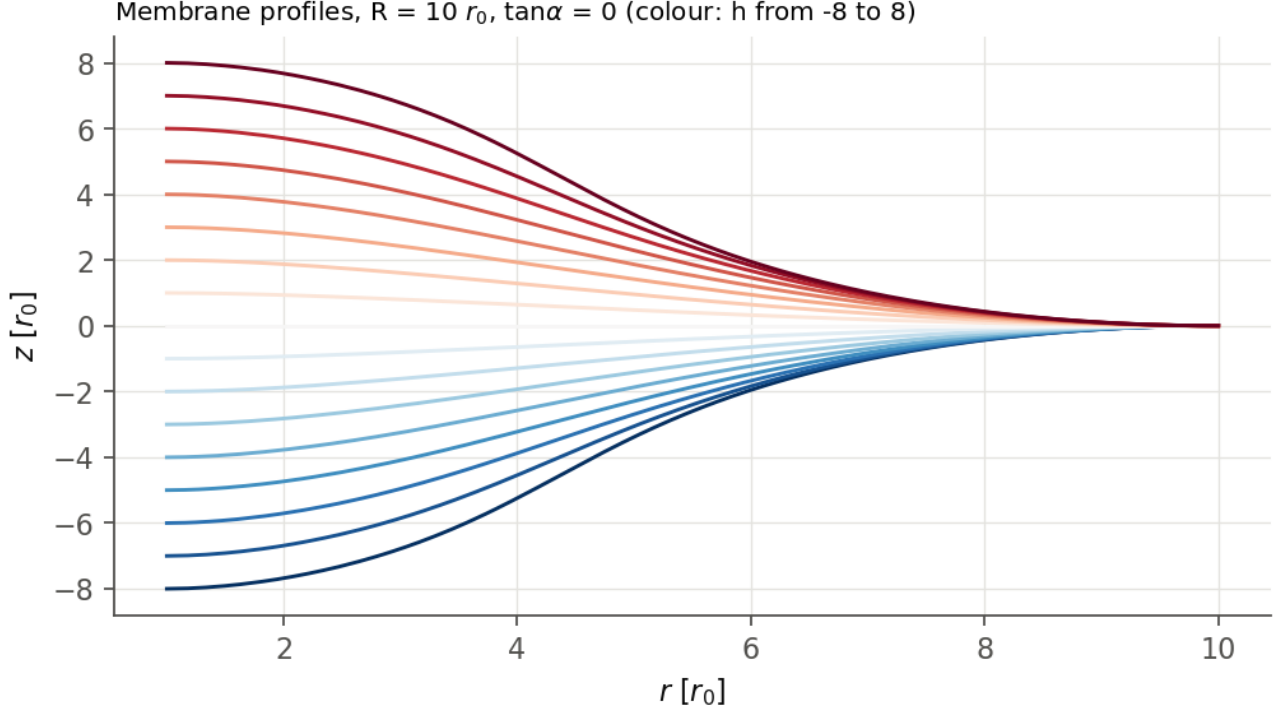


Figure 30: Symmetric zero-contact-slope ring profiles at outer radius ten. Upward and downward branches are related by reflection in the reference plane.

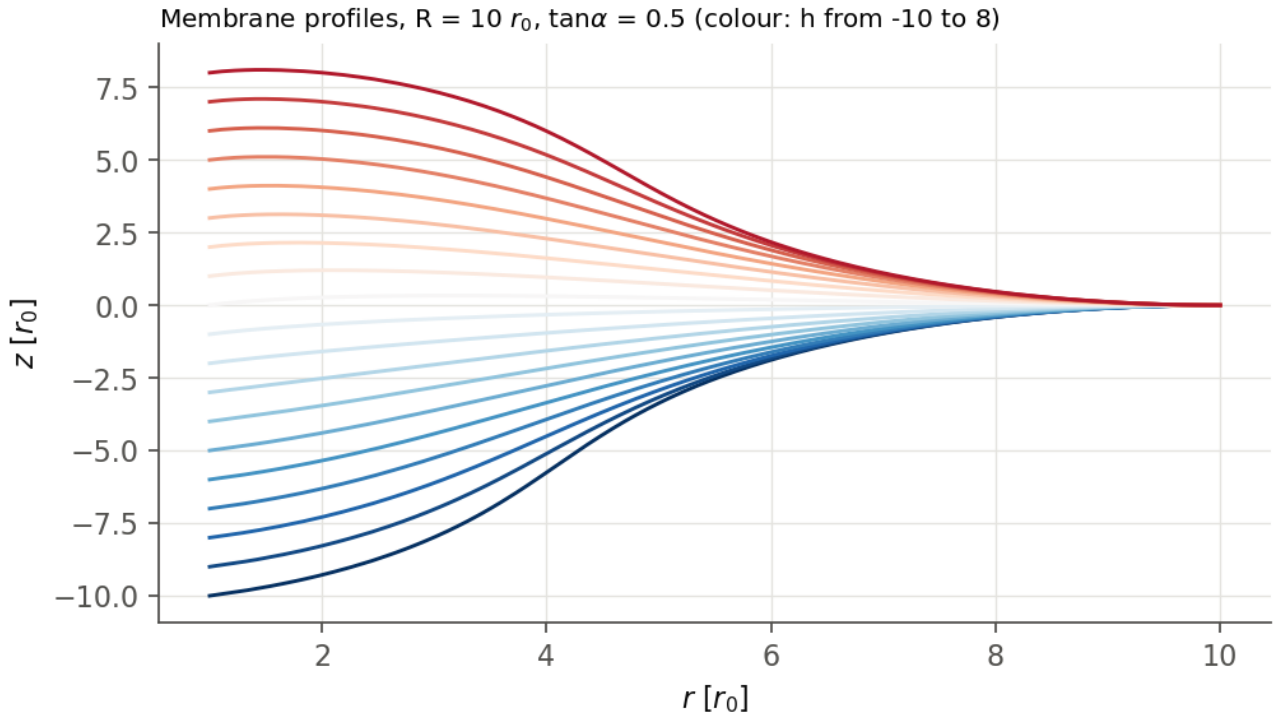


Figure 31: Ring profiles for outer radius ten and contact slope one half. Increasing imposed height produces a steep neck while the corresponding fixed-height spectrum remains stable in the sampled range.

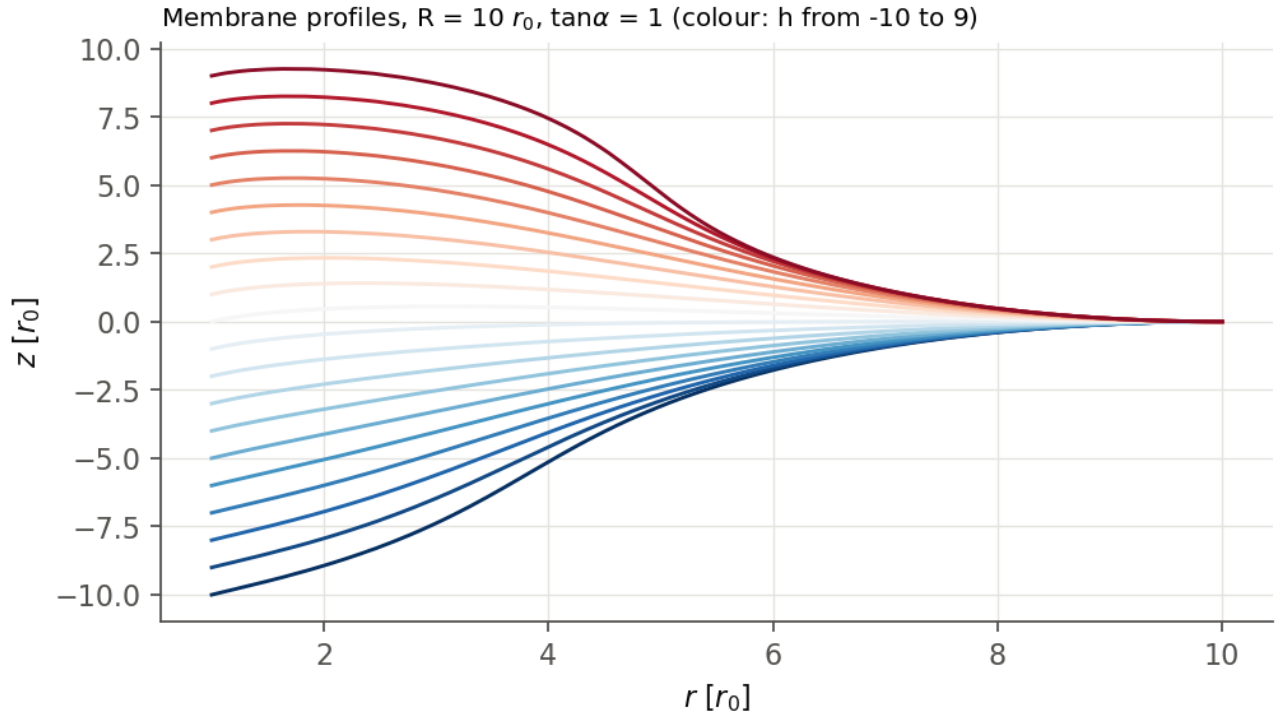


Figure 32: Ring profiles at outer radius ten and unit contact slope. The imposed slope biases the spontaneous height and the two force-control barriers.

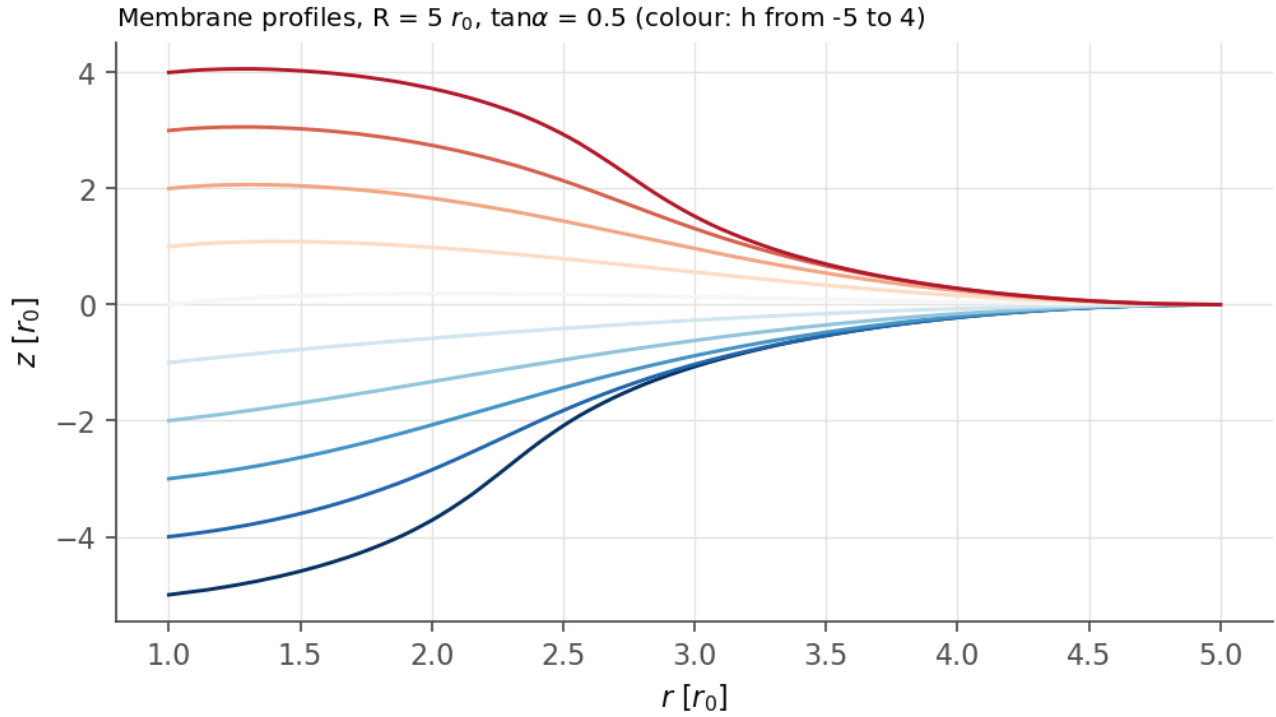


Figure 33: Strongly confined ring profiles at outer radius five. Large end slopes limit a height representation and distinguish the confined neck from a freely developed wide tube.

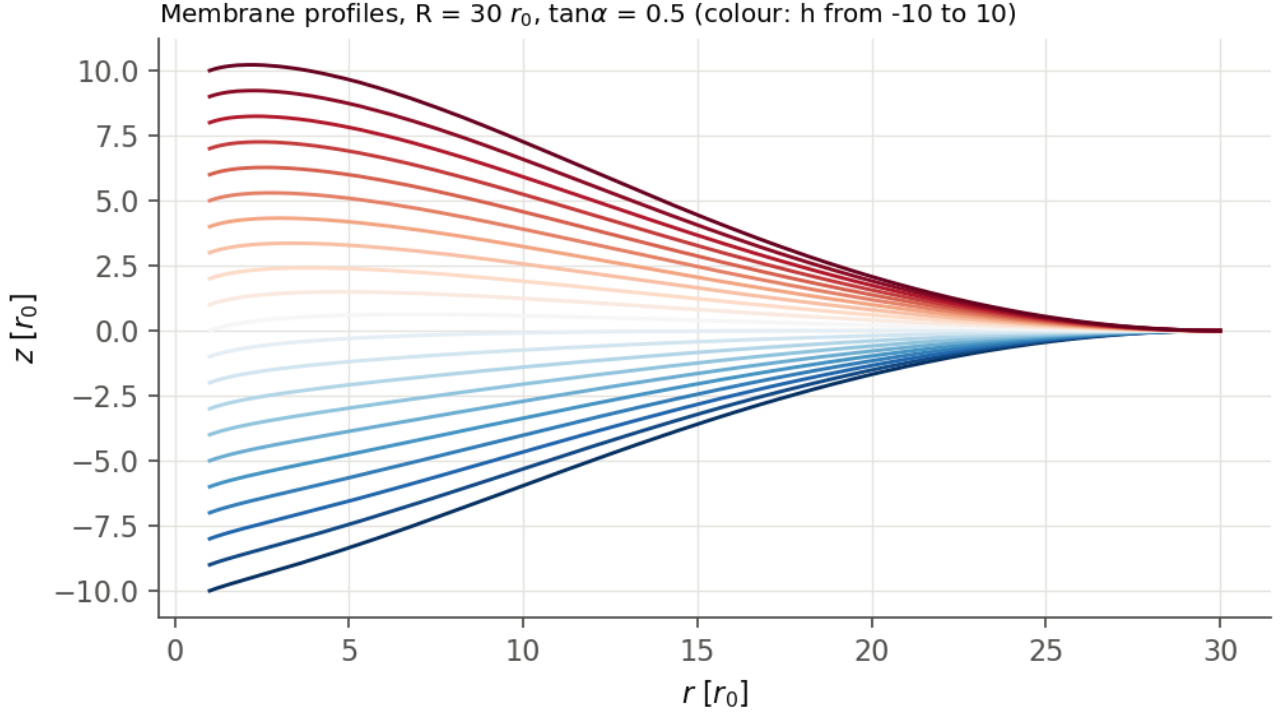


Figure 34: Profiles for outer radius thirty and contact slope one half. The displayed Monge window does not reach the long-tube force plateau.

6 Tether nucleation and continuation beyond a height graph

A force maximum is a nucleation barrier, whereas the force of an established tube is a propagation cost. Resolving both requires a representation that can follow an axisymmetric surface beyond the steep limit of a height graph.

6.1 Confinement-dependent pulling forces

For $r_0 = 10$ nm, $\kappa = 10k_B T$ at 300 K, and $\sigma_0 \simeq 10^{-6}$ N/m, the upward holding-force maximum is about 22 pN for $R = 50$ nm and 8.5 pN for $R = 100$ nm. The downward extrema are about 15 and 6.8 pN. These barriers exceed the long-tube force

$$P_t = 2\pi\sqrt{2\kappa\sigma_0} \simeq 1.8 \text{ pN}, \quad r_t = \sqrt{\kappa/(2\sigma_0)} \simeq 140 \text{ nm}. \quad (14)$$

An independent axisymmetric arclength solver reproduces the Monge forces and follows large-ring branches into tubes approaching this plateau [4]. Small rings, narrower than r_t , instead constrain a neck; their overshoot should not be described as an unconstrained long tube. Force control above the barrier drives extrusion, whereas an existing long tube resists with P_t , giving the familiar loading-unloading asymmetry.

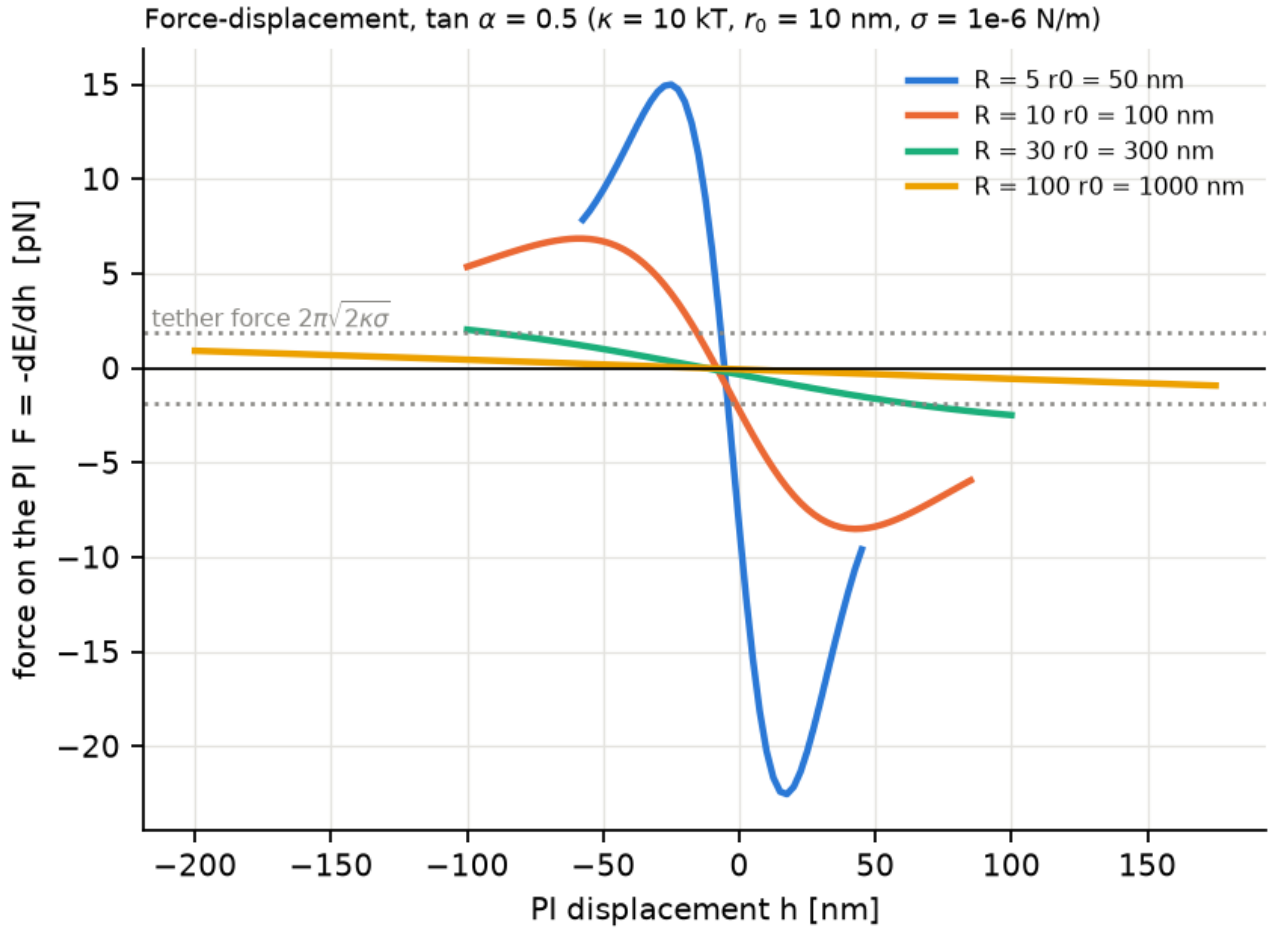


Figure 35: Mesh-converged force on the protein for $\tan \alpha = 0.5$, plotted with the repository convention $F_m = -dE/dh$. Upward holding-force maxima therefore appear as negative extrema. The dotted line denotes the magnitude of the long-tube force. Large rings have not reached their force extrema within this figure's displacement window.

6.2 Arclength formulation and endpoint conditions

To follow a no-flow axisymmetric surface beyond a height graph, use arclength s_a , radius r , height z , and tangent angle ψ . With $u = d\psi/ds_a$, a radial multiplier γ_r , and a constant vertical multiplier p_z , the implemented equations are

$$\psi' = u, \quad r' = \cos \psi, \quad z' = \sin \psi, \quad (15)$$

$$u' = -\frac{\cos \psi}{r} u + \frac{\sin \psi \cos \psi}{r^2} + \frac{\gamma_r \sin \psi - p_z \cos \psi}{\kappa r}, \quad (16)$$

$$\gamma_r' = \frac{\kappa}{2} \left(u^2 - \frac{\sin^2 \psi}{r^2} \right) + \sigma_0, \quad p_z' = 0. \quad (17)$$

Primes in this section mean d/ds_a . The free total arclength is selected by

$$\frac{\kappa r}{2} \left(u^2 - \frac{\sin^2 \psi}{r^2} \right) - \sigma_0 r + \gamma_r \cos \psi + p_z \sin \psi = 0. \quad (18)$$

The endpoints satisfy $(r, z, \psi) = (r_0, h, \arctan s)$ and $(R, 0, 0)$. The membrane force is $F_m = 2\pi p_z$, checked by energy differences and the Monge calculation. A scipy boundary-value solver with adaptive continuation reproduces the overlapping force branch to about 2×10^{-4} . This calculation establishes the large-ring tube radius and force plateau but is not a full two-dimensional stability computation of every post-overhang axisymmetric branch.

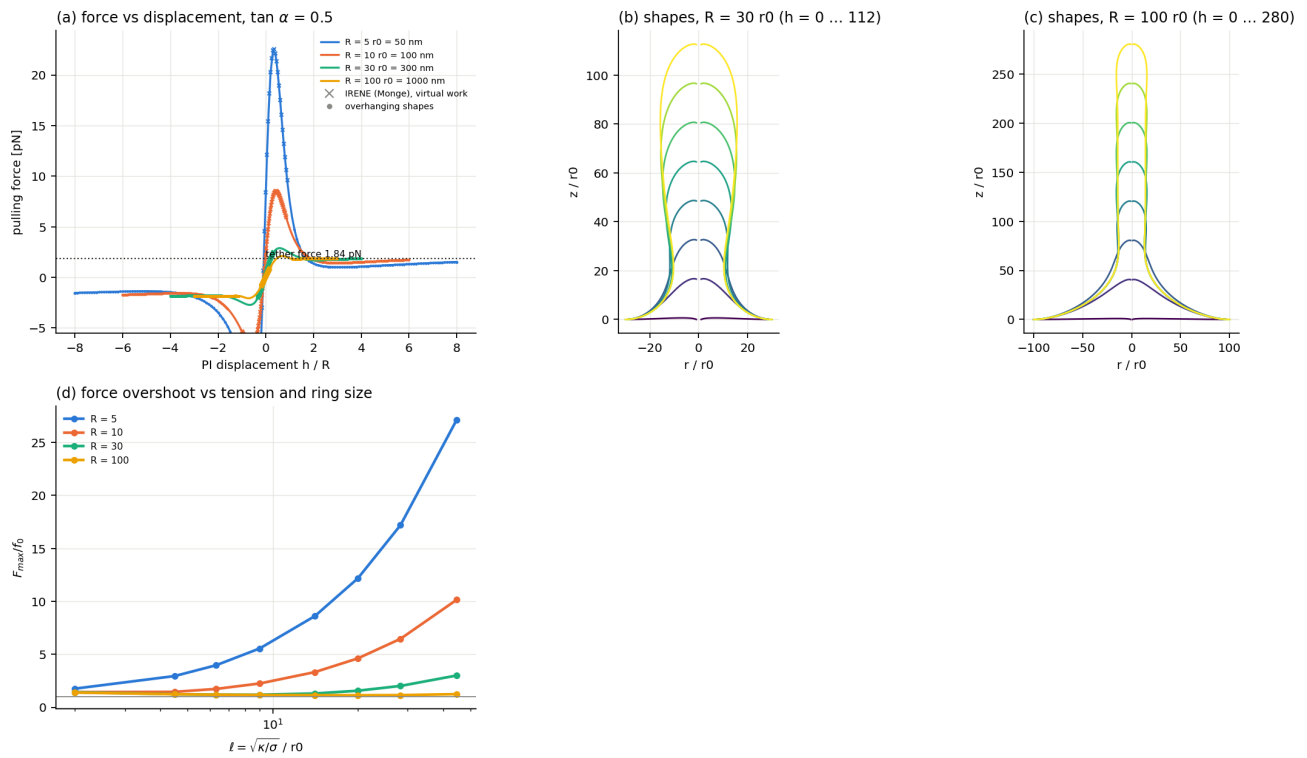
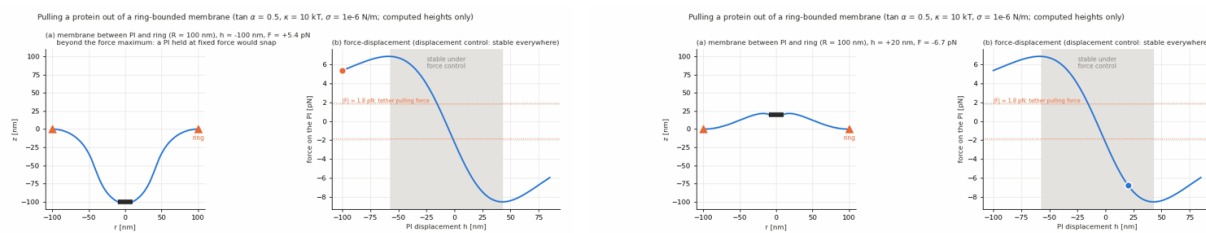
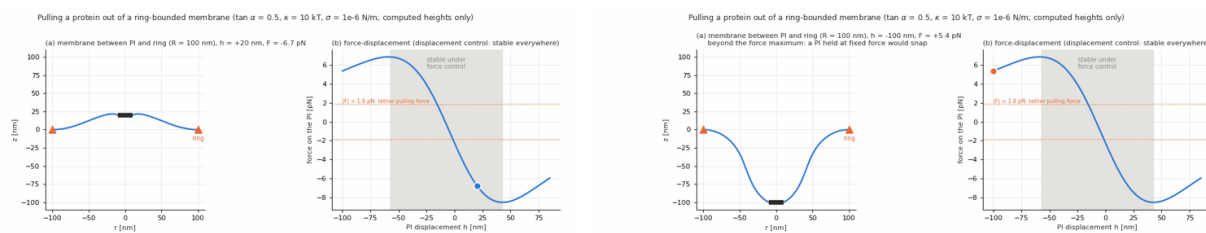


Figure 36: No-flow axisymmetric arclength continuation. Large rings develop tubes tending to the tether-force plateau; small rings constrain a neck and show a large force overshoot. This is distinct from flow-driven tube growth, which remains unfinished.



Frame 1/37

Frame 13/37



Frame 25/37

Frame 37/37

Figure 37: Ring-pulling profile sequence. The imposed height varies along a branch; force-control stability is determined separately by the virtual-work stiffness. Four ordered frames are selected by frame index; the spacing is not assumed uniform in physical time. The original GIF is included in the source package.

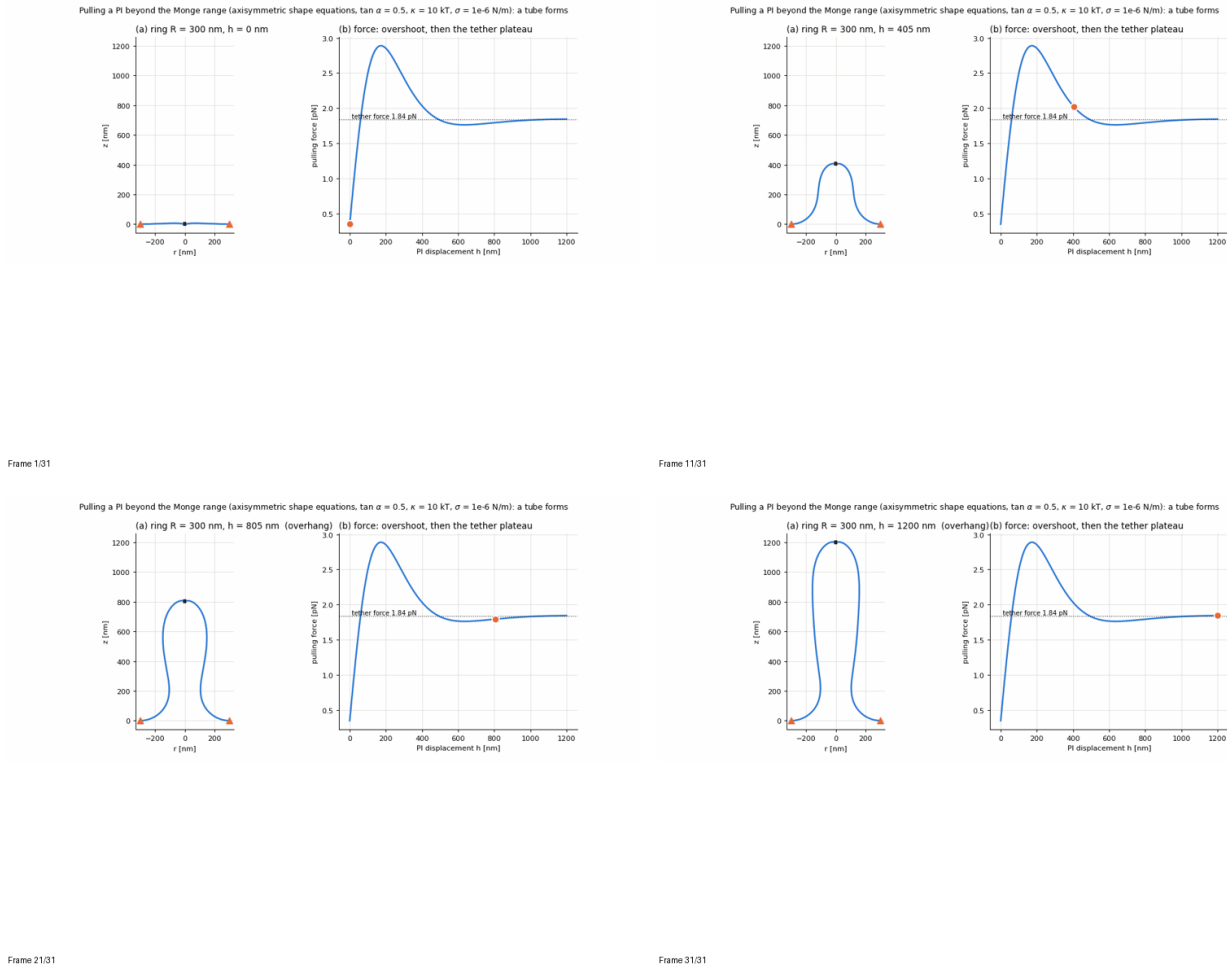


Figure 38: No-flow axisymmetric tube/neck continuation beyond the Monge range. The force plateau for a large ring agrees with tether theory. Four ordered frames are selected by frame index; the spacing is not assumed uniform in physical time. The original GIF is included in the source package.

The tube calculation completes the static comparison, but it does not provide a full two-dimensional spectrum for all arclength states. It also does not solve the later flow-driven folding problem: in that problem the shape is asymmetric, material flow persists, and maintaining a regular mesh is part of the dynamics.

7 From preliminary flow calculations to corrected onset predictions

The early flow calculations were useful because the eigensolver could predict the dynamics of the implemented equations. Their physical interpretation changed when the missing vertical protein condition was identified. The following sequence keeps that distinction explicit.

7.1 What the original-rim calculations established

The original-rim spectra placed onset near a drive of sixteen and displayed large smooth deformation for nonzero contact slope. Their stress fields already identified upstream tension loss. Those observations established a mechanism in that implementation, but their thresholds and deformation branches cannot be transferred to a rigid protein with the corrected force balance. The original-rim BDF2 comparison remains a valid check of linearization and time integration of that particular residual.

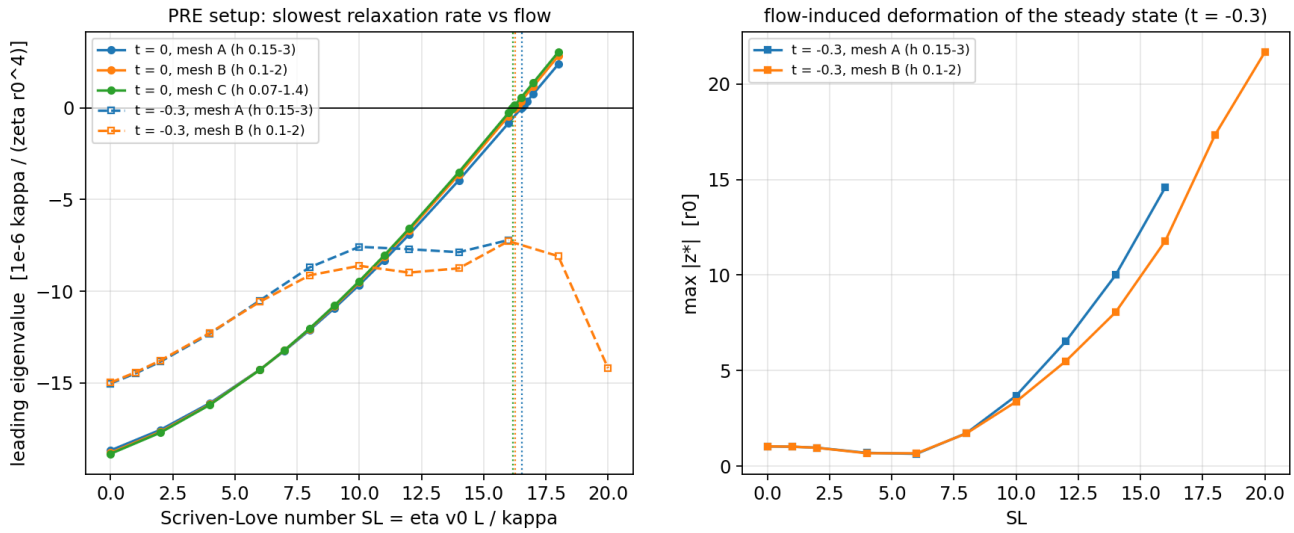


Figure 39: Superseded original-rim flow spectra and large imperfect deformation. The roughly sixteen threshold is a diagnostic of the original boundary-value problem; corrected thresholds are near twenty-seven in the reference geometry.

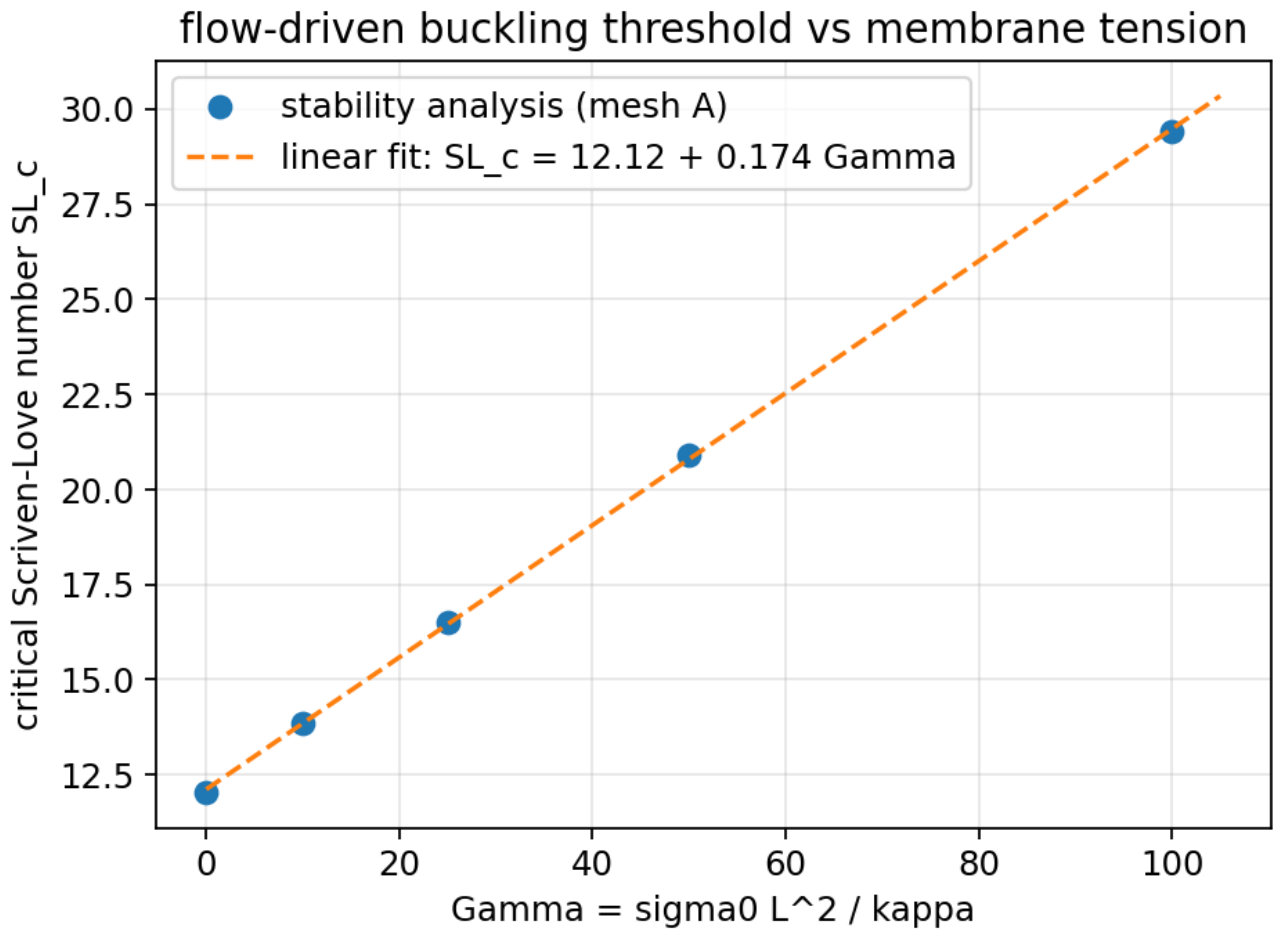


Figure 40: Superseded original-rim calculation. Its approximate linear threshold-versus-tension fit belongs to the implementation without vertical protein force balance and must not be used for the corrected physical model.

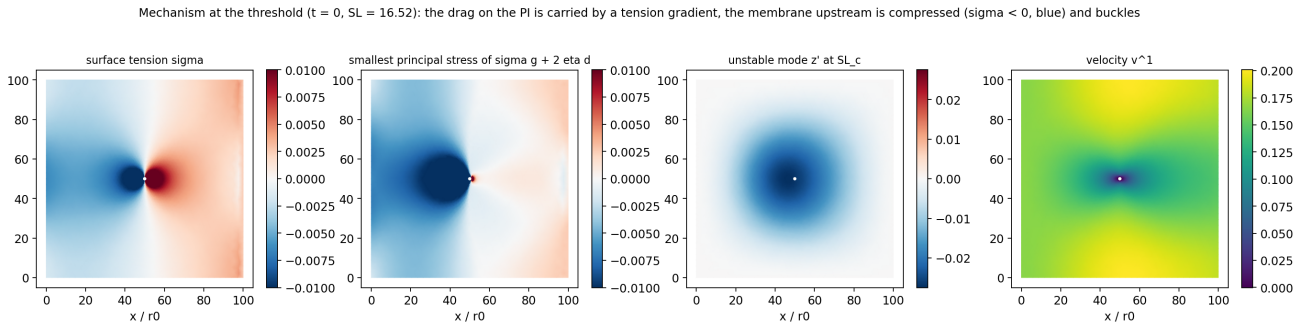


Figure 41: Superseded original-rim stress and eigenmode fields. They reveal upstream compression but place the mode nearer the protein than the corrected force-balanced problem.

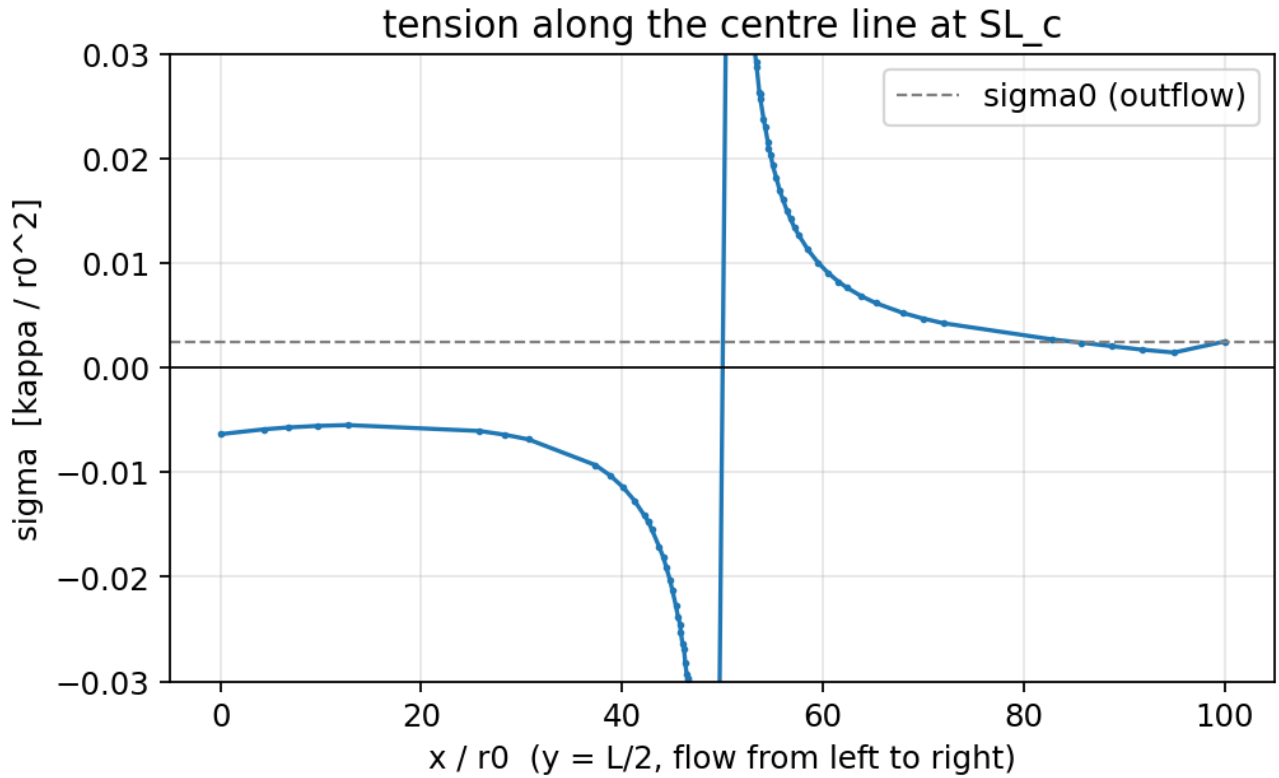


Figure 42: Original-rim flat base-flow tension along the centreline. The upstream tension decrease is mechanistic evidence; the onset value attached to this figure is superseded by the physical protein condition.

PRE setup with contact angle $t = -0.3$ (imperfect bifurcation): the flow-induced deformation is the buckling mode, growing smoothly

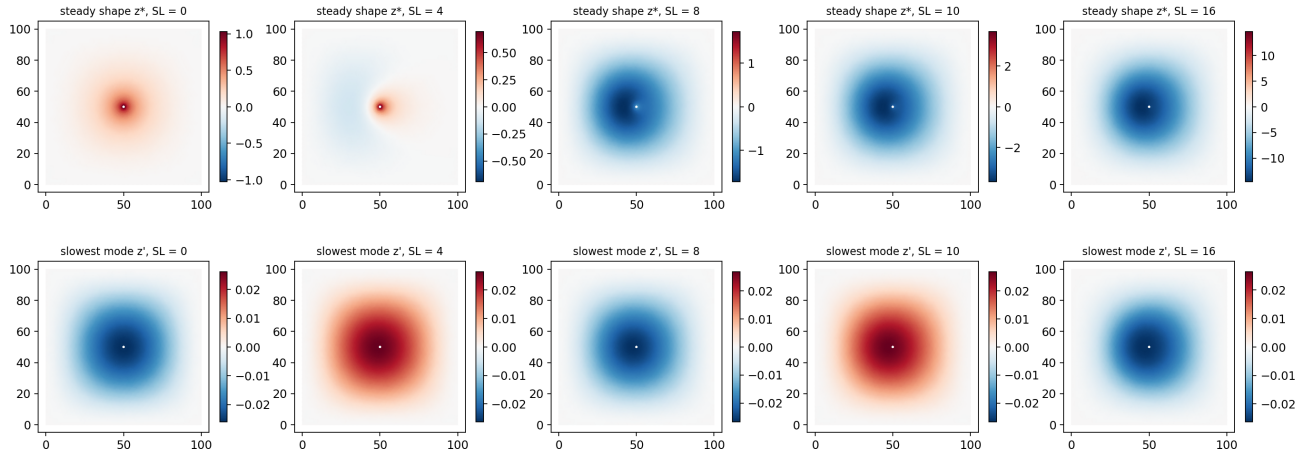


Figure 43: Superseded original-rim shapes and leading modes at nonzero contact slope. Their large smooth deformation is not evidence for a smooth transition with a physically force-free protein.

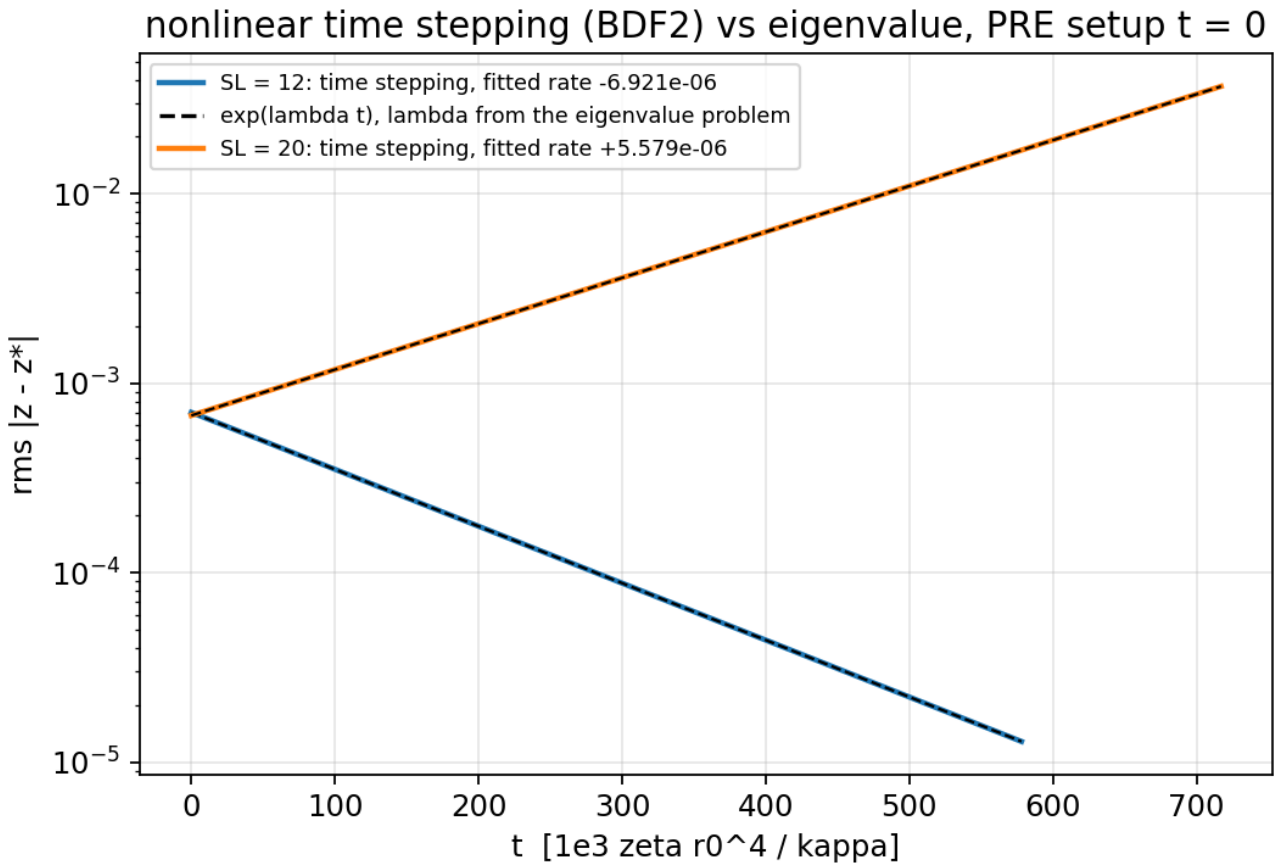


Figure 44: BDF2 versus eigenvalue growth and decay for the original-rim implementation. This checks that the numerical linearization predicts that implementation's dynamics, not that its rim condition is physically force-free.

7.2 An illustrative slip problem and its numerical limits

The slip-inclusion example explores inertial wave collisions rather than the overdamped low-inertia membrane onset. Its weak oscillatory growth and penalty sensitivity reveal why illustrative numerical examples must be separated from robust physical threshold claims. These tests also motivated explicit tracking of penalty and mesh effects.

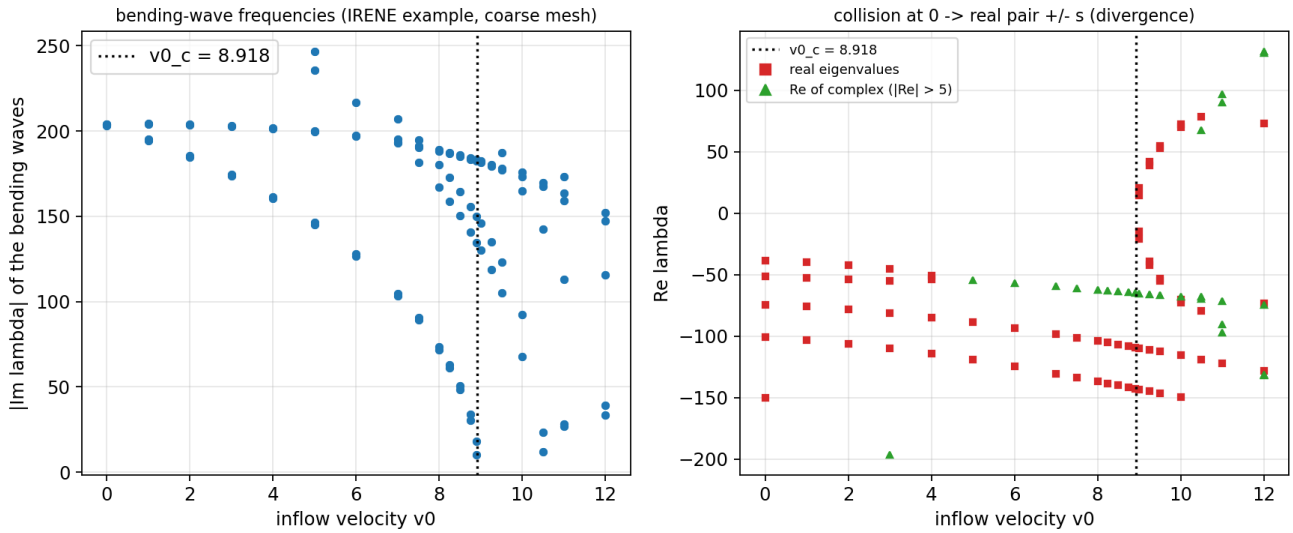


Figure 45: Inertial bending-wave collision in the slip-inclusion example. This is an illustrative high-inertia mechanism, not the overdamped lipid-membrane threshold.

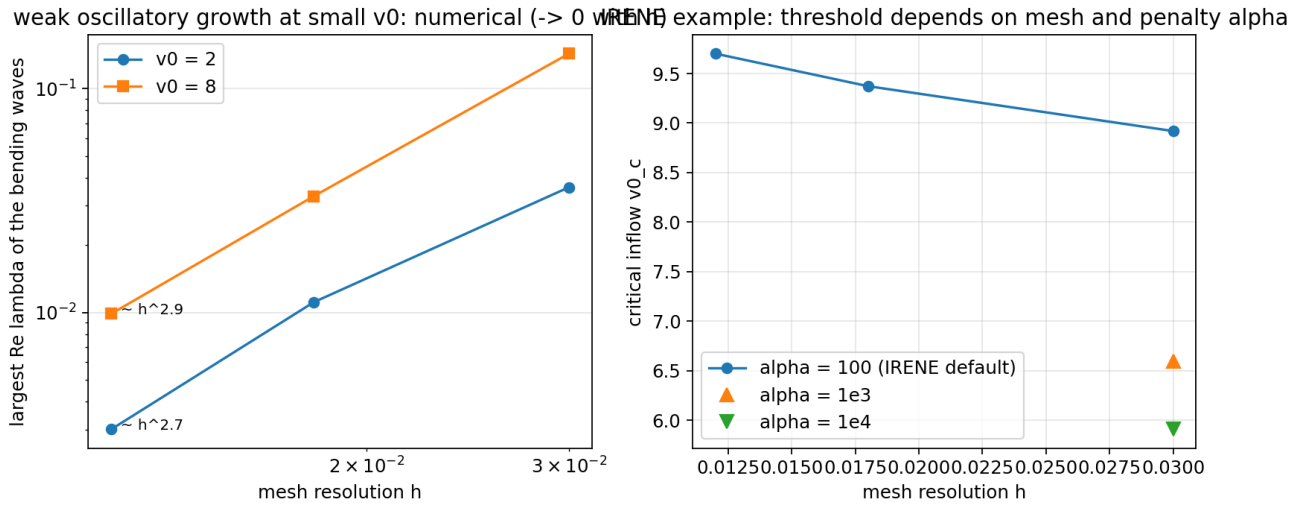


Figure 46: Slip-penalty example: weak oscillatory growth shrinks with mesh size, while the apparent threshold depends strongly on penalty strength. These results diagnose numerical and boundary-condition sensitivity and are excluded from quantitative core claims.

7.3 Corrected conditions and the quantitative verification ledger

Check	Recorded result and interpretation
Cylinder-wake onset	$Re_c = 46.46, 46.35, 46.32$ on coarse/medium/fine meshes; reference about 46.7.
Cylinder nonlinear dynamics	Growth rates agree within a few percent; frequencies within about 0.2% in the tested DNS comparisons.
Flow formulation at rest	Inertial/no-flow spectral comparison about 1.5%; an overdamped/inertial-limit check about 10^{-9} in the tested limit.
Ring force and leading eigenvalue	Refinement changes force by at most 3×10^{-4} and rate by 2.5×10^{-3} over the main $R = 5$ range; rate error rises near the steep end.
Full/plate threshold	0.1–0.4% across tested geometries; 27.449 vs 27.432 on the reference fine mesh.
BDF2/eigenvalue rate	Relative discrepancy down to 1.3×10^{-4} in the recorded normal-mode tests.
Vertical stress/virtual-work force	Approximately 5×10^{-5} in the ring check; cutoff variation near 10^{-6} for well-resolved equilibria.
Adjoint sensitivity/direct change	Agreement near 2×10^{-7} in the recorded check.

Table 1: Independent verification problems. Tolerances are not interchangeable confidence bounds on every nonlinear result.

Protein vertical condition	Mesh A	Mesh B	Mesh C	Plate C
Rigid force-free	27.504	27.465	27.449	27.432
Clamped	27.732	27.682	27.664	27.644
Original free-rim implementation	16.524	16.279	16.168	–

Table 2: Reference thresholds SL_c at $L = 100r_0$, $\Gamma = 25$, zero contact slope. The original row diagnoses a different boundary-value problem.

The reported 5×10^{-4} threshold convergence is consistent with the last relative mesh change, rather than an independently established exact error bound. Original-boundary tests varying penalty α from 100 to 1000 gave little change in their onset, but those tests cannot certify all corrected nonlinear branches. The slip-inclusion **square_a** example has a pronounced penalty dependence and is excluded from quantitative membrane thresholds. A final publication needs a corrected-condition penalty matrix, domain studies of flutter, and a pinned runtime/upstream IRENE revision.

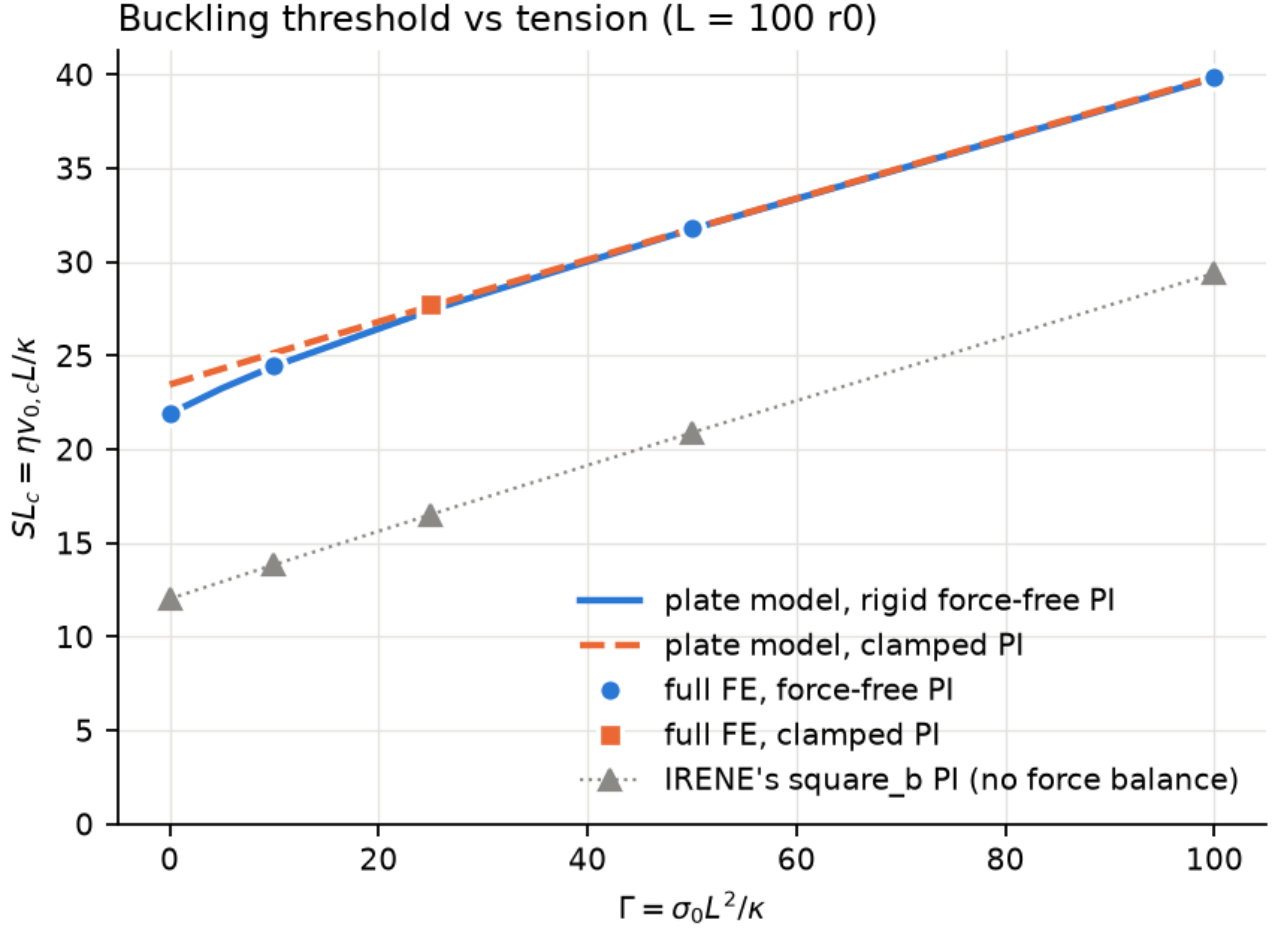


Figure 47: Critical drive with physical protein conditions. The rigid force-free and clamped formulations agree with their plate reductions. The lower grey curve uses the original free-rim condition without a vertical force equation and is retained only as a diagnostic. All thresholds refer to $L = 100r_0$.

8 Flow-driven buckling: compression and the plate reduction

With the physical vertical constraint imposed, the flat-state problem admits an especially clear reduction. It shows directly how tangential obstacle drag can destabilize a membrane without inertia.

8.1 Exact plate operator about the flat base state

At zero contact slope, the base membrane remains flat and the tangential problem is incompressible two-dimensional Stokes flow. Its tensile-positive stress is $\mathbf{T} = \sigma \mathbf{I} + 2\eta \mathbf{d}$, with $\mathbf{d} = \frac{1}{2}(\nabla \mathbf{v} + \nabla \mathbf{v}^T)$. At zero inertia the stress is linear in the imposed velocity: $\mathbf{T} = \sigma_0 \mathbf{I} + v_0 \mathbf{T}_1$. Normal and tangential perturbations decouple at first order. With no tangential body force, $\nabla \cdot \mathbf{T} = 0$, and the normal equation is

$$\zeta \partial_t z = -\kappa \Delta^2 z + \mathbf{T} : \nabla \nabla z = -\kappa \Delta^2 z + \nabla \cdot (\mathbf{T} \nabla z). \quad (19)$$

The operator is self-adjoint under the physical clamped or rigid force-free rim conditions. For destabilizing trial shapes the first crossing is equivalently

$$v_c = \min_z \frac{\kappa \int (\Delta z)^2 dx + \sigma_0 \int |\nabla z|^2 dx}{-\int \nabla z \cdot \mathbf{T}_1 \nabla z dx}. \quad (20)$$

This Rayleigh quotient is an exact flat-state reduction, not a small-amplitude fit to the full eigenvalues.

8.2 Independent mixed discretization and energetic stiffness

At the flat state the stress is $\mathbf{T} = \sigma_0 \mathbf{I} + v_0 \mathbf{T}_1$. Normal perturbations obey $\zeta z_t = -\kappa \Delta^2 z + T_{ij} z_{,ij}$. With no body force, integration by parts gives the quadratic stiffness

$$\mathcal{Q}[z] = \kappa \int (\Delta z)^2 + \sigma_0 \int |\nabla z|^2 + v_0 \int z_{,i} (T_1)_{ij} z_{,j}. \quad (21)$$

It is positive until the first generalized threshold. The physical rigid inclusion fixes its slope while allowing one uniform height variation. The natural global force balance makes the boundary work vanish for an admissible force-free displacement. A clamped height instead removes that displacement from the trial space.

The independent mixed plate problem introduces $m_\Delta = \Delta z$ through $\int m_\Delta q + \nabla z \cdot \nabla q = 0$, with zero normal slope as the natural constraint. Its normal equation includes bending, imposed tension, and the computed base-flow stress. Energy parts are evaluated on the critical eigenvector. With friction, non-conservative terms are retained and a real Rayleigh quotient no longer determines all possible onsets; the first complex crossing must also be searched.

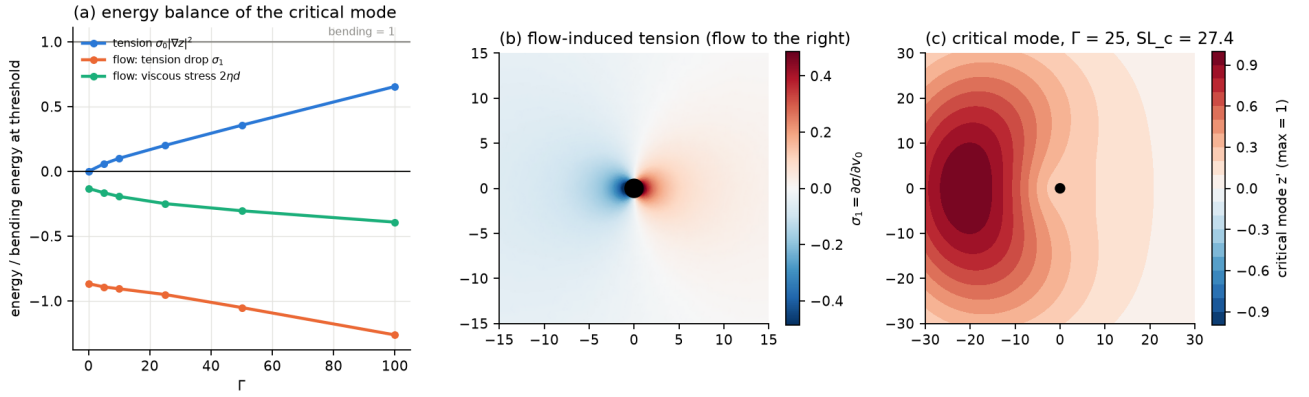


Figure 48: Energy decomposition, flow-induced tension, and the critical mode of the corrected flat-state problem. The tension falls upstream of the no-slip obstacle, and the critical deflection lies upstream rather than being centred on the protein. The negative flow contribution balances bending and the imposed tension.

8.3 Compression and critical-mode geometry

Obstacle drag is balanced by a tension gradient. The upstream tension drops enough to create a region of compressive principal stress, and the membrane buckles as a compressed plate. In the energy budget the tension drop supplies most of the destabilization; the viscous part of the stress supplies the remainder. The critical dome is centred approximately $20r_0$ upstream in the reference box. No inertia is needed for this mechanism.

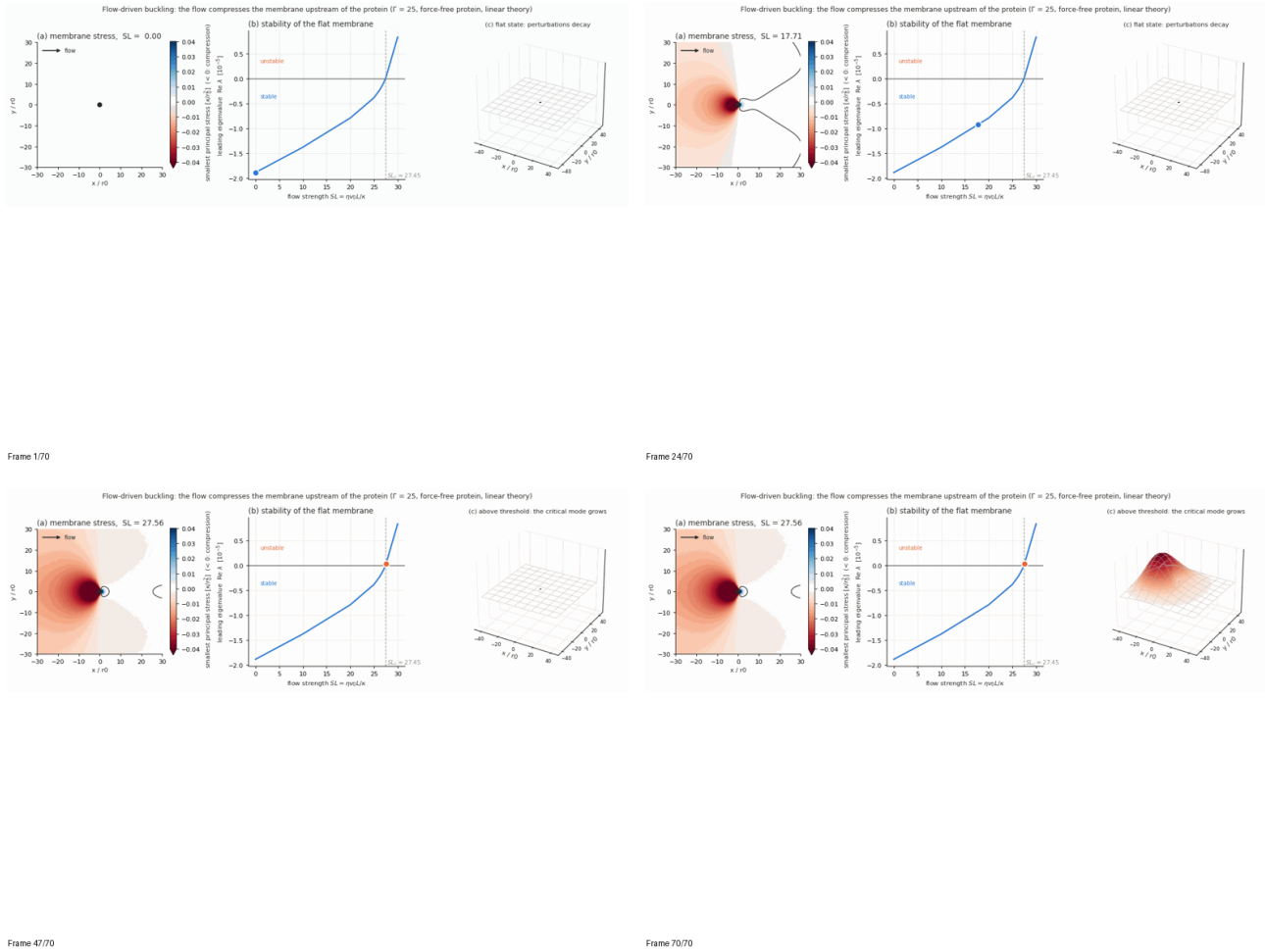


Figure 49: Critical-mode visualization of flow-driven buckling and upstream compression. This onset sequence belongs to the corrected study's visualization workflow; the static plots and force condition supply the quantitative threshold. Four ordered frames are selected by frame index; the spacing is not assumed uniform in physical time. The original GIF is included in the source package.

8.4 Beyond the first eigenmode

Increasing drive makes additional lobes and wrinkle-like normal modes available. The plate examples below illustrate how a larger compressed region admits more spatial structure. They describe linear modes above onset and sliding-membrane examples, not stable nonlinear wrinkle states around the reference single inclusion.

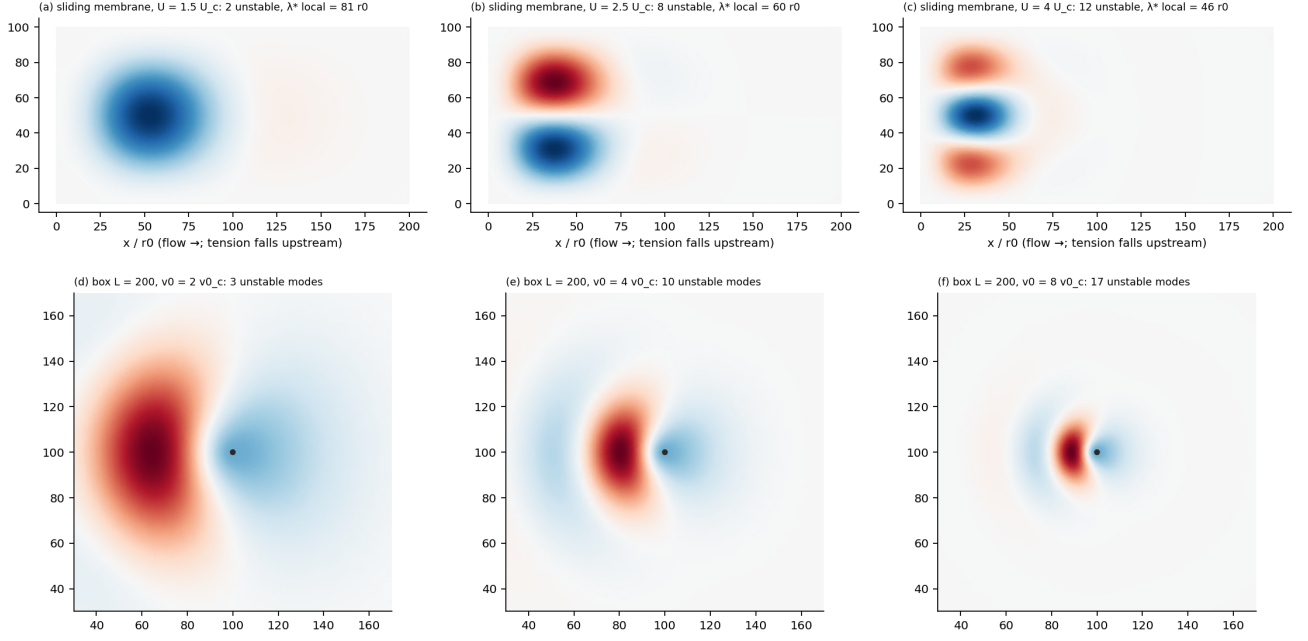


Figure 50: Plate-model modes well above onset and in sliding-membrane examples. Increasing compression permits multiple lobes and wrinkle trains; these modes do not establish a stable nonlinear pattern around a single protein.

9 Outer boundaries, domain size, and where forcing matters

The critical drive is a property of the entire boundary-value problem. Domain studies and adjoint sensitivity explain which aspects are geometric and which admit a local physical interpretation.

9.1 Size, width, and outer-height constraints

For the rigid force-free inclusion, successive meshes give $SL_c = 27.504, 27.465, 27.449$. At fixed $\Gamma = 25$, increasing L/r_0 from 50 to 100 to 200 gives 22.75, 27.50, 32.43: an approximately logarithmic increase over this tested range. At fixed dimensional tension, the same change instead gives 19.51, 27.50, 47.51, because Γ changes. Increasing the width from 50 to $200r_0$ changes the threshold from 21.86 to 31.44; holding only the inflow edge lowers it to 12.12.

These dependencies reflect the two-dimensional Stokes outer problem and the shape boundary conditions. Local Brinkman friction screens velocity disturbances but leaves a harmonic tension disturbance, so it does not remove finite-box dependence. The threshold 27.45 is therefore a reproducible reference value, not an inclusion-specific material constant. Adjoint calculations further identify a band 5–15 r_0 upstream where a tangential force most efficiently raises or lowers the threshold.

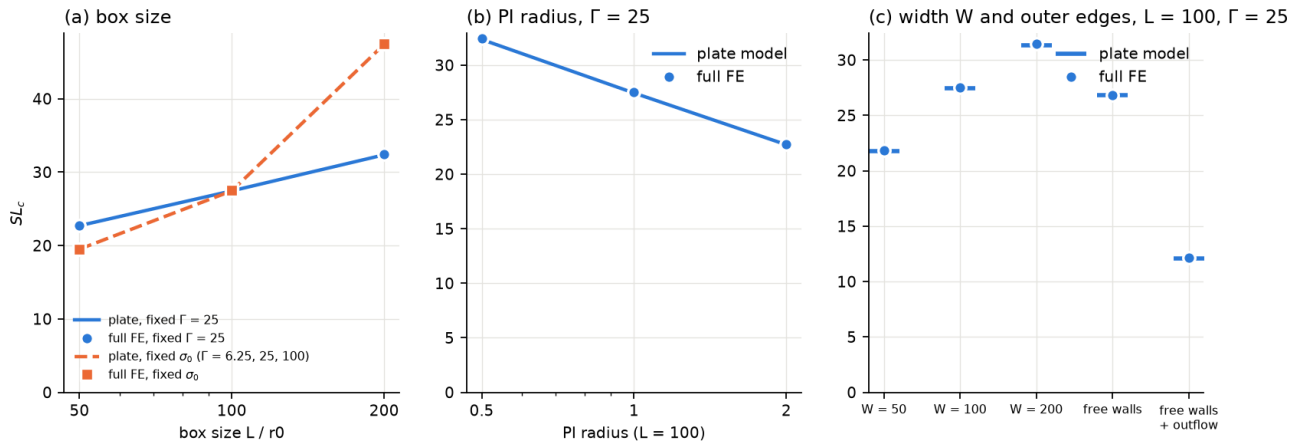


Figure 51: Corrected physical thresholds versus domain length, protein radius, width, and outer-height constraints. Fixed-tension and fixed-Gamma size sweeps are different experiments.

9.2 Tension and screening: what does and does not become local

At $\Gamma = 0, 10, 25, 50, 100$, the corrected fine-mesh force-free thresholds are approximately 21.91, 24.46, 27.45, 31.78, and 39.84. The older linear relation $SL_c \simeq 12.1 + 0.174\Gamma$ belongs to the original rim condition and is superseded. The corrected relation is not that line.

The size study explicitly compares fixed Γ with fixed σ_0 . At fixed Γ , doubling L/r_0 raises SL_c by about 4.8 over 50–200. This suggests a logarithmic dependence over the recorded interval; three domain sizes do not establish an asymptotic law for all L . Releasing only height on the lateral edges while retaining the imposed slope is also different from a completely free elastic edge. The report’s boundary comparisons refer to the implemented variants.

Brinkman coupling satisfies $\nabla \cdot \mathbf{T} = b(\mathbf{v} - v_0 \mathbf{e}_x)$. Taking a divergence in the flat incompressible problem leaves a harmonic tension perturbation. Velocity screening alone therefore cannot be invoked to eliminate the outer tension problem. A nonlocal three-dimensional-fluid coupling would be a different model.

9.3 Direct and adjoint forcing sensitivity

For $Ax = \lambda Bx$, let the left mode satisfy $y^\dagger A = \lambda y^\dagger B$ and normalize $y^\dagger Bx = 1$. For a parameter p ,

$$\lambda_p = y^\dagger (A_p - \lambda B_p)x, \quad (22)$$

including the base-state dependence of A . A tangential body force first perturbs the Stokes base state; a base-flow adjoint avoids one steady solve per force location. At a simple neutral crossing, $v_{c,p} = -\lambda_p/\lambda_v$. The tested normal direct and adjoint shapes nearly coincide without friction; the largest base-forcing sensitivity lies 5–15 r_0 upstream. A force with the flow there relieves compression and raises onset, and an opposite force lowers it. The archived direct perturbation agrees with the sensitivity prediction to about 2×10^{-7} .

Adjoint sensitivity at the threshold (clamped PI, mesh A, $v_0 = 0.2772$); adjoint check: predicted vs direct d lambda rel. error 2.0e-07

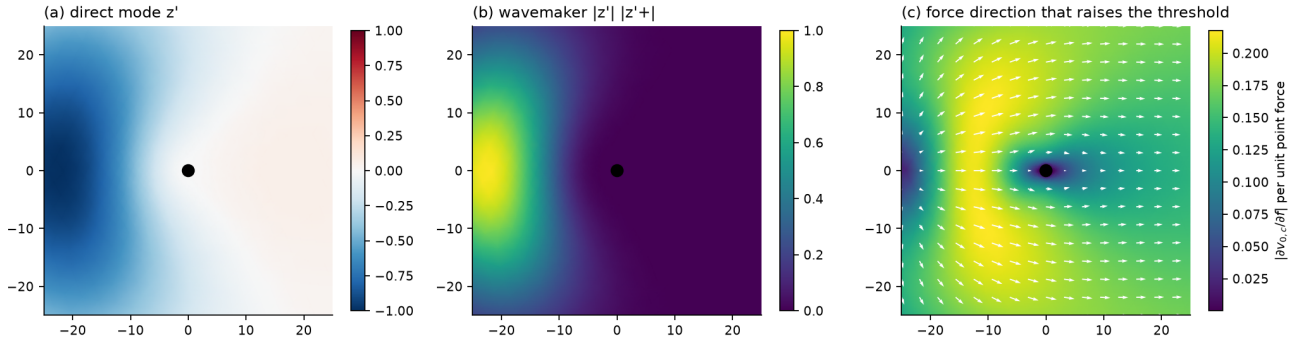


Figure 52: Normal direct/adjoint shape, wavemaker, and tangential-force direction that increases the reference threshold. The strong response is located upstream of the inclusion.

The sensitivity calculation connects an eigenvalue derivative to an intervention: forcing the upstream band with the flow reduces compression and raises the threshold, while opposite forcing advances instability. This is a local control statement about a specified global base problem, rather than evidence that the outer domain can be discarded.

10 Several anchored proteins and an approximate common drag scale

Once the corrected single-protein calculation and independent plate model agreed, the same machinery could distinguish arrangement-dependent critical velocity from the more compact force scale at onset.

10.1 Collective buckling of finite groups

The independently implemented plate model was applied to 24 arrangements of rigid force-free inclusions at $\Gamma = 25$. Adding proteins lowers the critical drive, especially when they obstruct the flow side by side. For a pair separated by $30r_0$, the threshold is 16.5 along the flow and 14.4 across it, compared with 27.44 for one protein. Five proteins across the flow at spacing $10r_0$ give $SL_c = 7.1$, whereas five in a line along the flow give 15.5. Streamwise shielding and transverse blockage therefore affect the critical velocity strongly.

(a) two PIs: threshold vs distance

(b) total anchoring force at threshold

(c) side_d10: $SL_c = 17.29$

(d) row5_d10: $SL_c = 7.06$

(e) cluster8: $SL_c = 11.26$

(f) tandem_d20: $SL_c = 19.08$

10.2 Total finite-patch drag versus dilute per-protein drag

$$F_{\text{drag,patch}} \sim \kappa/r_0 \quad \text{for the tested finite clusters.} \quad (23)$$

For dilute lattices, $L_c/r_0 = 56$ and 80 , the computed long-wave critical drag *per protein* is 1.09 and $1.00 \kappa/r_0$. The critical mean speed tends to approximately $0.24 \kappa/(\eta r_0)$. Dense lattices require larger per-protein drag: $18.6 \kappa/r_0$ at $\Phi = 8.7\%$. Thus the common force scale connects a finite patch's total drag with a dilute periodic lattice's drag per cell. It must not be applied as a per-protein criterion to every finite cluster or dense lattice. The nonlinear consequences of density are developed later in the collective-pattern sections.

11 A computed theory of subcritical snap-through

40

11.1 Continuation and the amplitude equation

The perfect, clamped-inclusion branch at zero slope bends toward smaller drive. With the critical mode normalized by $\max |z_c| = 1$ and shape amplitude A in units of r_0 , the computed amplitude equation is

$$\dot{A} = \lambda_v(v_0 - v_c)A + N_3 A^3 + H_s s + \dots, \quad (24)$$

where $\lambda_v = 3.3126 \times 10^{-4}$, $N_3 = 1.6505 \times 10^{-7} > 0$, and $H_s = -9.2488 \times 10^{-5}$ in repository units (mesh A, clamped inclusion). The coefficients follow from derivatives of the residual and direct/adjoint projections; no continuation data are fitted to obtain them. The nonzero equilibrium branch is

$$\frac{v_0}{v_c} = 1 - \frac{N_3}{\lambda_v v_c} A^2 = 1 - 0.0017966 A^2, \quad (25)$$

agreeing with 0.00179 from continuation. Its amplitude eigenvalue is positive, so this nearby buckled state is unstable. The direct cubic geometry term supplies about 96% of N_3 , and second-order flow/tension readjustment about 4%.

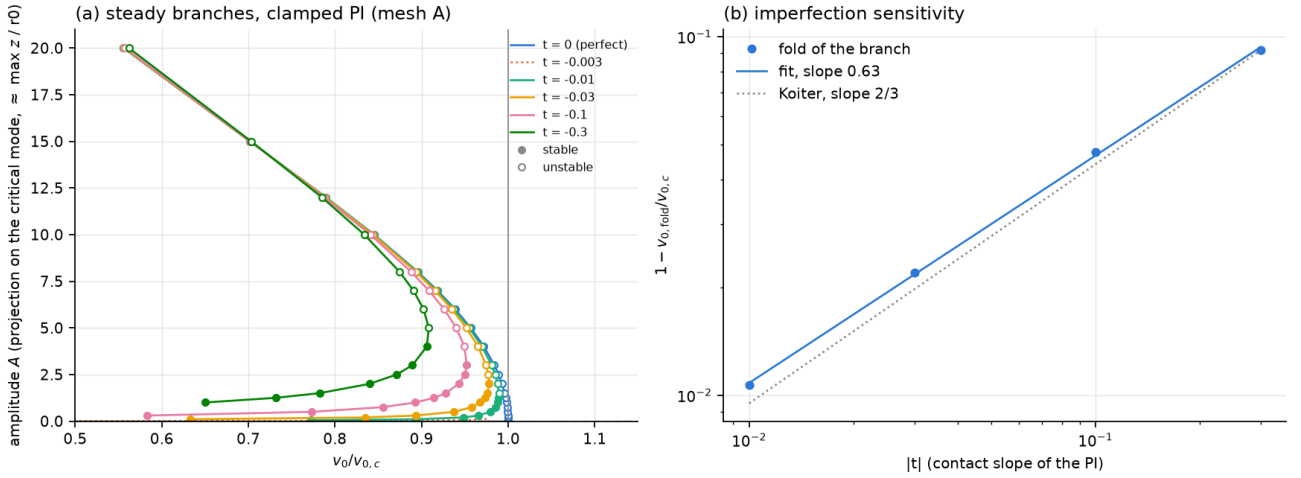


Figure 54: Corrected clamped-protein continuation. The perfect branch is subcritical and a nonzero contact slope produces a fold; the fitted imperfection exponent is close to two thirds. The later projected theory supplies an independently computed prefactor.

11.2 Residual derivatives and slaved second-order fields

Write $\Psi = \Psi_* + Ax_1 + A^2x_2 + \dots$ and expand the dynamic residual

$$\mathcal{R}(\Psi_* + d) = Jd + \frac{1}{2}\mathcal{R}_2(d, d) + \frac{1}{6}\mathcal{R}_3(d, d, d) + \dots \quad (26)$$

At the threshold, $Jx_1 = 0$, $y^T J = 0$, and $y^T Bx_1 = 1$. Up-down symmetry gives $y^T \mathcal{R}_2(x_1, x_1) = 0$. On the complementary subspace,

$$Jx_2 = -\frac{1}{2}\mathcal{R}_2(x_1, x_1), \quad N_3 = -y^T [\mathcal{R}_2(x_1, x_2) + \frac{1}{6}\mathcal{R}_3(x_1, x_1, x_1)]. \quad (27)$$

The source removes the critical projection from x_2 after solving; automatic residual differentiation supplies the higher derivatives. The contact-slope forcing is lifted into the boundary degrees of freedom and projected as $H_s = -y^T \mathcal{R}_s$. The critical-speed derivative is obtained from centred spectral differences.

For mesh A with a clamped protein, $v_c = 0.2773155$ in units $\kappa/(\eta r_0)$, $\lambda_v = 3.3126157 \times 10^{-4}$, $N_3 = 1.6504712 \times 10^{-7}$, $H_s = -9.2488061 \times 10^{-5}$. The coefficients depend on the mode normalization, friction/time units, and protein condition. They are not parameters that can be copied unchanged to the force-free or protein-coupled calculation.

11.3 Fold construction and Koiter sensitivity

At a fold, both the right-hand side of Eq. (24) and its derivative with respect to A vanish. Hence

$$A_f^3 = \frac{H_s s}{2N_3}, \quad 1 - \frac{v_f}{v_c} = \frac{3N_3}{\lambda_v v_c} \left| \frac{H_s s}{2N_3} \right|^{2/3} = 0.231 |s|^{2/3}. \quad (28)$$

The predicted shifts are 1.07, 2.23, 4.97, 10.3% for $|s| = 0.01, 0.03, 0.1, 0.3$; continuation gives 1.1, 2.2, 4.8, 9.2%. The departure at the largest slope is consistent with leaving the cubic regime. A force-free inclusion has slightly later folds: shifts of 1.8, 4.1, 8.1% for $|s| = 0.03, 0.1, 0.3$.

The coefficient 0.231 is computed from the projections, not obtained by fitting the exponent or folds. Continuation errors increase from below 1% in the intermediate small-slope case to about 11% at $|s| = 0.3$ when comparing the fold shift. This is the expected place to investigate higher-order terms, not to redefine the theory as exact at finite amplitude.

The cubic normal form has an unstable nonzero branch and contains no stabilizing large-amplitude term. Consequently it predicts a barrier and loss of the nearby stable imperfect branch, but cannot determine the ultimate attractor or the reverse-flow threshold. A demonstrated hysteresis loop needs a stabilized remote branch or a dynamic return experiment. Such evidence exists for ring/tube pulling and the intermediate-density slope model; it is incomplete for the post-snap single-protein flow overhang.

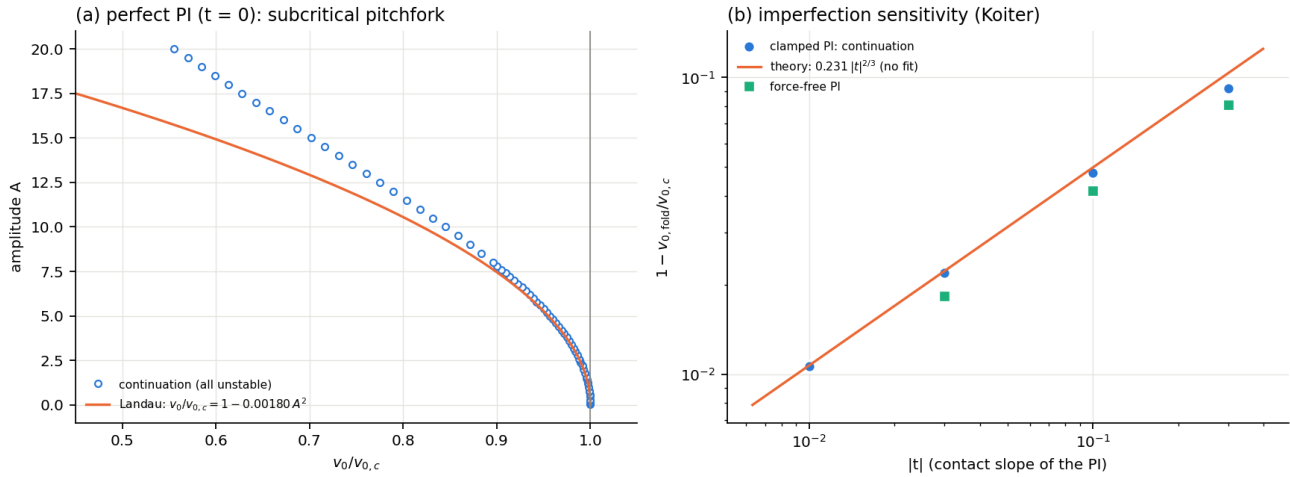


Figure 55: Weakly nonlinear predictions against full continuation. The perfect branch bends backward, establishing subcritical buckling. The imperfection law includes a coefficient computed from the equations. Clamped and force-free fold data are distinct; the displayed clamped theory is not a fitted force-free formula.

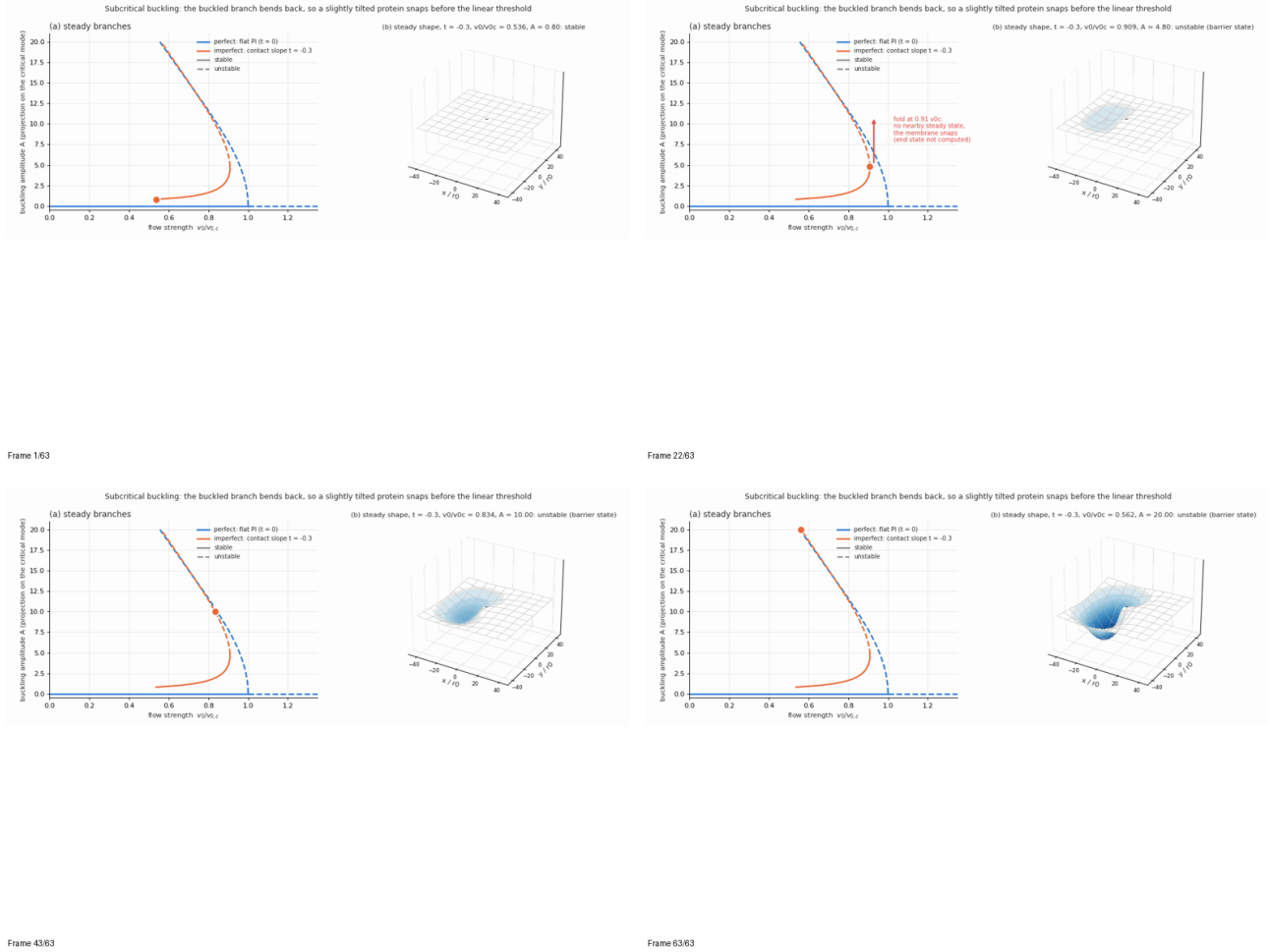


Figure 56: Branch visualization of the subcritical snap and its imperfection-sensitive fold. A continuation animation is a sequence of equilibria, not a physical-time simulation. Four ordered frames are selected by frame index; the spacing is not assumed uniform in physical time. The original GIF is included in the source package.

11.4 The force-free branch and the revised deformation scenario

For a force-free protein with $s = -0.3$, the rest height is $0.862r_0$. Its membrane remains near its rest shape up to about $SL = 23$, then snaps near 25.3. The original free-rim condition instead gives deflections above $10r_0$ already near $SL = 14$. The corrected branches therefore support an abrupt instability rather than the large smooth response of that implementation. Subcriticality establishes an imperfection-sensitive snap and a barrier; a closed steady loading–unloading loop for the post-snap single-protein flow state has not been resolved. It should not be inferred solely from the cubic equation.

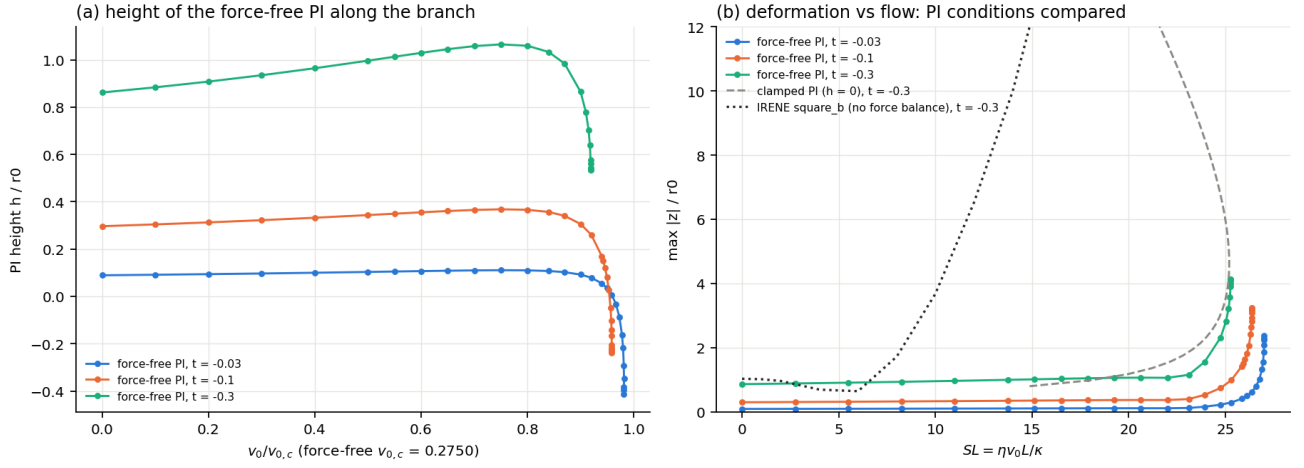


Figure 57: Nonlinear force-free branches for nonzero contact slopes, contrasted with clamped and original-rim conditions. The corrected free protein stays near its rest deformation before an imperfection-sensitive fold.

12 Nonlinear collapse, bottlenecks, and a protein that rises

Time integration answers a question left open by the cubic equation: what happens after the nearby branch disappears? The result is strongly asymmetric and differs qualitatively from a protein that follows a pit downward.

12.1 BDF2 stepping and a force-free height solve

The time residual is formed from the same weak equations using $(3\Psi_{n+1} - 4\Psi_n + \Psi_{n-1})/(2\Delta t)$ on dynamic fields. Initial growth is compared with the projected eigenmode amplitude. A secant solve on protein height closes the force-free condition at each step. If Newton fails, the step is reduced; rebuilding the linear solver after a failed factorization was required to avoid inherited MUMPS failure states.

At $s = -0.3$, force-free runs at $0.93v_c$ and $0.97v_c$ pass the fold near $0.919v_c$. They spend about 6×10^5 and 1.6×10^5 time units in the bottleneck before collapsing. The protein rises rather than falls; slopes eventually exceed eleven. Annular-force consistency is the relevant late-stage accuracy diagnostic, and it deteriorates before one could claim a converged terminal shape.

12.2 Clamped and force-free trajectories

BDF2 integration beyond the imperfect fold shows a long bottleneck followed by accelerating collapse. With a clamped inclusion the upstream pit reaches about $20\text{--}22r_0$ depth and the wall approaches vertical. With a force-free inclusion, a secant solve enforces zero vertical reaction at each time step. The protein rises to $7\text{--}8r_0$ on the downstream crest while the upstream pit deepens to approximately $-16.5r_0$. This reverses the early expectation that a free protein would simply follow the pit downward.

The drop from crest to pit is about $24r_0$ across $10r_0$, creating an S-shaped profile. Protein lift is established before the annular-force diagnostic degrades at large slope. The late Monge snapshots identify a geometric limit, not converged predictions of a terminal shape.

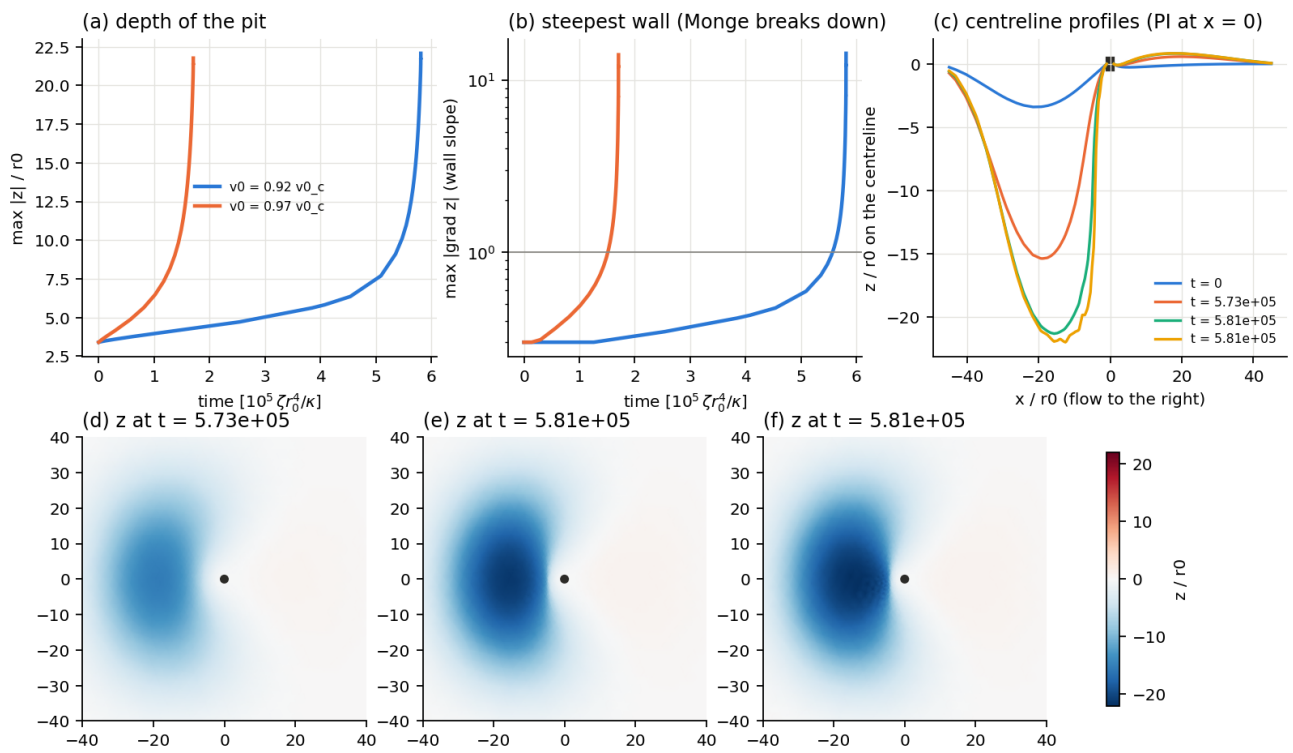


Figure 58: Clamped-protein time stepping after the fold: bottleneck, deepening upstream pit, and rapid slope growth. The final steep shapes are at the Monge limit, not resolved steady attractors.

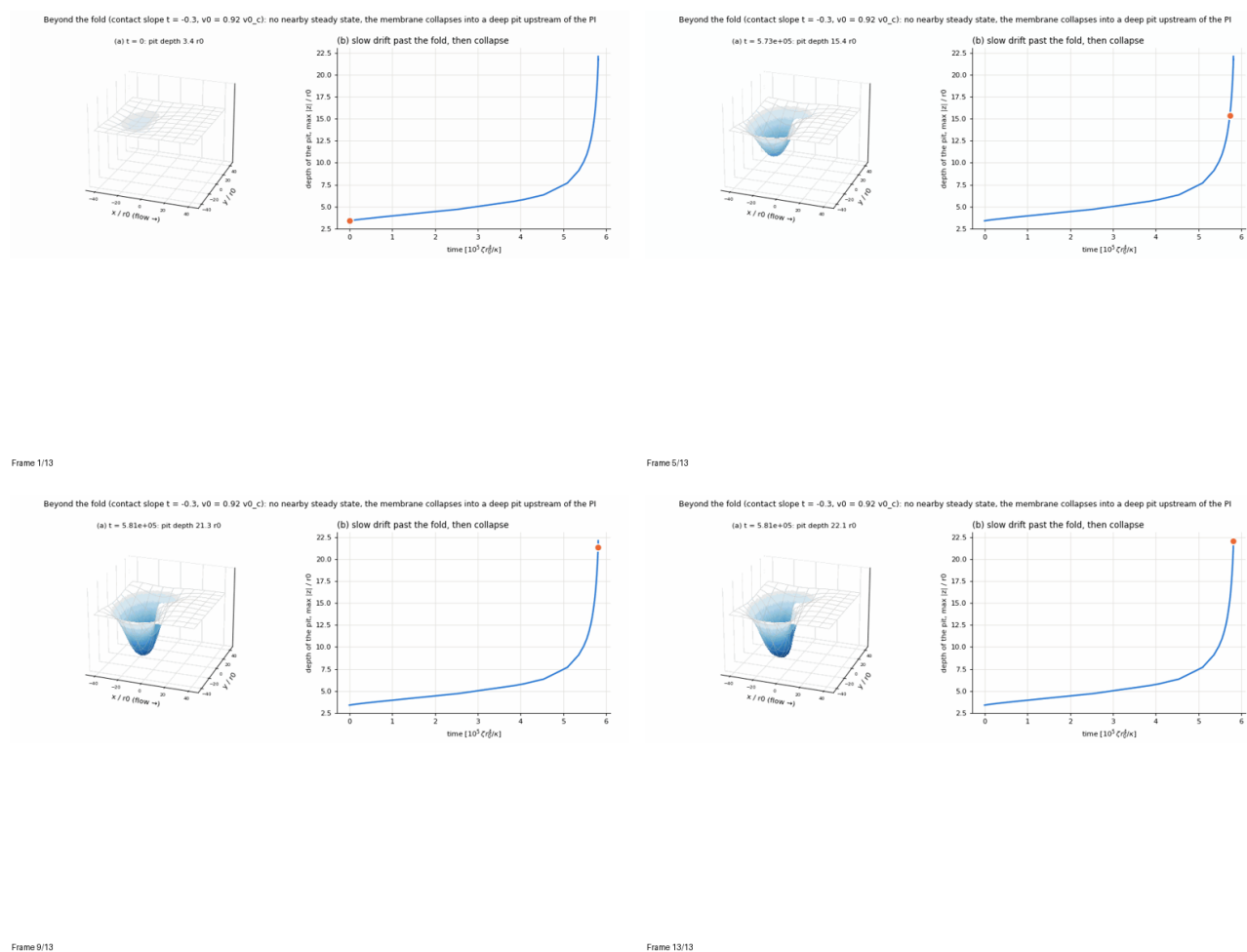


Figure 59: Clamped-protein post-fold time integration: an upstream pit grows and the wall approaches vertical. Four ordered frames are selected by frame index; the spacing is not assumed uniform in physical time. The original GIF is included in the source package.

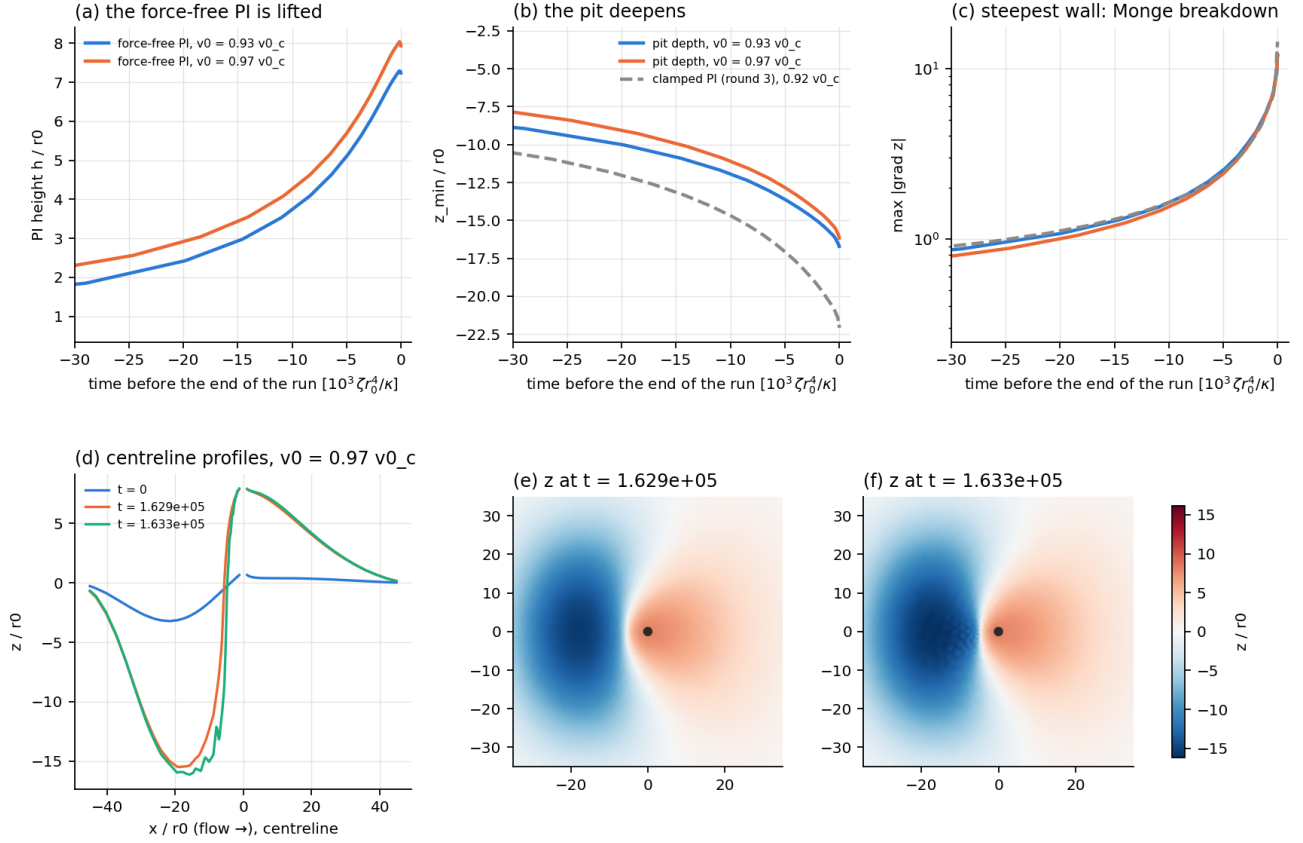


Figure 60: Force-free protein lift during upstream collapse. The protein rises onto a downstream crest and the wall steepens; cutoff-force consistency deteriorates at the last slopes.

The bottleneck is consistent with passage near an imperfect fold, whereas the subsequent accelerating collapse leaves the small-amplitude expansion. The geometric and annular-force diagnostics must therefore accompany late snapshots; a larger apparent depth is not automatically a more accurate physical endpoint.

13 Parametric surfaces, overhangs, and the limits of remeshing

A height graph fails when the membrane wall becomes vertical. The parametric extension tests whether the observed steepening is a representation limit or the beginning of an actual overturning surface.

13.1 General frame, mesh gauge, and comparison with Monge dynamics

The surface is generalized to $\mathbf{X}(\mathbf{u}, t)$ with a mixed frame $\mathbf{E} = \nabla_{\mathbf{u}} \mathbf{X}$. IRENE's geometric and force forms are reused with this frame. Kinematics is $\partial_t \mathbf{X} = w\mathbf{n} + a^i \mathbf{e}_i$, where $\mathbf{a}$ chooses tangential mesh motion rather than material transport. A regularized quasi-Monge gauge allows parametrization points to slide as the wall turns vertical. The protein's force is evaluated as a consistent vertical reaction of the volume force balances.

From a common snapshot at slope 1.7, the two descriptions agree closely in pit depth up to slope seven; protein heights differ by about 7%, reflecting the sensitivity of the force-free height to the force discretization. The parametric calculation reaches $\min n_z \simeq -0.67$, a genuine overhang, with the protein at $h \simeq 10.5\text{--}11r_0$ and pit depth $-23.6r_0$. The membrane undercuts the elevated disc. This supports the onset of a flow-driven invagination within the model.

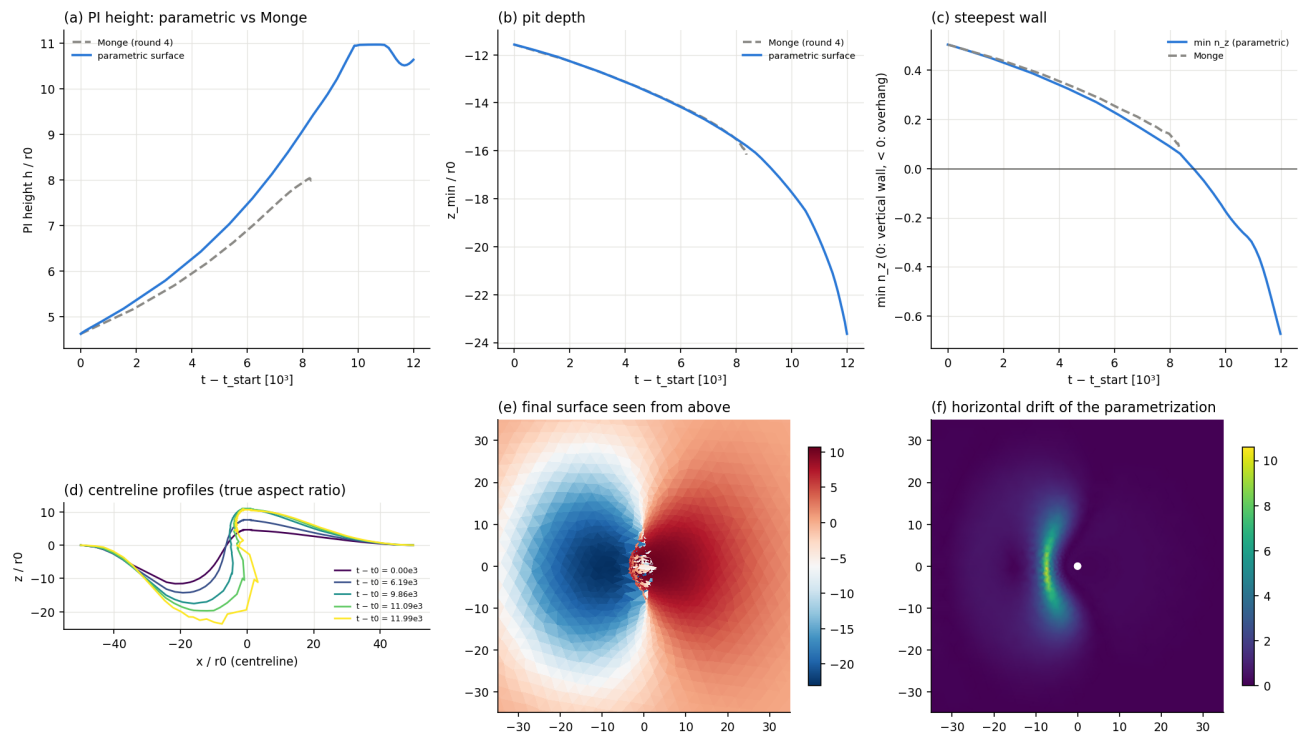


Figure 61: Continuation of the force-free post-snap state on a parametric surface, compared with the Monge calculation. True-aspect-ratio centreline profiles show the wall turning over beneath the elevated protein. The final large-deformation state is limited by mesh distortion; it is not a completed tube.

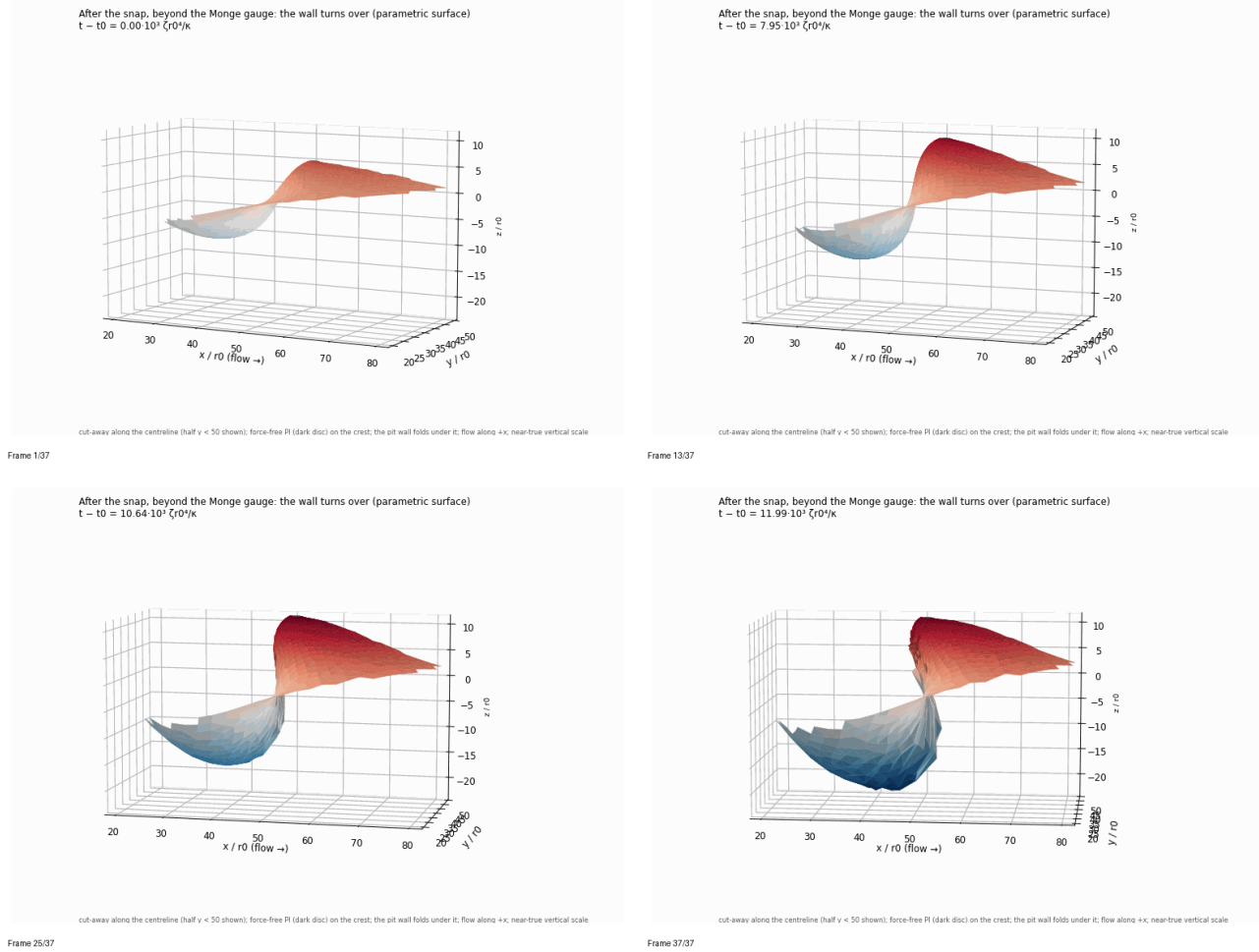


Figure 62: Parametric continuation of the force-free post-snap wall through vertical into an overhang underneath the lifted disc. The final state is distortion-limited. Four ordered frames are selected by frame index; the spacing is not assumed uniform in physical time. The original GIF is included in the source package.

13.2 Mesh regularity and transfer experiments

Use $\mathbf{X}(\mathbf{u}, t)$ and $\mathbf{E}_i = \partial_i \mathbf{X}$ as independent mixed fields. The metric, normal, and curvature now follow from the general frame. Tangential parametrization velocity a^i is a gauge in $\mathbf{X}_t = w\mathbf{n} + a^i \mathbf{e}_i$. A regularized quasi-Monge choice minimizes horizontal mesh displacement while permitting sliding along steep walls. It is distinct from the material tangential velocity v^i .

The production run changes its gauge regularization from 0.05 to 0.5 near a very steep wall and reaches negative vertical normal. The result is an overhang, with mesh stretch about forty and anisotropy later diagnosed near seventy-nine. The later remeshing study uses log-polar harmonic coordinates because an ordinary map drives nodes into the hole. Pinning $r < 2.5r_0$ gives an injective map and reduces anisotropy to about ten, but transfers do not converge in the new discretization. An identity transfer converges, showing that the transfer machinery itself is not the only obstruction. Coarse-facet curvature, gauge changes, and a poorly resolved fold preclude a reliable tube continuation. The proposed remedy is weak-form ALE smoothing or early remeshing with C^1 -consistent transfer.

13.3 Established invagination and unresolved tube evolution

The final wall occupies a few distorted elements, with local area stretch approaching forty. Later harmonic reparametrization and remeshing reduce distortion but do not yield a convergent continuation to a tube. A regular ALE formulation with integrated mesh smoothing, or earlier remeshing with curvature-consistent transfer, remains necessary. Self-contact, scission, and a physical endpoint are outside the demonstrated result. The axisymmetric *no-flow* tube calculation in the no-flow arclength calculation must not be confused with a successfully grown *flow-driven* tube.

The negative vertical normal in Figure 61 demonstrates an overhang. It does not by itself demonstrate a stable

invagination, continued tube growth, self-contact, or scission. The remeshing experiments identify a concrete next methodological task: preserving curvature and force consistency while redistributing the discretization before the wall is carried by only a few facets.

14 Tangential friction and a saturated source of upstream waves

Friction changes more than the shape-relaxation time. Its in-plane body force changes the base stress and introduces a follower contribution to the normal operator. This creates an oscillatory route to instability that the conservative no-friction plate cannot support.

14.1 Follower forcing and the divergence–flutter map

For a tangential body force $\mathbf{f}$, in-plane balance gives $\nabla \cdot \mathbf{T} + \mathbf{f} = 0$. The normal stress acts as $\mathbf{T} : \nabla \nabla z$, so the flat-state equation becomes

$$\zeta z_t = -\kappa \Delta^2 z + \nabla \cdot (\mathbf{T} \nabla z) + \mathbf{f} \cdot \nabla z. \quad (29)$$

For Brinkman coupling, $\mathbf{f} = b(v_0 \mathbf{e}_x - \mathbf{v})$. The last term breaks self-adjointness and permits a complex pair to cross first. Merely inserting a friction-dependent stress into the divergence-form operator would miss this follower term.

A map in (ℓ_b, Γ) identifies flutter between static-divergence sectors. For $\Gamma = 25$, flutter occurs first over approximately $3.2 < \ell_b/r_0 < 13$; increasing tension narrows the window. Eleven comparisons of the plate and full IRENE calculations agree within 0.3%, including instability type. Strong and weak friction recover static onset in the tested reference boxes.

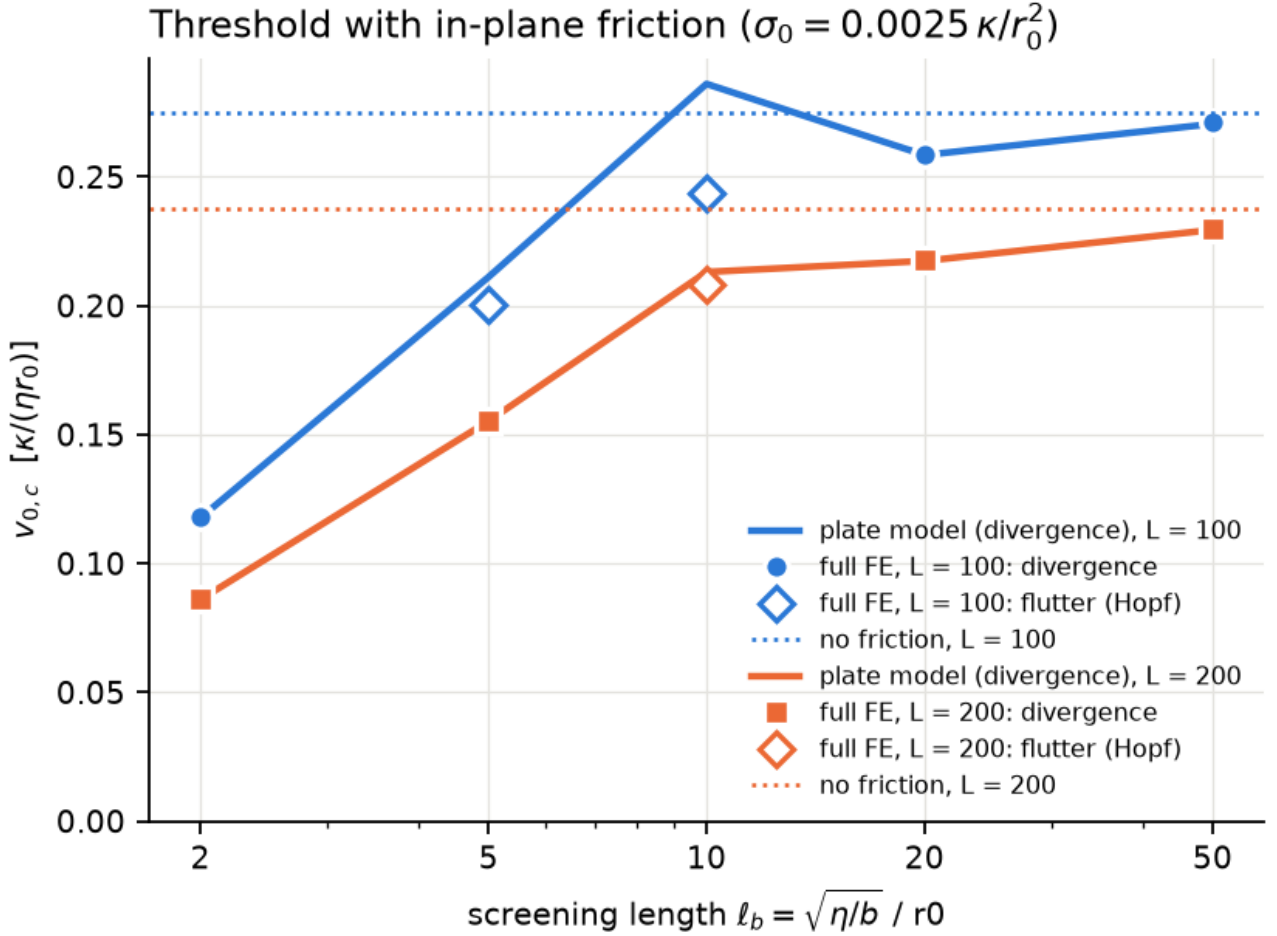


Figure 63: Early friction comparison showing divergence thresholds and full-equation flutter crossings. The real plate threshold curve alone is not the first onset when a complex pair crosses earlier; the subsequent divergence–flutter map resolves this.

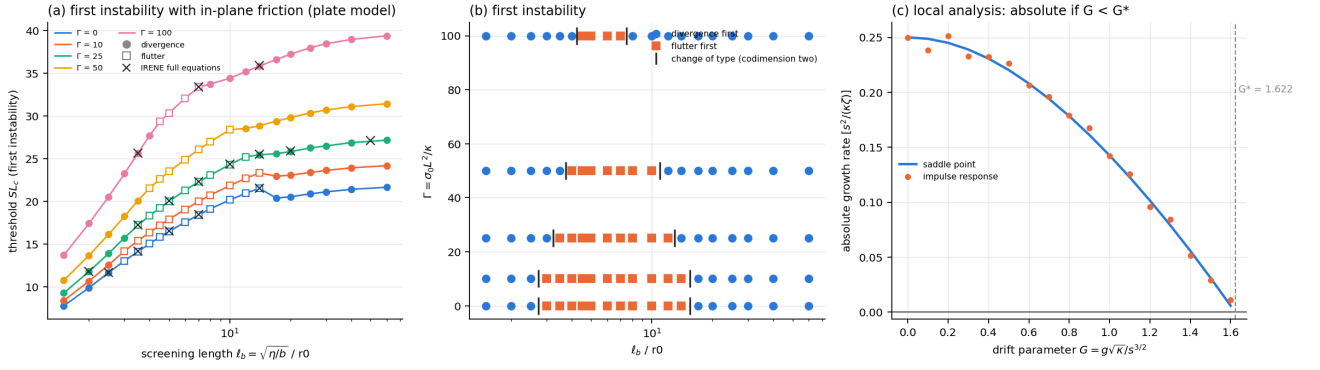


Figure 64: Divergence/flutter onset map versus screening length and tension, with full-equation checks. The local absolute-instability panel is a guide to the global wave source; near onset the local wavelength is comparable to the compressed region.

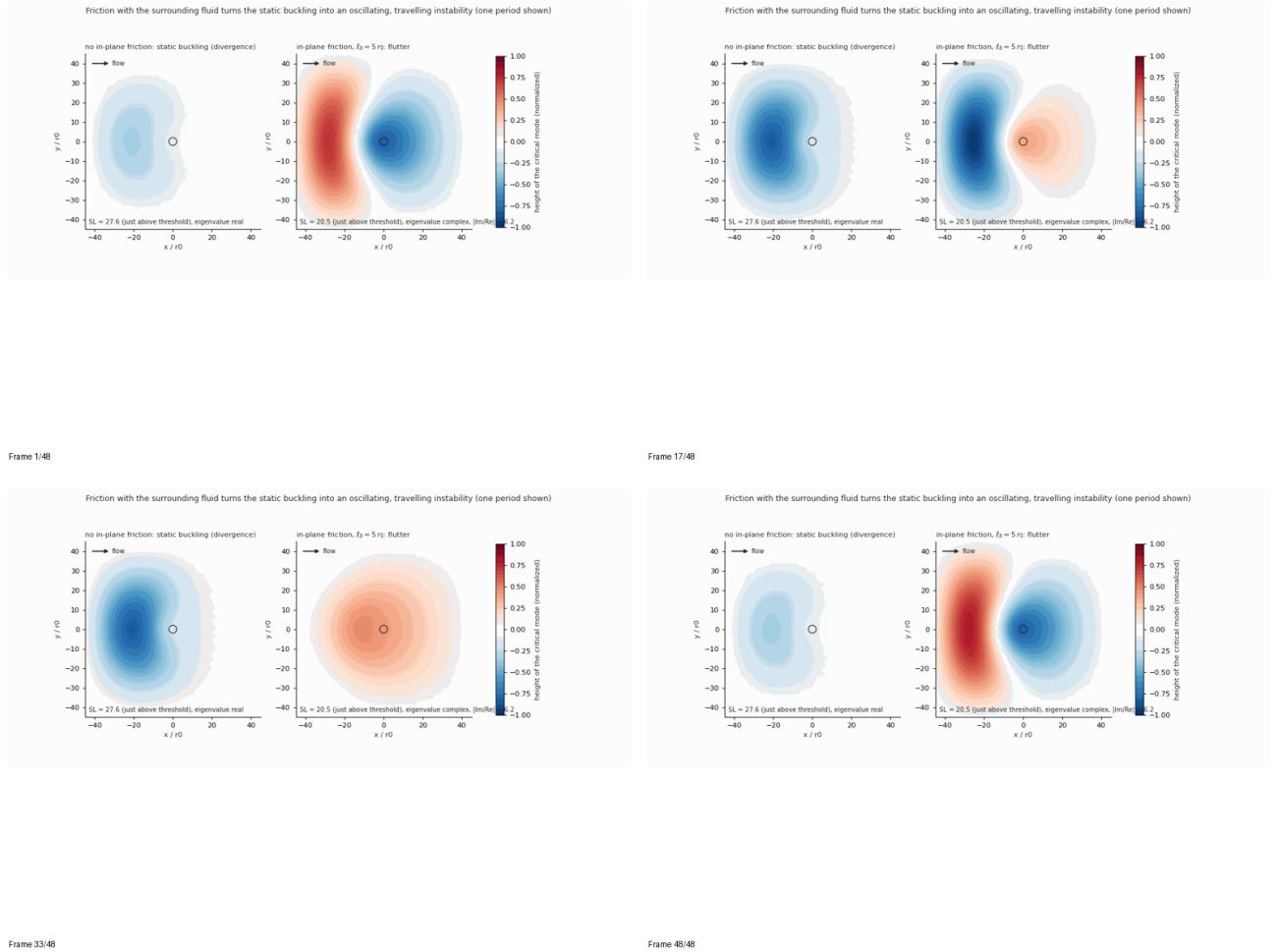


Figure 65: Real divergence mode versus oscillatory flutter mode. The travelling phase of the complex mode is distinct from a nonlinear saturated cycle. Four ordered frames are selected by frame index; the spacing is not assumed uniform in physical time. The original GIF is included in the source package.

14.2 Nonlinear saturation and the below-threshold restart

At $l_b = 7r_0$, the flat-state Hopf threshold is $SL_c = 22.30$, with $\omega_c = 7.96 \times 10^{-6} \tau_n^{-1}$ and period $7.9 \times 10^5 \tau_n$. At 1.05, 1.10, 1.20 times the critical velocity, the measured upstream amplitudes are 3.9 ± 0.3 , 6.5 ± 0.6 , and approximately $11r_0$. The protein itself oscillates by roughly $\pm 3-4r_0$, emitting crests that travel upstream and decay over about $30r_0$.

At $1.05v_c$, the initial growth rate is 3.3×10^{-6} , compared with 3.7×10^{-6} inferred from the linear slope. A below-threshold kick decays. More decisively, restarting the saturated cycle at $0.97v_c$ reduces its amplitude to

about $0.01r_0$ after three periods. No hysteresis is observed in this test, unlike the barrier-controlled ring and subcritical lattice states.

These runs support a supercritical, saturating flutter regime. The amplitudes increase faster than the leading square-root law, and fitted nonlinear coefficients depend on the sampled drive, indicating higher-order effects. Flutter is the stable protein-attached wave source demonstrated in the full flowing-membrane dynamics. Its dimensional period remains proportional to the chosen normal friction; a local drag model cannot supply a quantitatively established solvent-wave frequency without an appropriate bulk-fluid calculation.

At $\ell_b = 7r_0$, $\Gamma = 25$, the nonlinear calculations use roughly twenty-eight BDF2 steps per linear period. They show sustained limit cycles and decay after reducing the drive to $0.97v_c$. This establishes stable saturation in the tested runs. It does not replace time-step refinement, a near-onset adjoint Hopf coefficient, or a Floquet spectrum for all possible perturbations. The effective cubic damping inferred from finite amplitudes varies between runs; the available amplitudes already involve higher-order nonlinearities.

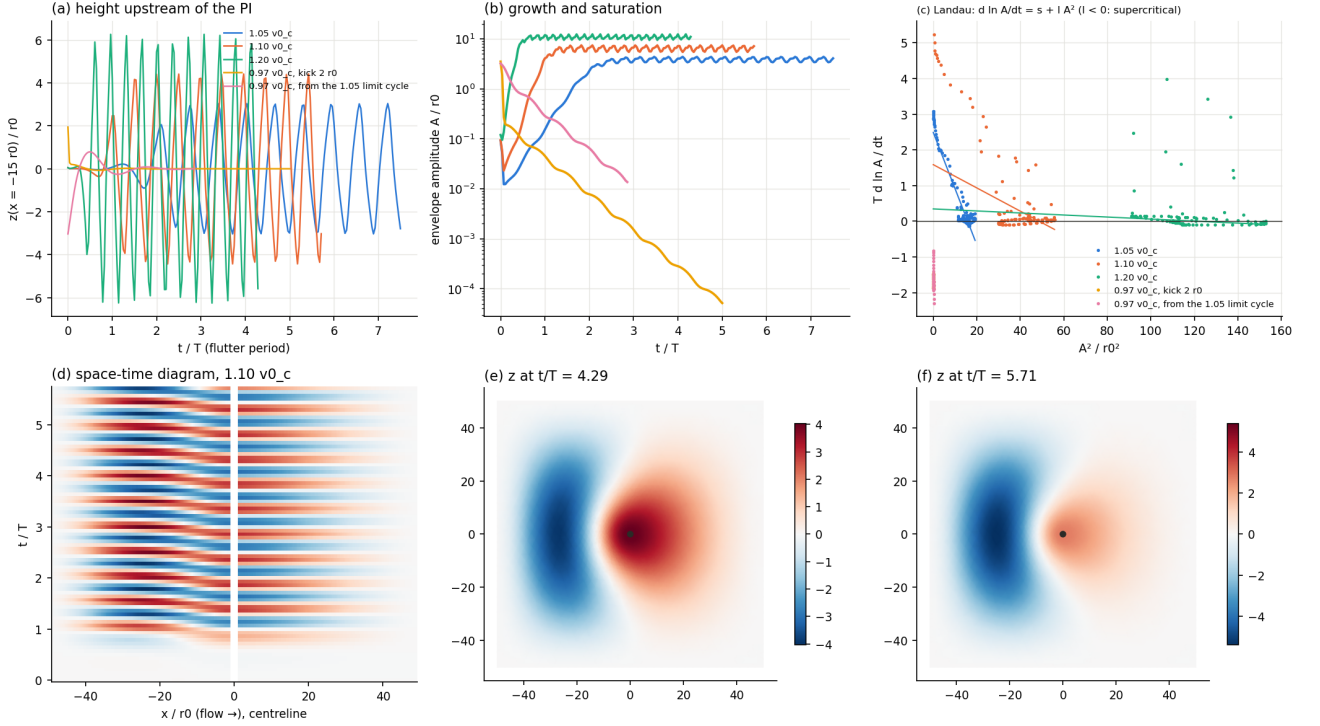


Figure 66: Nonlinear flutter at $\ell_b = 7r_0$, $\Gamma = 25$, and a rigid force-free protein. Growth saturates above onset, and the space-time bands reveal upstream propagation. The negative effective nonlinear growth contribution supports a stabilizing Hopf transition; these finite-amplitude runs are not a precision measurement of a single cubic coefficient.

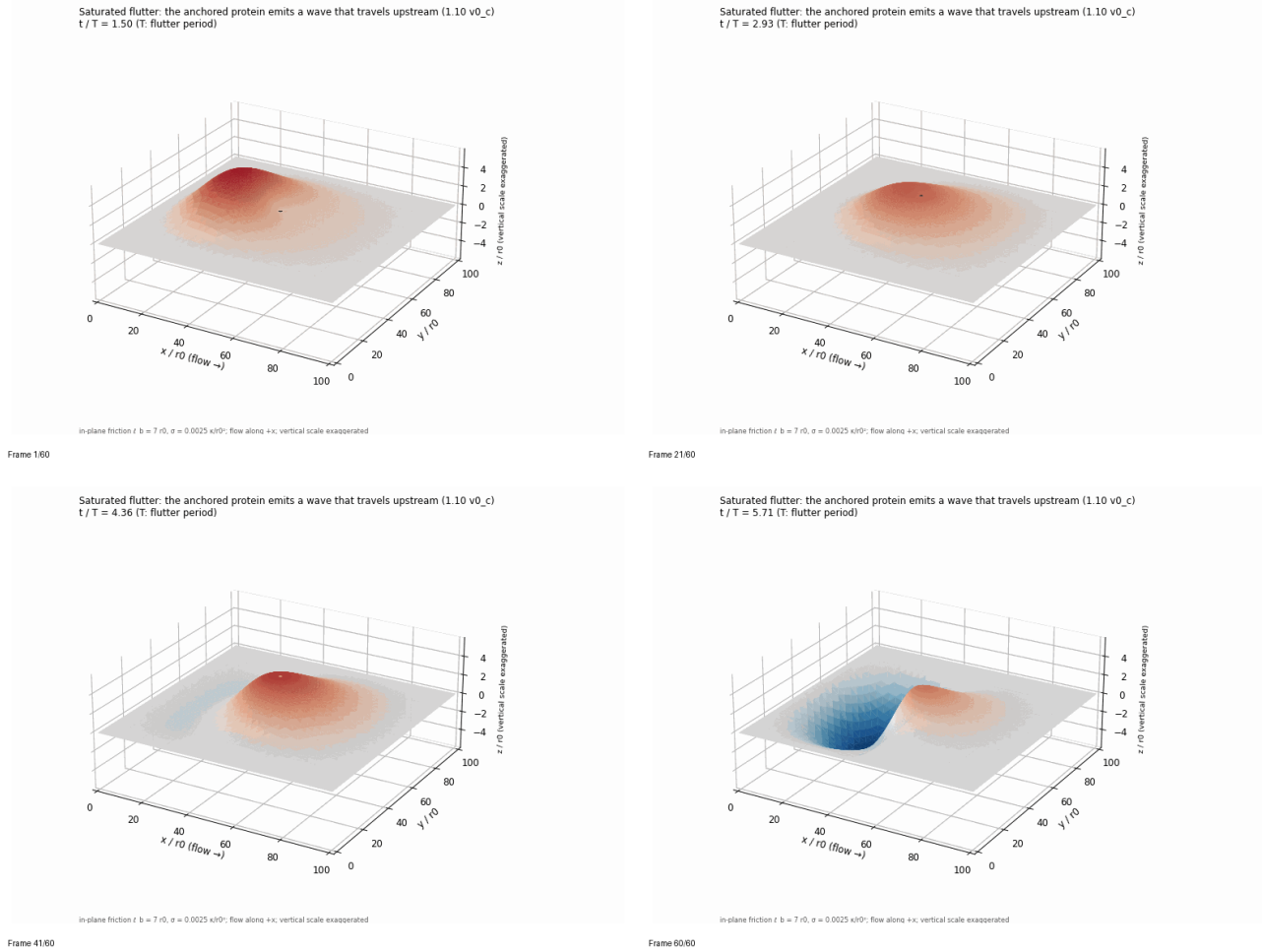


Figure 67: Nonlinear friction-driven flutter: protein oscillation and a sustained source of upstream membrane waves. Four ordered frames are selected by frame index; the spacing is not assumed uniform in physical time. The original GIF is included in the source package.

Flutter supplies the stable, protein-attached dynamic state demonstrated in the full flowing-membrane calculations. Its stability is established by the tested time trajectories and return below threshold; broader perturbation stability and physical-frequency calibration remain separate checks.

15 Consistent equations for mobile curvature proteins

The next extension replaces purely anchored geometric inclusions by a mobile composition field coupled to spontaneous curvature. It adds a conserved relaxation time and allows membrane shape to feed back on protein transport.

15.1 Energy, density convention, and nonlinear virtual power

The mobile field φ is a signed deviation from a reference protein area fraction, not a nonnegative absolute density. The code does not specify enough biological calibration to infer an absolute concentration from a value such as -0.36 . Mean curvature and protein spontaneous curvature have a sign tied to the chosen normal. A large negative deviation correlated with a negative-curvature pit is an unambiguous mechanochemical correlation in this convention; interpreting it as enrichment requires identifying which signed species/deviation corresponds to added curvature proteins. The report therefore avoids converting the plotted deviation directly into a positive molecular concentration.

The consistent free energy is

$$G = \int [2\kappa(\mathcal{H} - C\varphi)^2 + g_p(\varphi) + \frac{1}{2}\epsilon_p|\nabla_s\varphi|^2]dA, \quad g_p = \frac{1}{2}\chi\varphi^2 + \frac{1}{4}\beta\varphi^4. \quad (30)$$

The chemical potential is $m = -4\kappa C(\mathcal{H} - C\varphi) + g'_p(\varphi) - \epsilon_p\Delta_s\varphi$. In a vertically moving Monge coordinate frame,

$\dot{z} = w\sqrt{g}$ and $a^i = g^{ij}\omega_j\dot{z}$, so

$$\partial_t\varphi + (v^i - a^i)\partial_i\varphi = M\Delta_s m + S_Q - \gamma s_{\text{out}}(x)\varphi. \quad (31)$$

Away from open reservoirs/sources/sinks this is material conservation on an incompressible surface. Omitting the mesh-frame velocity a^i changes nonlinear transport.

The virtual-power protein normal coupling contains

$$2\kappa\{\Delta_s(\mathcal{H} - C\varphi) + 2(\mathcal{H} - C\varphi)[\mathcal{H}^2 + \mathcal{H}C\varphi - \mathcal{K}]\} - 2\mathcal{H}[\sigma + g_p(\varphi)] + \epsilon_p[b^{ij}\varphi_{,i}\varphi_{,j} - \mathcal{H}|\nabla_s\varphi|^2], \quad (32)$$

with the force/residual orientation taken consistently with IRENE. The tangential internal term is $-m\nabla_s\varphi$ in the consistent residual. Its alternative Gibbs–Duhem representation differs from $\varphi\nabla_sm$ by a gradient that can be absorbed into the incompressibility tension, provided the accompanying normal and boundary terms use the same convention. The earlier forms were exact at linear order only; they must not be used to claim nonlinear energy consistency.

At rest, the consistent runs reproduce the linear thresholds and give a monotonic energy decrease from 24.992 to about 23.90 for the documented labyrinth calculation. A quartic coefficient $\beta = 1$ saturates this demixing state. Protein-number changes in open-domain tests are accounted for by reservoir exchange rather than assumed to vanish.

15.2 Analytical dispersion and relative advection

Set $\kappa = \zeta = 1$, $C_0 = 2C$, and linearize a uniform flat membrane advected at speed U . For a Fourier wavevector (k_x, k_y) , $k^2 = k_x^2 + k_y^2$, the matrix is

$$\begin{pmatrix} -(k^4 + \sigma_0 k^2) & -C_0 k^2 \\ -MC_0 k^4 & -Mk^2(C_0^2 + \chi + \epsilon_p k^2) - iUk_x \end{pmatrix}. \quad (33)$$

At zero flow the determinant changes sign when $\chi + \epsilon_p k^2 = -\sigma_0 C_0^2 / (k^2 + \sigma_0)$. If $C_0 > \sqrt{\epsilon_p \sigma_0}$, minimization gives

$$\chi_c = \epsilon_p \sigma_0 - 2C_0 \sqrt{\epsilon_p \sigma_0}, \quad k_*^2 = C_0 \sqrt{\sigma_0 / \epsilon_p} - \sigma_0. \quad (34)$$

For $C_0 = 0.5$, $\epsilon_p = 1$, $\sigma_0 = 0.0025$, the wavelength is about $42r_0$ and $\chi_c = -0.0475$; the finite-box onset is -0.0506 . This analytical criterion recovers the classic curvature-composition mechanism [7], while advection detunes wavevectors along the flow. Transverse wavevectors avoid that relative phase drift, explaining stripes parallel to the flow in the uniform/local problem. The lattice terraces, by contrast, have ridges *across* the flow: these are different patterns in different fields.

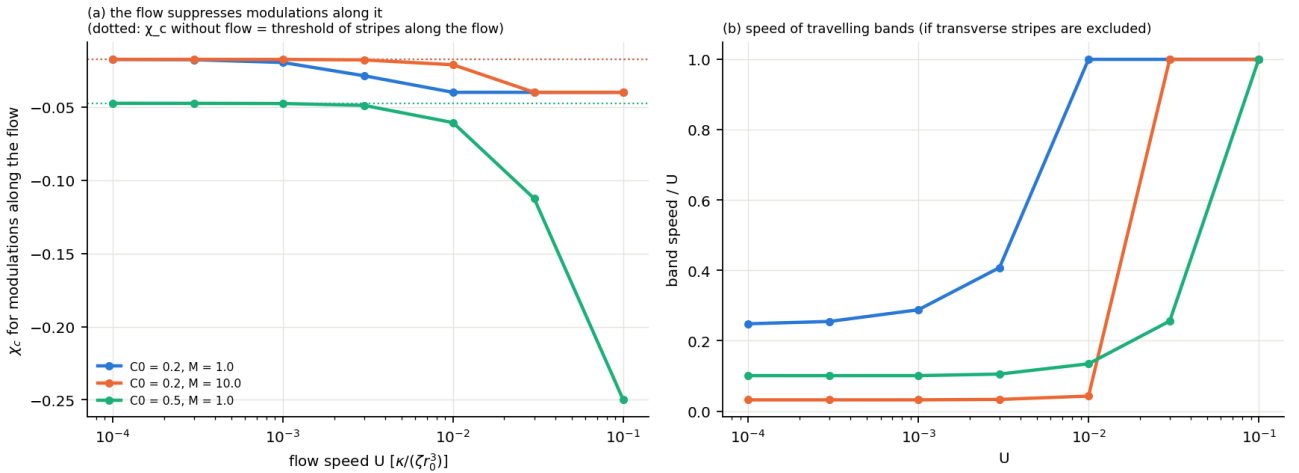


Figure 68: Uniform-plane curvature-protein dispersion under relative drift. Transverse wavevectors remain least affected, giving stripes parallel to the flow; travelling longitudinal bands require suppressing those transverse modes.

16 Demixing, transported domains, and the treatment of outflow

The nonlinear protein implementation was checked first without flow, where free-energy decay is a useful consistency test, and then in open domains. The open boundary matters because material can leave the observed patch before it has time to demix.

16.1 Energy relaxation, domains, and a source-driven plume

The consistent model reproduces its no-flow linear threshold and relaxes into modulated domains with a decreasing free energy. In flow, domains elongate along the stream, while a rim source creates an advected curvature-associated wake. A quartic local term can saturate demixing in the resting membrane; that saturation cannot simply be assumed to arrest a flow-driven shape collapse.

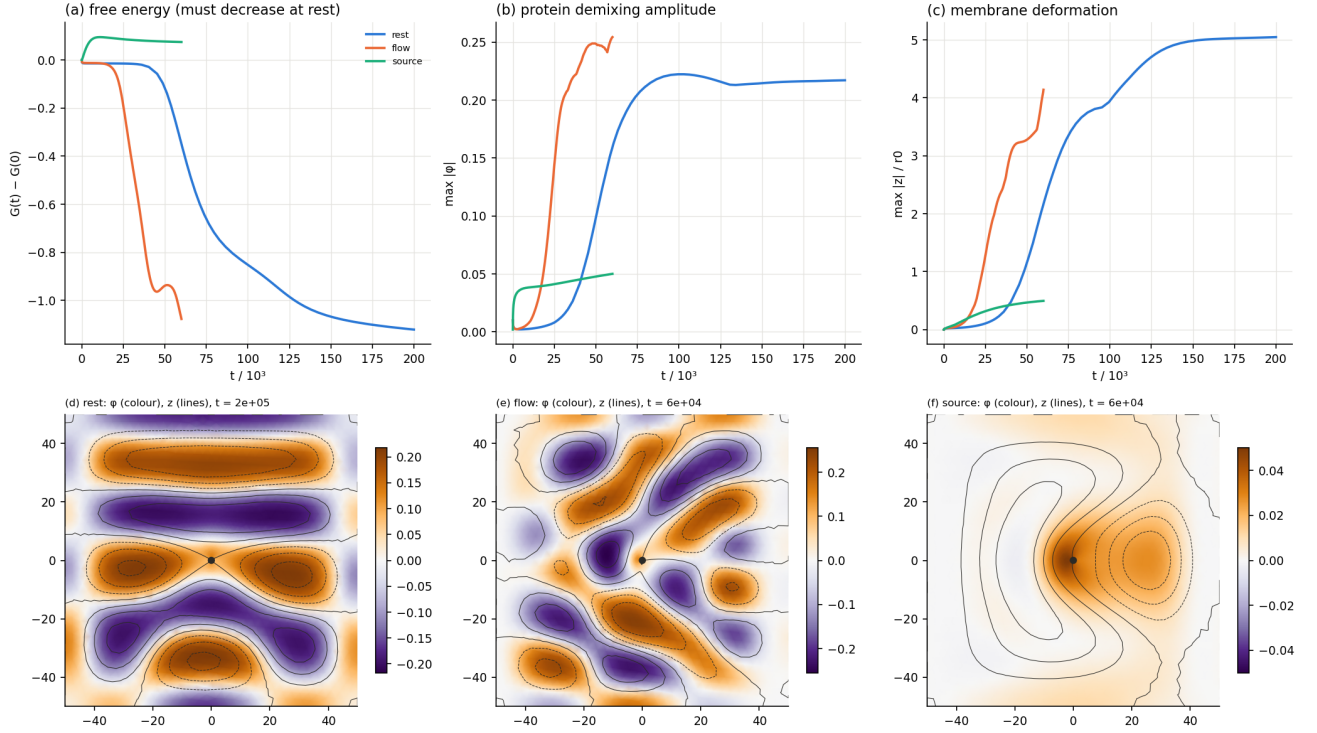


Figure 69: Consistent nonlinear mobile-protein dynamics: no-flow energy relaxation, amplitude saturation, flow-elongated domains, and a source-driven plume. Open reservoirs/sinks mean global protein conservation must include boundary exchange.

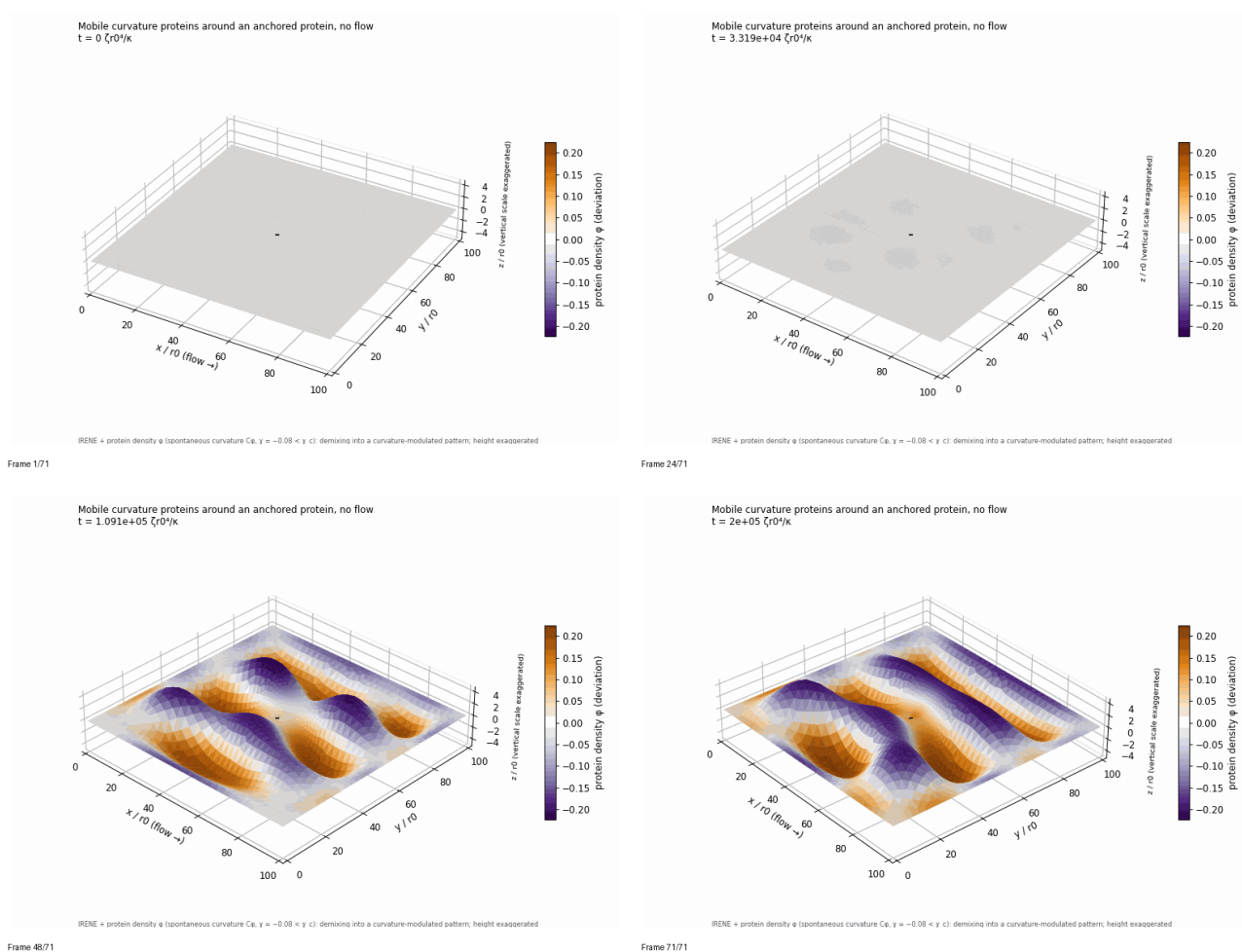


Figure 70: Consistent nonlinear curvature-protein demixing at rest. A modulated labyrinth forms while the free energy decreases. Four ordered frames are selected by frame index; the spacing is not assumed uniform in physical time. The original GIF is included in the source package.

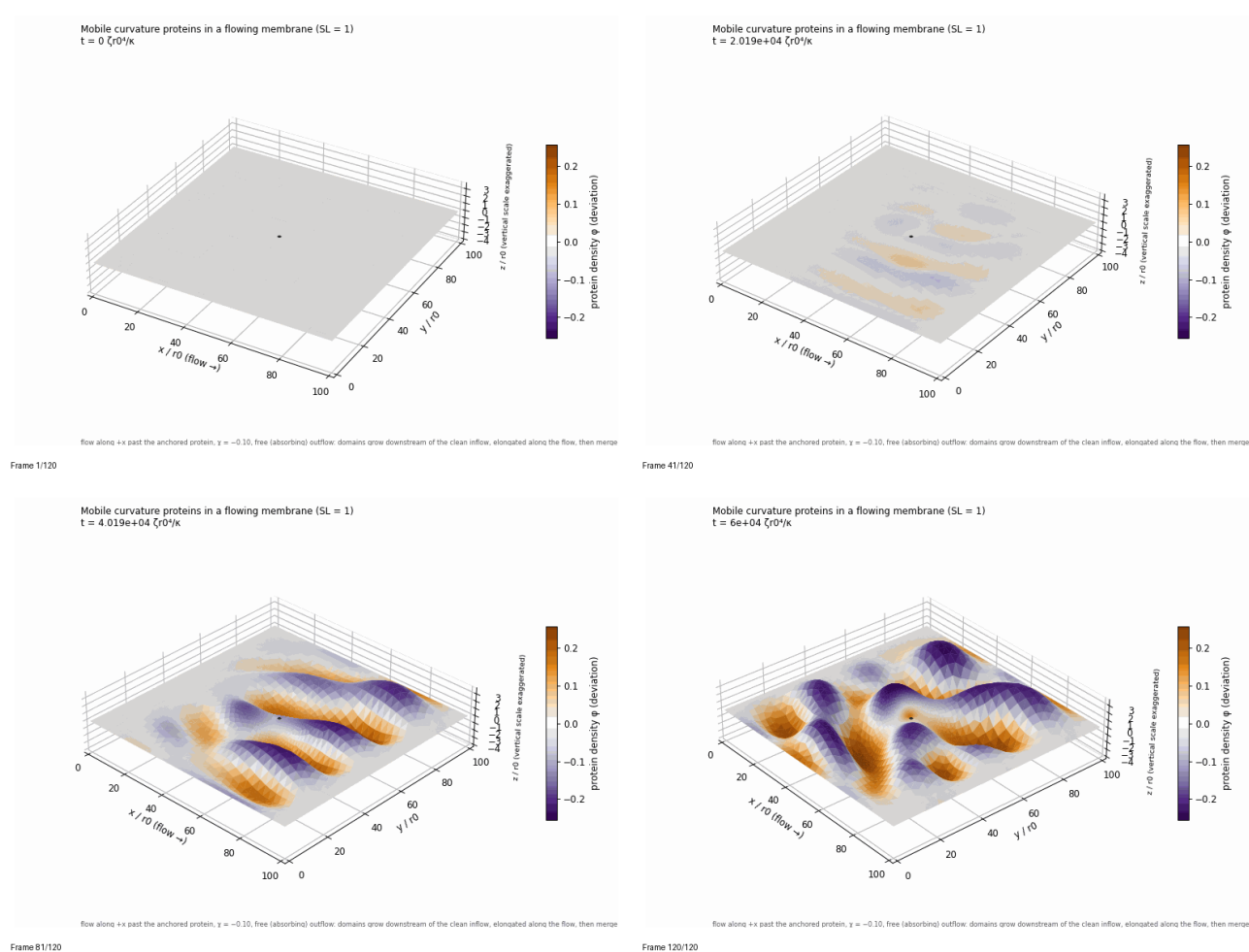


Figure 71: Mobile proteins in open flow. Downstream domains elongate along the flow and the membrane follows their curvature modulation. Four ordered frames are selected by frame index; the spacing is not assumed uniform in physical time. The original GIF is included in the source package.

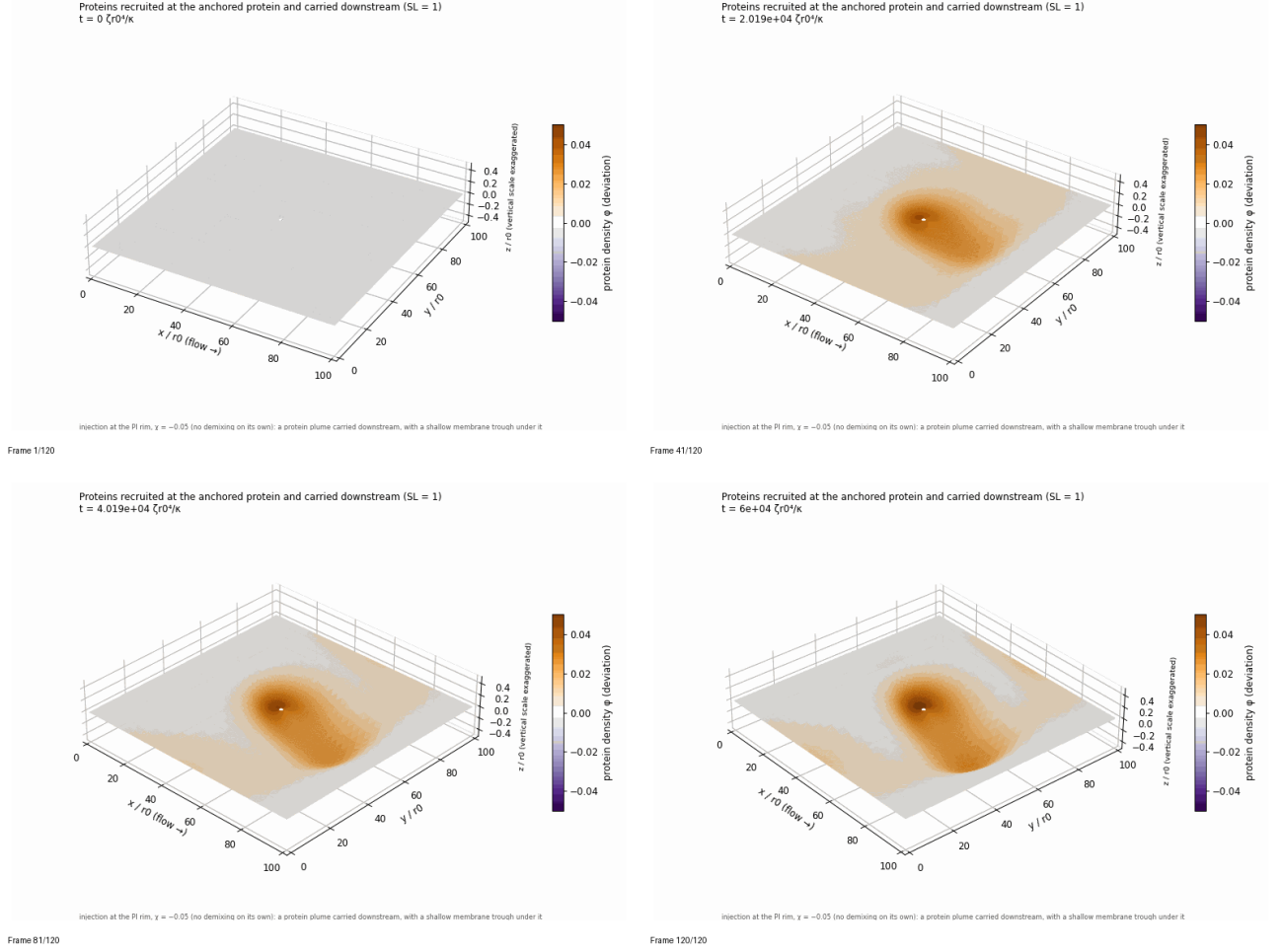


Figure 72: Rim-source protein plume carried downstream with a shallow associated trough. The final run uses absorbing outflow. Four ordered frames are selected by frame index; the spacing is not assumed uniform in physical time. The original GIF is included in the source package.

16.2 Transport scales and boundary artifacts

A useful scale-dependent comparison is $Pe_\ell = U\ell/D_k$, with $D_k = M(C_0^2 + \chi + \epsilon_p k^2)$ for the stable uncoupled protein relaxation at $k \sim 1/\ell$. Equivalently compare the measured coupled relaxation time with ℓ/U near a neutral mode. There is no single constant diffusivity valid for every wavevector or for negative stiffness beyond a spinodal. Mobility scans $M = 1, 10, 100, 1000$ consequently represent transport regimes for the fixed parameters, not four universal Péclet values.

The inflow fixes $\varphi = 0$. A no-diffusive-flux outflow can retain an edge mode; a passive sponge raising χ produces a chemical barrier whose drift scales with $M\nabla\chi$. The final absorbing sponge preserves material coefficients and adds a sink with $\gamma = 0.05$ over the last $15r_0$. For $M = 1$ it gives $\chi_c = -0.051, -0.083, -0.124, -0.177, -0.406$ at $SL = 0, 2, 5, 10, 20$. The older no-flux edge mode at $SL = 10$ and apparent disappearance of demixing at 20 are not the final free-outflow result. Sponge width/rate sensitivity remains an appropriate publication robustness check.

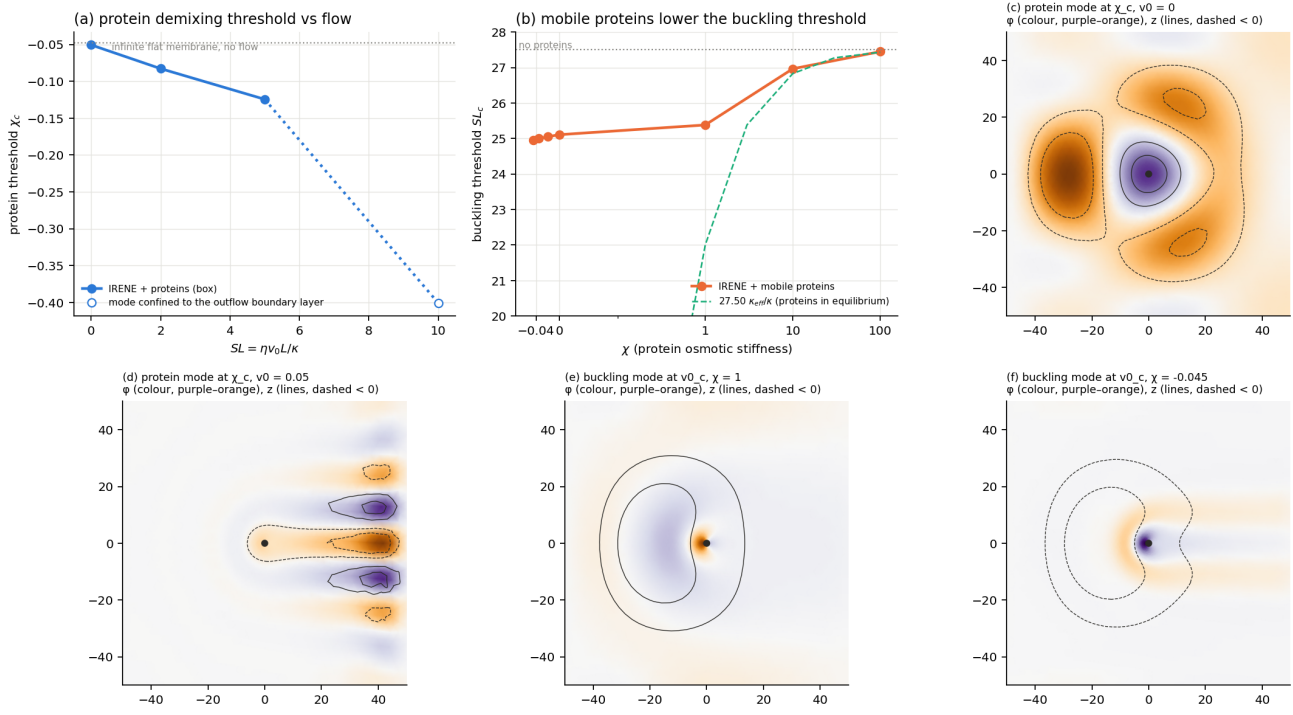


Figure 73: Earlier mobile-protein thresholds with no-flux outflow. The high-flow demixing mode is an outflow-layer artifact. The final absorbing-outflow mobility study replaces those high-flow thresholds.

16.3 Final mobility regimes with absorbing outflow

For $C = 0.25$, $\epsilon_p = 1$, and $\sigma_0 = 0.0025$ in repository units, the no-flow box threshold is $\chi_c = -0.0506$, close to the infinite-plane value -0.0475 [7]. A scale-dependent Péclet number compares residence time ℓ/v_0 with diffusive relaxation of a protein mode. Mobility M is the scanned parameter, not itself a Péclet number.

At $M = 1$, proteins traverse the patch before relaxing into the buckle. With free outflow, χ_c falls to -0.406 at $SL = 20$, and buckling levels off near $SL_c = 25$, only 8–9% below the protein-free value. At $M \geq 100$, the fields cooperate: for $\chi = -0.03$, the threshold reaches 3.2, about 8.6 times lower than 27.5. At $M \simeq 10$, a complex pair describes travelling mechanochemical modes.

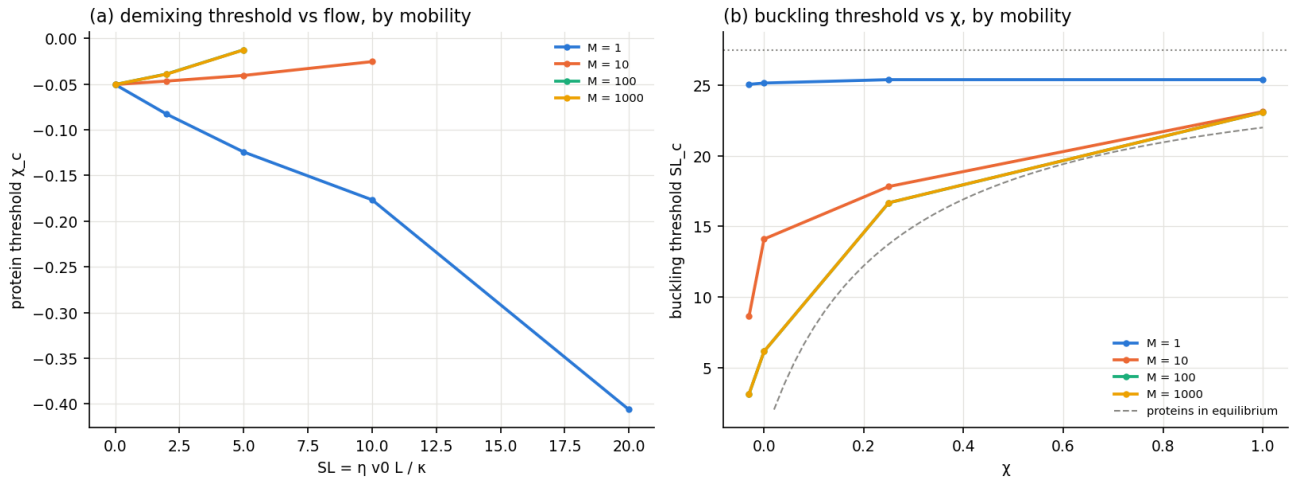


Figure 74: Mobility dependence of demixing and buckling with absorbing outflow. Slow diffusion suppresses global demixing and changes the buckling threshold modestly. Fast diffusion permits curvature relaxation and merges the two instability sectors. Intermediate mobility produces travelling modes. These sweeps use the rigid force-free flat-state condition; the separate codimension-two analysis uses a clamped protein.

Figure 74 establishes three regimes for the final force-free sweep. Slow mobility lets advection flush the patch and suppresses demixing; high mobility permits curvature and composition to cooperate and greatly lowers the threshold; intermediate mobility admits travelling modes. These trends concern a coupled, scale-dependent relaxation process rather than a unique scalar diffusion constant.

17 Mechanochemical mode merger and nonlinear runaway

Intermediate mobility makes the mode structure more than a competition between two unrelated instabilities. A pair of real modes merges into a travelling pair, linking shape buckling and composition dynamics in one spectral family.

17.1 Boundary-specific onsets and destabilizing cubic coefficients

The final mobility sweep uses a rigid force-free flat-state protein; the nonlinear time-step, codimension-two, and source-imperfection analyses use a clamped protein. They illustrate related mechanisms but their detailed thresholds must not be merged into a single quantitative phase boundary.

For $M = 10$, a clamped protein, $C = 0.25$, $\beta = 1$, and absorbing outflow, static onsets occur at $SL = 13.85$ and 13.16 for $\chi = 0.10$ and 0.075 . Their static cubic coefficients are $+7.20 \times 10^{-8}$ and $+6.23 \times 10^{-8}$. At $\chi = 0$, the Hopf onset is 17.83 , $\omega = 5.41 \times 10^{-5}$, with $\text{Re } d = +1.49 \times 10^{-6}$. The evaluated near-Hopf point $(\chi, SL) = (0.05, 19)$ gives $\text{Re } d \simeq +1.46 \times 10^{-6}$, but its linear real part remains slightly negative; it is not an independently refined exact onset. Positive coefficients in $\dot{A} = \mu A + aA^3$ and $\dot{Z} = (\nu + i\omega)Z + d|Z|^2Z$ indicate subcriticality.

Time integration at $(\chi, SL) = (0.1, 16)$ yields exponential growth near 4.9×10^{-6} and a pit approximately $14r_0$ deep; at $(0.05, 20)$ a travelling oscillation gives way to collapse near $|z| \simeq r_0$. The observations agree with destabilizing cubic terms. They do not determine where the runaway ends.

17.2 A candidate degenerate double-zero organizing centre

The static window near $\chi = 0.05$ closes, and two real eigenvalues coalesce into a complex pair. Near $(\chi, SL) = (0.064, 18)$ the recorded eigenvalues are about $+6.8 \times 10^{-7}$ and -2.8×10^{-6} . These are close to zero, not exactly zero. A two-mode Jordan-basis reduction, checked by $Y^T B W = I$ and companion-form $Y^T A W$, motivates the reflection-symmetric normal form

$$\ddot{x} = \mu_1 x + \mu_2 \dot{x} + ax^3 + bx^2 \dot{x} + \dots \quad (35)$$

At nearby evaluated points $(0.06436, 17.989)$ and $(0.08674, 19.989)$, $a = -9.05 \times 10^{-13}$ and $+6.73 \times 10^{-13}$, while $b = -1.36 \times 10^{-6}$ and -5.10×10^{-7} . The exceptionally small, sign-changing a suggests a degenerate Bogdanov-Takens-type point. A precise codimension assessment requires locating the double zero and cubic degeneracy with parameter, mesh, and normalization sensitivity; the evidence should not be presented as an exactly classified point.

The archived cubic unfolding script assumes $a < 0, b < 0$ and predicts saturating sectors on that side. It cannot establish saturation where the directly measured primary coefficient is positive, or at $a = 0$ where quintic terms decide the unfolding. The possible narrow supercritical static interval remains open. Strided real/imaginary NumPy views assigned to FEniCS vectors were found to scramble complex-mode values; the corrected code uses contiguous copies, and pre-fix complex-mode results were discarded in the project analysis.

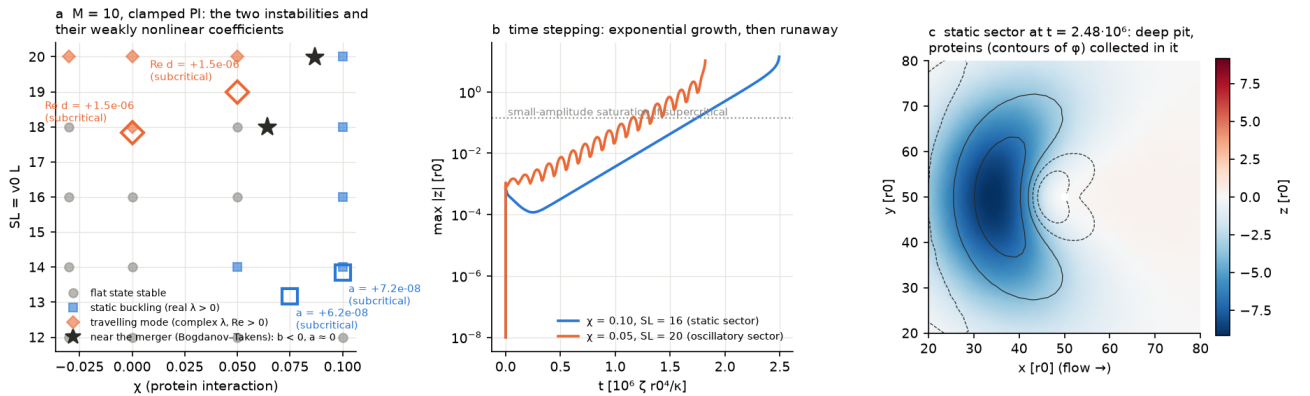


Figure 75: Clamped-protein, intermediate-mobility eigenvalue sectors and nonlinear runaway. The nearly vanishing eigenvalues motivate a degenerate Bogdanov-Takens-type interpretation; an exact double-zero location and quintic unfolding remain unresolved.

17.3 What the nonlinear trajectories resolve

The static-sector and oscillatory-sector trajectories below both proceed toward a deep curvature-associated pit. In the oscillatory case the initial growing travelling pattern does not become the friction-driven saturated wave source described earlier. The finite-time trajectories establish runaway in the computed window, but neither its endpoint nor an exact double-zero classification. The signed composition convention remains essential when relating the pit to molecular enrichment.

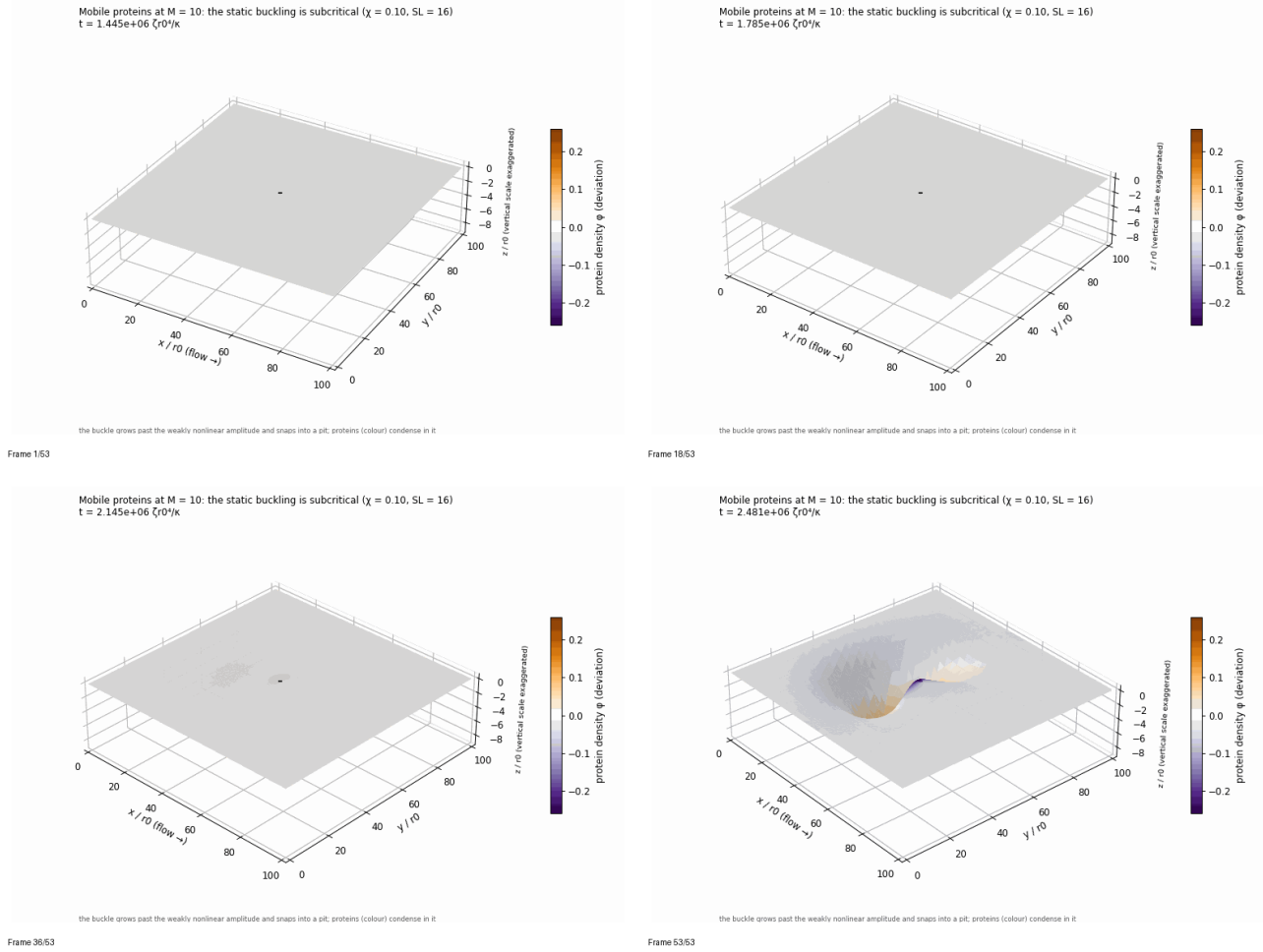


Figure 76: Intermediate-mobility clamped-protein static-sector runaway. Shape and signed density deviation grow into a deep pit; no terminal attractor is resolved. Four ordered frames are selected by frame index; the spacing is not assumed uniform in physical time. The original GIF is included in the source package.

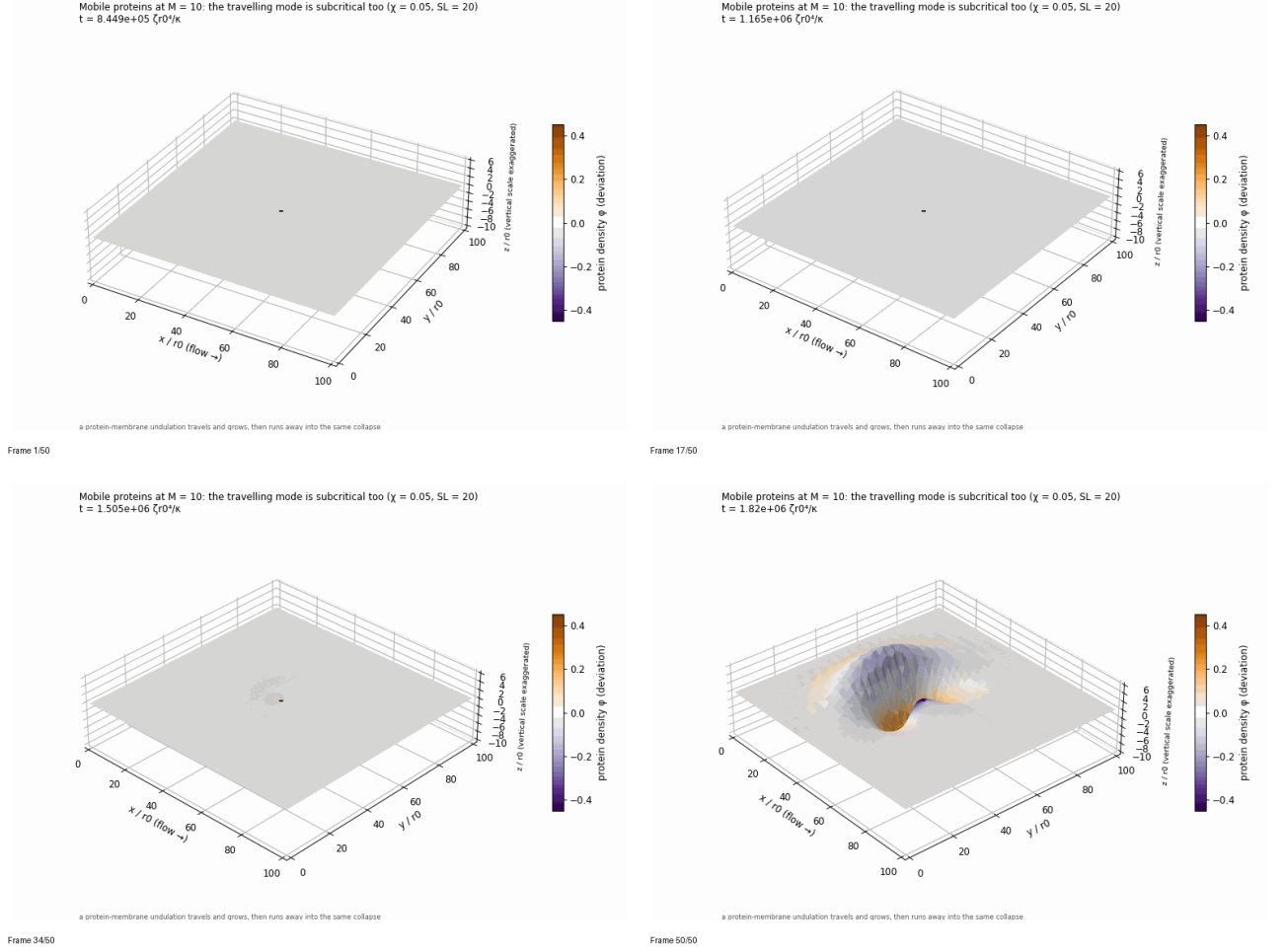


Figure 77: Intermediate-mobility oscillatory-sector runaway. A growing travelling mode evolves into collapse rather than a saturated wave source. Four ordered frames are selected by frame index; the spacing is not assumed uniform in physical time. The original GIF is included in the source package.

18 A recruited protein wake acts as an imperfection

A rim source breaks the reflection symmetry of the perfect buckling problem. Its effect can therefore be compared directly with the geometric contact-slope imperfection rather than described as a separate smoothing mechanism.

18.1 Projected forcing, fold shifts, and snap sharpness

For a rim source with total rate Q , projection gives

$$\dot{A} = \lambda_v(v - v_c)A + aA^3 + H_Q Q + H_s s + \dots \quad (36)$$

At $M = 1$, $\chi = -0.05$, $C = 0.25$, $\beta = 1$, a clamped protein gives $v_c = 0.2439$, $a \simeq 2.0 \times 10^{-7}$, and $H_Q \simeq 3.2 \times 10^{-4}$. With $C = 0$, the implementation reproduces the no-protein reduction to printed precision and $H_Q = 0$. The second-order readjustment contributes about 91% of a , compared with 4% without proteins.

The source fold has $A_f^3 = H_Q Q / (2a)$ and $1 - v_f/v_c = 3aA_f^2 / (\lambda_v v_c)$. The predicted/continued shifts are 0.036/0.037, 0.075/0.0815, and 0.167/0.204 for $Q = 0.01, 0.03, 0.1$. The larger discrepancy at $Q = 0.1$ coincides with a $4.3r_0$ fold deflection outside the smallest-amplitude regime. The coefficient of the normalized backward branch is about 0.0030 versus 0.0018, an increase of roughly 65–67%. The raw cubic coefficient increases by about 21%, not 65%. Contact slope and source add at leading projection order with signs determined by the mode orientation; cancellation has not been separately demonstrated in the archived nonlinear runs.

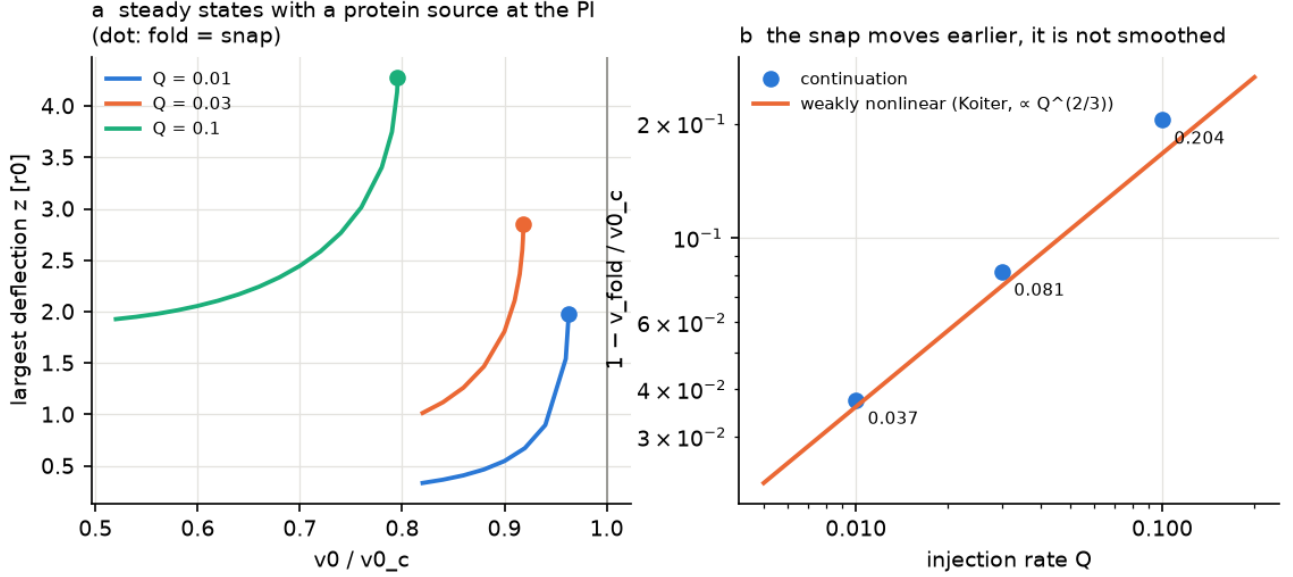


Figure 78: Source-induced folds and the Koiter prediction. Recruitment is an imperfection that advances rather than removes the snap. The largest-source case is outside the smallest-amplitude cubic regime.

The fold data in Figure 78 show that the source advances the fold according to the same two-thirds law. The distinction between the raw cubic and the normalized backward-branch coefficient prevents an apparent discrepancy in the phrase “sharper snap”: the latter changes by about 65 percent, while the raw cubic changes by about 21 percent. At the largest source rate the deflection is already outside the smallest-amplitude regime, explaining why the cubic fold prediction underestimates the observed shift.

19 Periodic anchored arrays and long-wave collective onset

A periodic lattice replaces the arbitrary distant boundary of an isolated protein by a physical cell density. The translational zero mode makes the long-wave expansion, rather than the smallest returned eigenvalue alone, the correct onset diagnostic.

19.1 Bloch perturbations and density-controlled thresholds

Periodic cells contain one rigid horizontal force-free disc and have mean stress σ_0 . Flow is driven by a uniform tangential force or friction relative to a moving outer fluid. For $z' = e^{i\mathbf{q}\cdot\mathbf{x}}p(\mathbf{x})$, covariant Fourier shifts are implemented through $\nabla \mapsto \nabla + i\mathbf{q}$ in the plate problem, with real/imaginary doubled fields. At $\mathbf{q} = 0$, uniform vertical translation is neutral. An onset criterion therefore tracks the q^2 coefficient, not the unavoidable zero eigenvalue.

The small- q longitudinal band has

$$\lambda(q) = -\frac{\sigma_{\text{eff}}}{\zeta}q^2 - \frac{K_{\text{eff}}}{\zeta}q^4 - icq + \dots \quad (37)$$

An array-drag benchmark agrees with its reference values to about 3×10^{-4} . The periodic problem is recovered at $q = 0$, and reciprocal-lattice periodicity is checked to about 10^{-3} . These are finite-resolution tests. Very dilute thresholds for $L_c \geq 40r_0$ come from the dedicated long-wave calculation, not solely the earlier finite- q grid; the latter can select a nonzero grid wavevector even when the true onset is long-wave.

L_c/r_0	Φ	$U_c\eta r_0/\kappa$	$F_{\text{drag},c}r_0/\kappa$	K_{eff}/κ
6	0.087	1.00	18.6	1.8
8	0.049	0.74	10.3	1.3
14	0.016	0.45	4.1	1.05
28	0.0040	0.28	1.75	1.04
56	0.0010	0.24	1.09	1.15
80	0.0005	0.24	1.00	1.3

Table 3: Long-wave lattice thresholds at $\sigma_0 = 0.0025\kappa/r_0^2$. Force is per disc. Dense-lattice values do not obey a $1 \times \kappa/r_0$ criterion.

Different drives give close thresholds in their shared tested range, but strong screening changes the very dilute regime: at $\ell_b = 5r_0$, $L_c/r_0 = 56, 80$, the critical speed rises to about 0.6, 0.5. Removing arbitrary box dependence does not remove density, drive, screening, or finite-patch dependence.

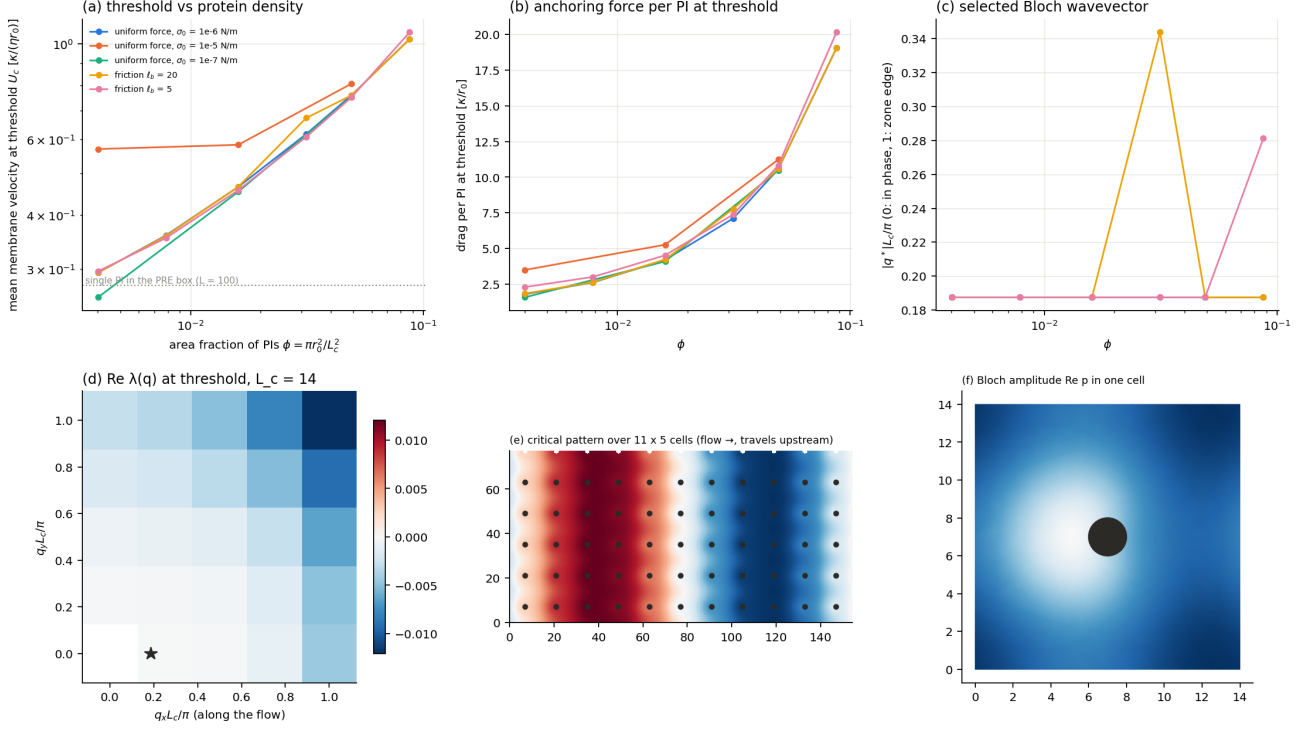


Figure 79: Periodic array spectra and patterns. Early finite-wavevector scans are interpreted with the dedicated later long-wave coefficients: a neutral uniform translation exists at zero wavevector, and the longitudinal stiffness controls onset.

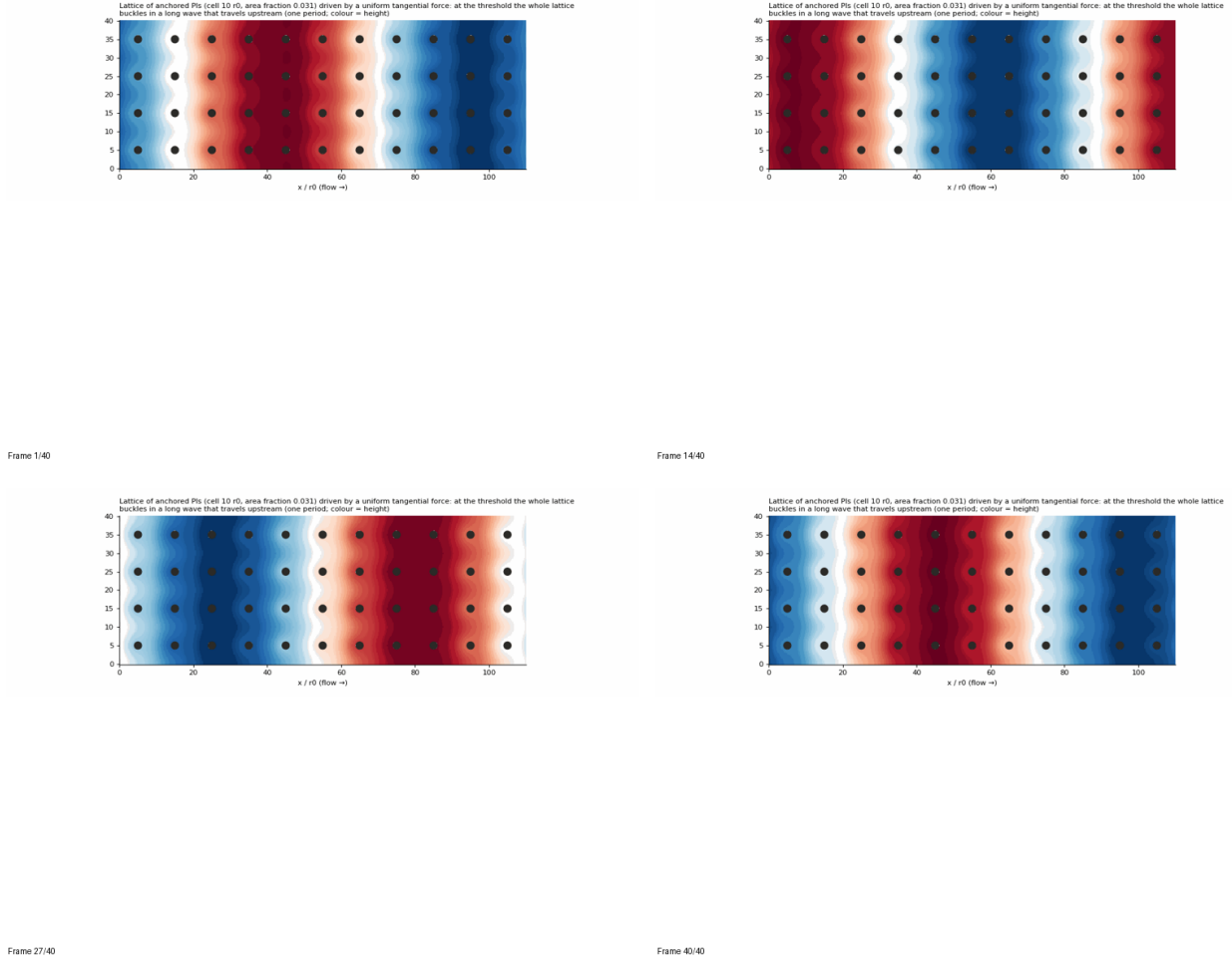


Figure 80: Collective Bloch-wave pattern of a periodic protein array. This visualizes the long-wave instability and upstream phase drift, not a nonlinear terrace simulation. Four ordered frames are selected by frame index; the spacing is not assumed uniform in physical time. The original GIF is included in the source package.

The dilute limit links the per-cell drag to the finite-cluster scale, while the dense limit requires much larger drag per inclusion. The finite-wavevector images preserve the early spectral exploration; the dedicated long-wave calculation determines the later dilute thresholds and avoids mistaking the inevitable translation mode for buckling.

20 Nonlinear lattice stresses and one-dimensional terraces

Linear Bloch coefficients do not determine whether a collective instability saturates. Nonlinear tilted-cell calculations provide an approximate stress law for the slowly varying membrane slope.

20.1 Tilted cells, rigid-rim geometry, and frozen-flow closure

The static tilted-cell ansatz is $z = ux + p(x, y)$ with periodic p . A rigid horizontal disc at x_c requires $p = h - u(x - x_c)$ on its rim, not merely $p = h$. The height is a Real unknown and the normal force balance uses the nonlinear membrane geometry. The flat in-plane flow and tension are frozen. The macroscopic vertical stress $J(u)$ is averaged over strips away from the disc.

At rest, $J(u)/u$ agrees at small slope with $2E/(A_{\text{cell}}u^2)$ and with the normalized Bloch stiffness. A 5–10% mismatch remains between the driven static-cell linear coefficient and the Bloch threshold, because the drifting dynamic cell and flow readjustment are omitted. The nonlinear coefficients are therefore a homogenized approximation derived from the model, not a parameter-free exact reduction at arbitrary drive.

The small-slope fitting window $u \leq 0.1$ gives cubic coefficients approximately $+1.68, -0.08, -0.56, -0.76, -0.86$ at $L_c/r_0 = 6, 6.5, 7, 7.5, 8$. This identifies a sign change near $\Phi = 0.075$. Cubic/quintic fits predict continuous terraces for 8.7%, hysteretic steep terraces near 4.9%, and unsaturated growth for $\Phi \leq 1.6\%$. Extrapolated folds at 3.1% involve slopes beyond the computed fit window and require qualification.

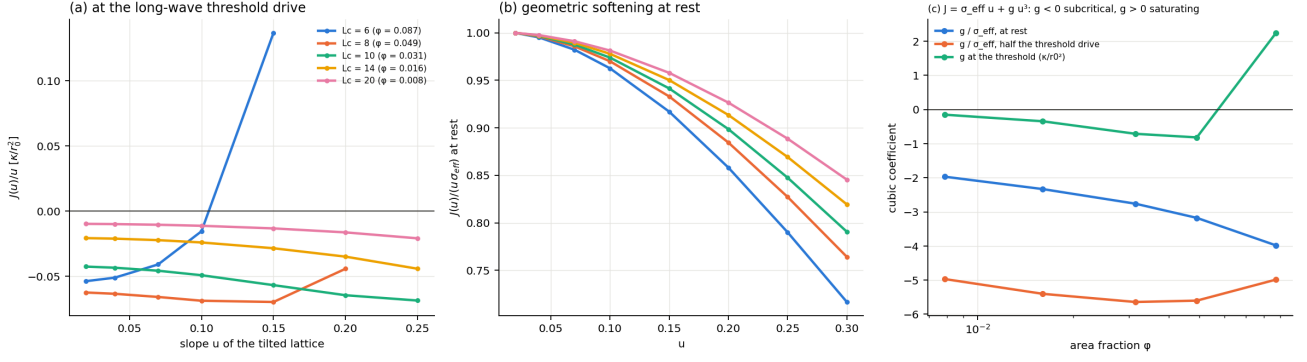


Figure 81: Nonlinear tilted-cell stress and geometric softening. Frozen flat-state flow is used; the small-slope cubic changes sign between intermediate and dense arrays, while the driven linear coefficient differs modestly from the dynamic Bloch value.

20.2 A convected slope equation and density-dependent criticality

For mean lattice height Z and slope $u = Z_x$, Bloch coefficients and nonlinear tilted-cell stresses give

$$\zeta Z_t = \partial_x J(u) - K_{\text{eff}} Z_{xxx} - \zeta c Z_x, \quad J(u) = \sigma_{\text{eff}} u + g_3 u^3 + g_5 u^5. \quad (38)$$

Writing the relative drive as $\delta = D/D_{\text{lw}} - 1$ gives $\sigma_{\text{eff}} = -\delta \sigma_{\text{rest}}$. Differentiation yields a convected Cahn–Hilliard-type equation for the slope. Drift affects disturbance survival in a finite patch.

Small-slope cell fits place the sign change of g_3 near $\Phi_* \simeq 0.075$. At $\Phi = 8.7\%$, stable slopes emerge continuously in the one-dimensional reduction and form terraces. At $\Phi = 4.9\%$, steep terraces persist below onset with hysteresis; at $\Phi \leq 1.6\%$, the computed terms do not saturate growth. These are reduced-equation results with frozen flat-state flow.

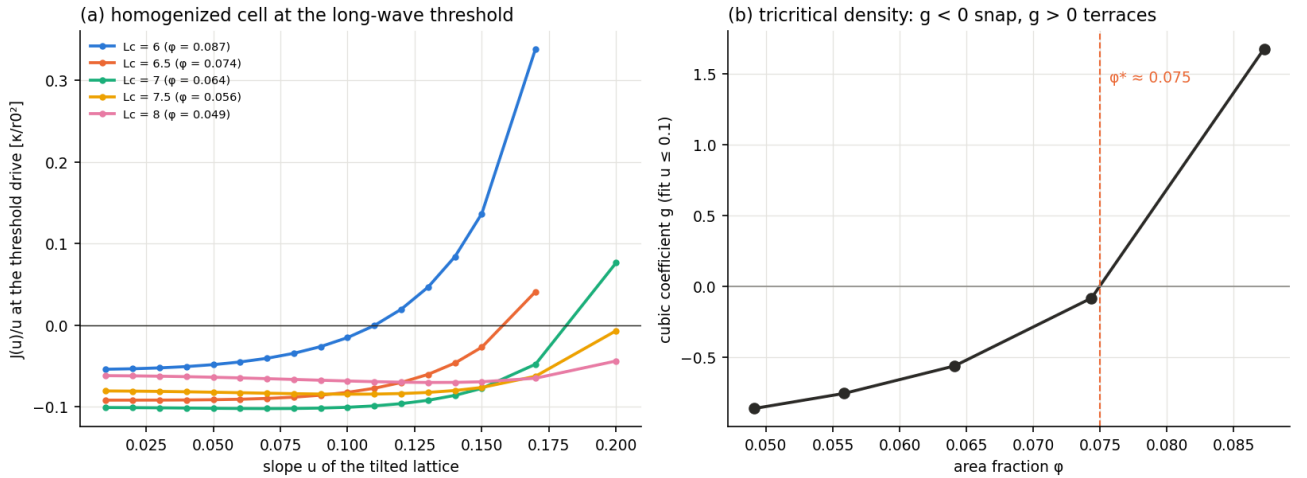


Figure 82: Small-slope nonlinear-cell fits locating the cubic sign change near anchored-disc area fraction 7.5 percent. This onset diagnostic uses a narrower fitting window than the later two-dimensional finite-amplitude polynomial.

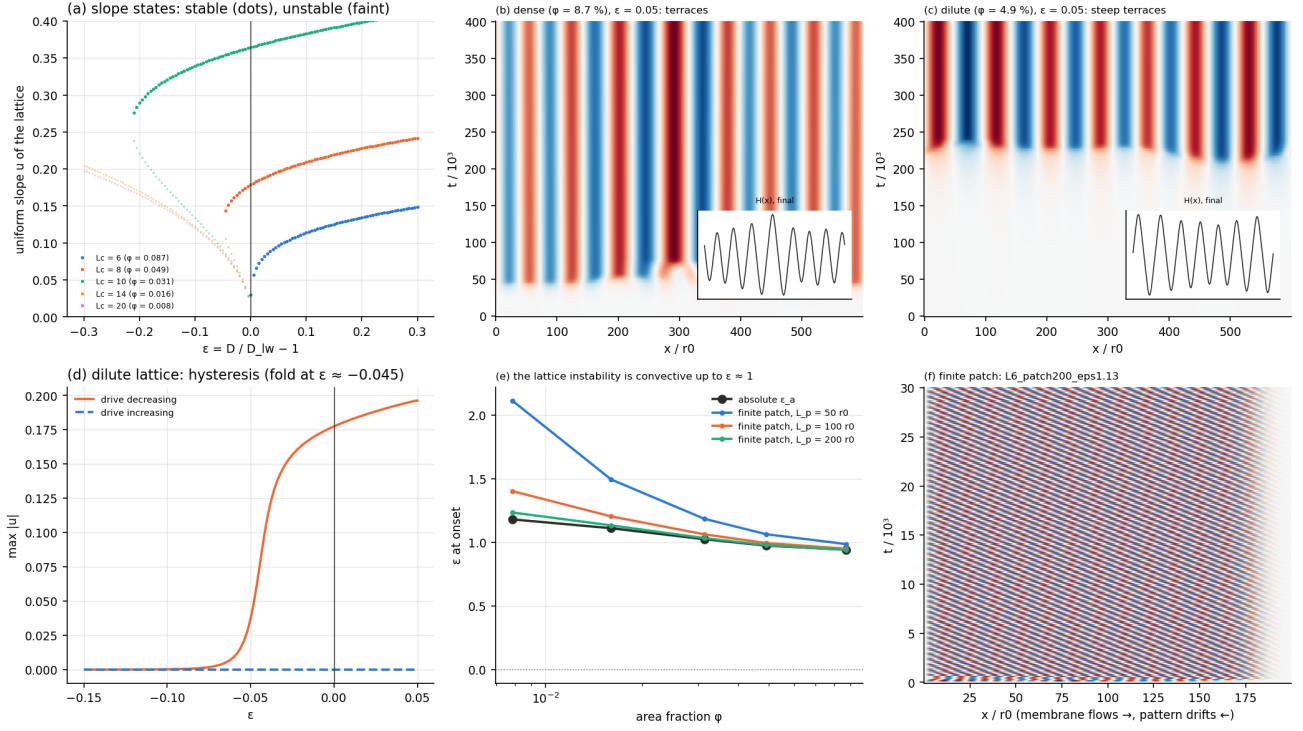


Figure 83: One-dimensional homogenized lattice dynamics: continuous or hysteretic slope states, coarsening terraces, and finite-patch drift. The absolute-threshold estimate uses a long-wave model near the edge of its spatial validity.

The dense one-dimensional reduction has continuous terrace formation, the intermediate-density equation has steep hysteretic terraces, and sufficiently dilute closures do not arrest growth. These are outcomes of the fitted homogenized equation. The small-slope fitting window determines onset criticality; extrapolating its polynomial far beyond that window requires a separate accuracy assessment.

21 Two-dimensional terraces and finite patches

Two questions remain after one-dimensional terrace formation: do ridges survive transverse perturbations, and can a pattern stay in a finite patch while drifting upstream? These tests connect local saturation to spatial persistence.

21.1 Anisotropic coefficients and transverse stability

For $Z(x, y)$ and $\mathbf{u} = \nabla Z$, the implemented equation is

$$\begin{aligned} \zeta Z_t &= \partial_x J_x + \partial_y J_y - (K_{xx} \partial_x^4 + K_m \partial_x^2 \partial_y^2 + K_{yy} \partial_y^4) Z - \zeta c Z_x, \\ J_x &= u_x (\sigma_{xx} + a_1 u_x^2 + a_2 u_y^2 + a_3 u_x^4 + a_4 u_x^2 u_y^2 + a_5 u_y^4), \\ J_y &= u_y (\sigma_{yy} + b_1 u_y^2 + b_2 u_x^2 + b_3 u_y^4 + b_4 u_x^2 u_y^2 + b_5 u_x^4). \end{aligned} \quad (39)$$

Bloch bands provide anisotropic linear terms; cell tilts at 0, 30, 45, 60, 90 degrees provide nonlinear terms. The q^2 coefficient obeys $\sigma(\theta) = \sigma_{xx} \cos^2 \theta + \sigma_{yy} \sin^2 \theta$. At onset, $\sigma_{yy} \simeq 0.677$ for $L_c = 6$ and 0.255 for 8. The corresponding (K_{xx}, K_m, K_{yy}) are (1.80, 7.0, 0.23) and (1.28, 5.0, 0.41).

An important distinction emerges from the archived coefficients: the two-dimensional fit uses slopes up to 0.2 and gives $a_1 = -7.76$, $a_3 = 811$ at $L_c = 6$, unlike the positive small-slope one-dimensional cubic. It still produces stable finite slopes near 0.112 and 0.136 at relative drives 0.05 and 0.2. This demonstrates finite-amplitude terrace stability in that fitted equation, not an independent verification of the supercritical onset or tricritical density. Onset criticality must use a controlled shrinking slope window; a global polynomial fit can trade cubic against quintic terms on a steeply stiffening branch.

For a terrace $\mathbf{u} = (u_s, 0)$, transverse stiffness is $D_{yy} = \sigma_{yy} + b_2 u_s^2 + b_5 u_s^4$. It remains positive in the tested cases: approximately 0.62–0.56 for the dense lattice and 0.12–0.07 for the 4.9% lattice. The two-dimensional runs relax to straight ridges along y . A local seed spreads laterally at 0.08865, compared with $2\sqrt{\sigma_{yy} \lambda_{\max}} = 0.08369$ in the

normalized units. This pulled-front estimate neglects transverse fourth-order terms in the leading envelope; the roughly 6% agreement is empirical support in the tested regime.

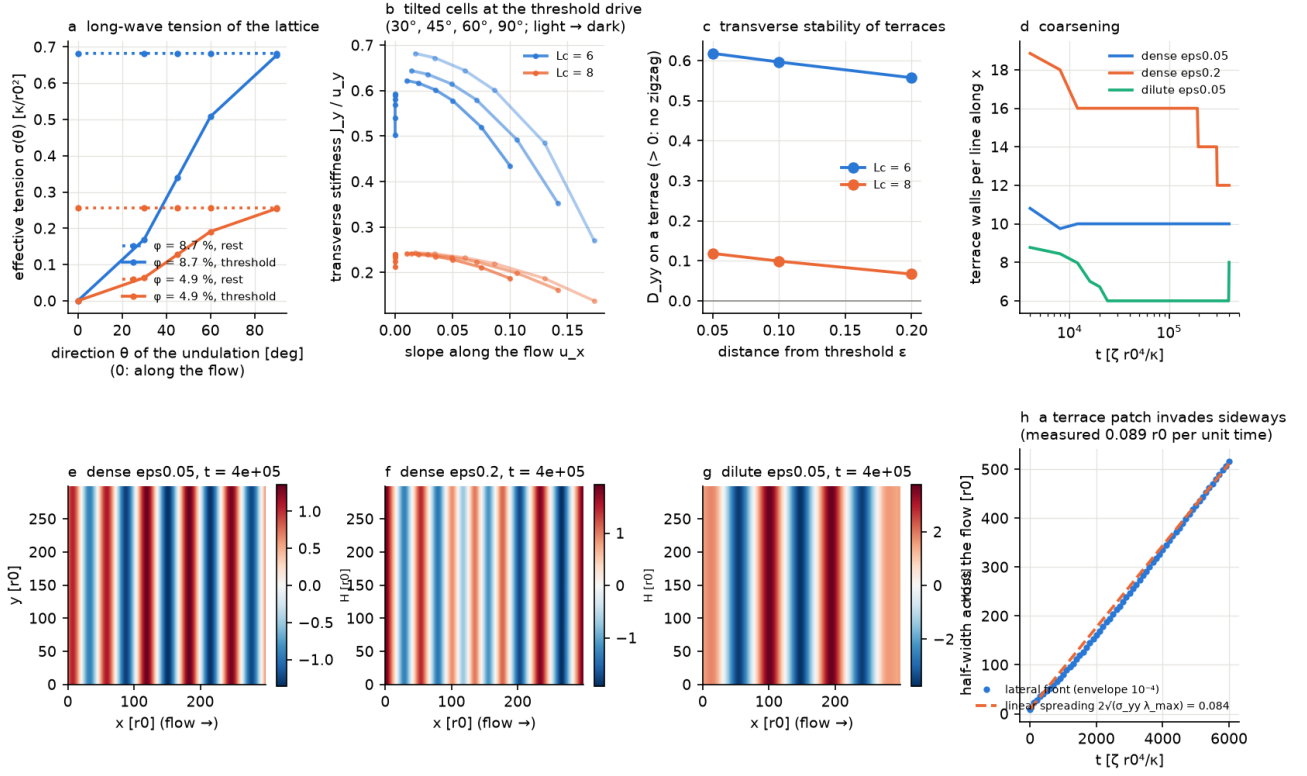


Figure 84: Two-dimensional lattice reduction. Long-wave stiffness is softened along the flow but remains positive transversely. Straight terrace ridges run across the flow. Positive transverse terrace stiffness excludes a zigzag in the tested dense and intermediate-density cases. A localized patch spreads laterally at a speed close to the linear pulled-front prediction.

21.2 Convective versus absolute instability

For $\sigma_{\text{eff}} < 0$, rescale the one-dimensional dispersion to $\Lambda(k) = k^2 - k^4 - iCk$, with

$$C = \frac{\zeta|c|\sqrt{K_{\text{eff}}}}{|\sigma_{\text{eff}}|^{3/2}}. \quad (40)$$

A saddle/pinch satisfying $\partial_k \Lambda = 0$ and $\text{Re } \Lambda = 0$ gives $C_a = 1.622$. The impulse response is absolutely unstable for smaller scaled drift, and convective for larger drift [8]. If $\sigma_{\text{eff}} = -\delta\sigma_{\text{rest}}$ and the drift is held at its reference value, then

$$\delta_a = \frac{1}{\sigma_{\text{rest}}} \left(\frac{\zeta|c|\sqrt{K_{\text{eff}}}}{C_a} \right)^{2/3}. \quad (41)$$

The repository takes $\zeta = 1$ and obtains approximately 0.94–1.18. Clamped reduced-model patches at dense-lattice lengths 50, 100, 200 r_0 approach 0.99, 0.95, 0.94. Very long non-normal patch spectra are contaminated by round-off and are excluded.

At $\delta \simeq 1$, the intrinsic length $\sqrt{K_{\text{eff}}/|\sigma_{\text{eff}}|}$ is only 1.6–2 r_0 , below the cell size. Freezing the drift and extending a long-wave expansion this far are limitations. Accordingly the approximately doubled onset is a physically motivated drift trend of the reduced equation, requiring finite-cell/full-model confirmation before quantitative use.

The dense and intermediate-density results therefore answer distinct questions: the nonlinear closure permits persistent terrace slopes, the transverse calculation tests ridge stability, and the patch calculation tests whether drift carries disturbances out of the observed region. A future finite-cell patch calculation must preserve all three aspects.

22 Physical scales, synthesis, and reproducibility

The calculations now connect a verified numerical method to a physical mechanism, a snap-through law, and two qualitatively different nonlinear outcomes: collapse in the unscreened reference problem and saturated waves in the tested friction window. The mechanochemical and lattice extensions add transport and collective organization while retaining identifiable limits.

22.1 Dimensional force, velocity, and time scales

Using $k_B T = 4.14 \times 10^{-21}$ J at 300 K, $\kappa = 10k_B T$, $r_0 = 10$ nm, $\eta = 10^{-8}$ Pa s m gives

$$\frac{\kappa}{r_0} = 4.14 \text{ pN}, \quad \frac{\kappa}{\eta r_0} = 414 \mu\text{m/s}, \quad \frac{\kappa}{\eta L} = 4.14 \mu\text{m/s} \quad (L = 1 \mu\text{m}). \quad (42)$$

Thus $SL_c = 27.45$ maps to $113.6 \mu\text{m/s}$. The 4 pN drag criterion is specific to the low-tension reference and dilute/finite-cluster limits. Higher tension raises the single-protein drag substantially; the archived physical tables give about 42 pN at 10^{-4} N/m. Normal rates use $\tau_n = \zeta r_0^4 / \kappa$ independently of the tangential velocity unit.

A solvent drag is nonlocal and wavelength-dependent; replacing it by a constant ζ gives only an order-of-magnitude frequency conversion. The approximate millisecond estimate in the interpretation is not a verified biological flutter frequency. Likewise pN forces compatible with molecular processes do not imply that the reference $100 \mu\text{m/s}$ membrane flow is routine in cells.

22.2 Established conclusions and remaining scientific questions

The strongest conclusions are the control-ensemble distinction, physical-condition reference thresholds, compression mechanism, independently computed subcritical coefficient and small-imperfection law, and saturated flutter in the tested configuration. Mobile-protein transport regimes and the nonlinear runaway are demonstrated within the open-domain model. The double-zero classification, tiny-coefficient sign near its merger, and any gentle interval require refinement and quintic terms. Density-dependent terraces are results of homogenized equations with frozen flow; full nonlinear lattice validation is still needed. The overhang is established, but its final depth and tube evolution are limited by distortion and force consistency.

Before a journal submission, the project checklist calls for agreement on scope/authorship, joint review of the rim/traction/force diagnostics, corrected-condition penalty and domain robustness, and a reproducible environment. This report completes the synthesis and editorial review; it does not replace those outstanding scientific and collaborative decisions.

A core manuscript can be organized around corrected physical force balance, static control ensembles, compression-driven buckling, the approximate drag scale, the computed imperfection law, and flutter. Mobile composition, mode merger, wake forcing, and collective terrace dynamics provide a coherent follow-up. The present integrated report keeps their common framework visible while preserving the boundary-condition, approximation, and evidence distinctions needed to separate those manuscripts responsibly.

22.3 Implementation and evidence map

The report uses the complete interpretation and paper plan, all round-status notes and the README, the generic eigensolver, ring and flow problem interfaces, force current, plate/base-flow library, weakly nonlinear projection, nonlinear protein forms, two-dimensional slope closure, axisymmetric tube equations, codimension-two/unfolding code, plotting scripts, summaries, and selected archived raw checks. Legacy 2025 teaching/solver branches provide context but do not supply the new numerical claims. The computational sweeps were not rerun during report preparation; numerical values are audited against committed outputs and code conventions.

Result	Principal implementation under Ruben/membrane_stability/	Archived evidence
Generic spectrum	irene_addon/modules/stability/linear_ stability.py	Cylinder stability/README.md, spectra and DNS figures
Ring force/stability	irene_addon/stability/no_flow/ring_problem. py, force_displacement_stability.py	results/summary.json, results/checks/
Force consistency	check_force_energy.py, check_force_exact.py, modules/stability/vertical_force.py within add-on	Exact-force logs; cutoff checks

Result	Principal implementation under Ruben/membrane_stability/	Archived evidence
Corrected flow onset	irene_addon/stability/flow/flow_problem.py, flow_critical.py, flow_plate_model.py	results/flow2/, flow2_summary.json
Snap theory/branches	flow/flow_landau.py, flow_branch.py, flow_forcefree.py under add-on stability	results/round3/, round3_summary.json
Collapse and lift	flow/flow_snap.py, flow_snap_ff.py	round3_summary.json, round4_summary.json
Overhang	irene_addon/stability/parametric/param_ problem.py, param_snap.py	results/round5/param_ff_v0. 97r/; round 5 fold summary
Flutter	plate/study_flutter_map.py, flow/flow_flutter.py	Round 3 map; round 5 flutter summary/time series
Multiple proteins	plate/study_multi.py	Round 3 multi data/summary
Ring tubes	axisymmetric/ring_tube.py	Round 3 tube branches and physical tables
Lattice spectrum	plate/study_lattice.py, lattice_longwave.py	Round 3 cell/long-wave outputs
Lattice nonlinear closure	plate/nonlinear_cell.py, slope_equation.py, slope_equation_2d.py	Round 4–6 coefficients and simulations
Mobile proteins	irene_addon/stability/flow_phi/flow_problem. py, phi_stability.py, phi_dynamics.py	Round 4–5 thresholds and dynamics
Wake and merger	flow_phi/imperfection.py, codim2.py, bt_unfolding.py	round6_summary.json and round 6 raw outputs

Table 4: Implementation-to-evidence map. Directory prefixes are shortened where the surrounding entry supplies them; exact source paths are also recorded in the separate evidence audit.

22.4 Reproduction of numerical results and document sources

The plotting entry points are `plot_membrane_results.py`, `plot_flow_results.py`, `plot_flow_v2.py`, and `plot_round3.py` through `plot_round6.py`. They read committed `results/` files. Some scripts retain defaults for the original runtime under `/tmp/claude-0/`; new reproductions must supply portable paths or adapt those defaults. This report copies the existing raster figures unchanged and records checksums and first-addition dates in `figure_manifest.json`.

Full IRENE runs require a compatible external IRENE checkout via `IRENE_ROOT`; the add-on imports its modules and solver forms. The historical root solver files of `Copy_Stage` are not a substitute for pinning the external module layout expected by the add-on. Dependencies include classic FEniCS/dolfin, UFL (with the provided legacy shim when appropriate), PETSc/SLEPc, MUMPS, NumPy/SciPy, gmsh, meshio, and plotting tools. Pinning the exact upstream revision and dependency versions is an outstanding reproducibility deliverable.

Representative commands are preserved in the project README for flat spectra, ring branches, physical flow onset, plate reductions, and sensitivities. Their runs are potentially expensive and were not launched to prepare the report. The source package includes a simple LaTeX build script; compilation itself needs no FEniCS installation.

The figure manifest records original source paths, first-addition dates, dimensions, frame counts, checksums, and each integrated figure number. All original image bytes are preserved. The GIF panels are ordered illustrative samples, not newly measured physical-time series. The earlier main report and supplement are retained unchanged; this document is a new synthesis with one continuous set of sections, equations, and figures. Authorship and affiliations must be supplied after the collaborative scope is agreed.

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